Supplementary Material: FL-EHDS: A Privacy-Preserving Federated Learning Framework for the European Health Data Space

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***Abstract*—This document provides supplementary material for the FL-EHDS paper, including complete algorithm pseudocode for all framework components, extended experimental figures, detailed algorithm comparison analysis with effect sizes (rank-biserial correlation), 10-seed statistical validation, advanced FL paradigm descriptions, infrastructure component specifications, extended EHDS interoperability details, clinical imaging experiment configurations with Chest X-ray baseline results, 10-seed Wilcoxon validation on imaging (90 experiments confirming Ditto's superiority and HPFL's modality-dependent reversal), Byzantine robustness on medical imaging (108 experiments with Krum defense), local epochs sweep, scalability analysis up to K=100 clients, noise-as-regularization analysis for small-cohort studies, Article 71 opt-out impact with policy recommendations, an extended threat model with attack taxonomy and defense mapping, and a detailed framework positioning analysis. The open-source reference implementation (∼40K lines, 159 modules) is available at https://github.com/FabioLiberti/FL-EHDS-FLICS2026.**

I. PRISMA Flow Diagram

II. Algorithm Pseudocode

This section provides formal algorithmic descriptions of all FL-EHDS framework components. Each algorithm is presented with: (1) a contextual explanation of *why* the component is needed in the EHDS regulatory context; (2) the formal pseudocode; and (3) practical considerations for deployment. The algorithms are organized following the data flow through the three-layer architecture: governance validation (Layer 1), privacy-preserving aggregation (Layer 2), and local data processing (Layer 3).

Reading guide: Algorithms S1–S4 form the core FL-EHDS training pipeline. Algorithms S5–S6 address EHDS-specific challenges (non-IID data and citizen opt-out). Algorithms S7–S8 handle data preprocessing and privacy budget management.

A. FedAvg with EHDS Compliance

Algorithm S1 presents the core federated averaging procedure adapted for EHDS regulatory requirements, operating in a client-server architecture where the central aggregator coordinates training across distributed hospital nodes within a Secure Processing Environment.

Key Design Decisions:

- **ValidatePermit**: Before each round, the HDAB-issued permit is verified against temporal bounds and Article 53 permitted purposes.
- **SelectParticipants**: Configurable client selection—full participation or sampling for large federations.
- **FilterOptedOut**: Records from citizens who exercised Article 71 opt-out rights are excluded *before* gradient computation.
- **Weighted Aggregation**: Gradients weighted by local dataset size (n_h), following original FedAvg [13].
- **ClipGradient**: L2-norm clipping bounds individual contributions, providing sensitivity bounds for DP.

Relationship to subsequent components: The `ClipGradient` operation in Algorithm S1 establishes a bounded sensitivity C for each client's contribution. This bound is the prerequisite for Algorithm S2 (Gaussian DP), which calibrates noise proportional to C. Meanwhile, `ValidatePermit` invokes Algorithm S3 (Permit Validation) and `FilterOptedOut` invokes Algorithm S6 (Opt-Out Filtering).

B. Gaussian Differential Privacy Mechanism

Algorithm S2 implements the Gaussian mechanism for differential privacy, applied at the aggregation server after receiving clipped gradients.

Mathematical Foundation: The noise scale $\sigma = C \cdot \sqrt{2\ln(1.25/\delta)}/\varepsilon$ guarantees (ε, δ)-DP. The cumulative privacy expenditure is tracked using Rényi DP (RDP) [26] composition, providing 5–6× tighter bounds than naive composition.

Practical Considerations:

- $\varepsilon = 10$: moderate noise, <2pp accuracy drop on PTB-XL and Cardiovascular (see Table S-X in Section XV)
- $\varepsilon = 1$: strong privacy; personalized methods (Ditto, HPFL) retain 87–89% on PTB-XL while FedAvg collapses to 52%
- The ε selection must be negotiated with HDABs during permit approval

Integration with privacy budget: The ε consumed by Algorithm S2 in each round is tracked by Algorithm S8 (RDP

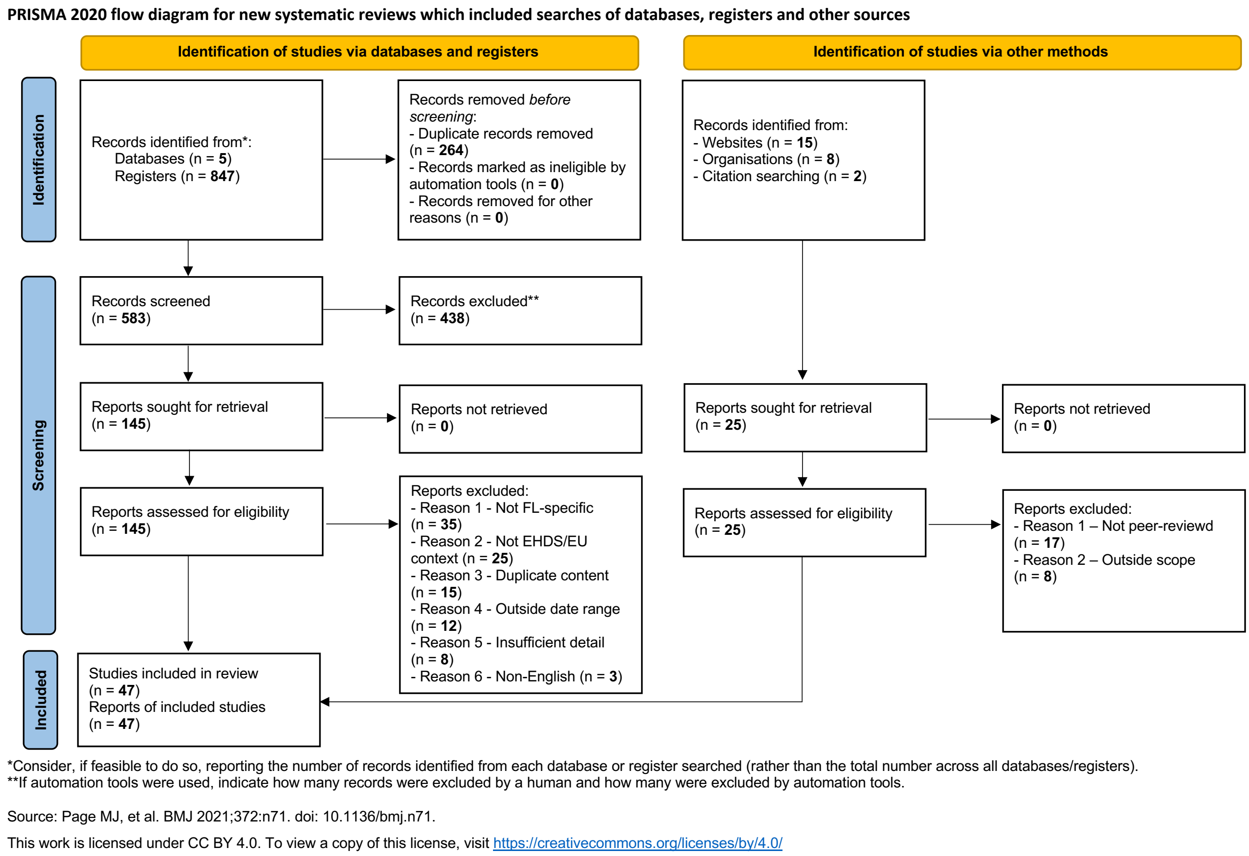

Fig. S-1. PRISMA 2020 flow diagram for the systematic review. Database searches across PubMed, IEEE Xplore, Scopus, Web of Science, and arXiv identified 847 records; after deduplication (264 removed) and screening, 47 studies met inclusion criteria (2022–2026, FL/EHDS focus, peer-reviewed or recognized institutional origin). Additional records from institutional websites (n=15), organisations (n=8), and citation searching (n=2) were assessed but did not contribute to the final inclusion set. Adapted from Page et al. (BMJ 2021;372:n71), licensed under CC BY 4.0.

Privacy Budget Accountant). If the cumulative budget exceeds the threshold approved in the HDAB data permit, training is automatically terminated. The next algorithm (S3) formalizes the permit validation that authorizes each round.

C. HDAB Permit Validation

Algorithm S3 ensures all FL operations comply with the data permit issued by HDABs. Under EHDS Article 53, secondary use of health data is only lawful for specifically enumerated purposes (scientific research, public health surveillance, AI training for health). The permit validation module is invoked at the beginning of every FL round—not just at training start—to guarantee continuous compliance even if a permit is revoked mid-study or its temporal validity expires. Each validation event is logged as a GDPR Article 30 processing record, creating an immutable audit trail that regulators can inspect.

EHDS Governance Role: This algorithm is the enforcement point between the Governance Layer (Layer 1) and the FL Orchestration Layer (Layer 2). Without it, a permit expiring at round 15 of a 20-round study would allow unauthorized data processing for rounds 16–20.

Failure modes: Each exception type triggers a different response: `PermitExpiredError` terminates the entire study; `PurposeMismatchError` indicates a configuration error requiring researcher intervention; `UnauthorizedCategoryError` may allow continued training on the authorized subset of categories. All failure events are logged for regulatory audit. Once a round passes permit validation, the aggregation server collects client gradients using the secure protocol described next.

D. Secure Aggregation Protocol

Even though FL prevents raw data sharing, the gradient updates themselves can leak patient information: Zhu et al. [21] demonstrate that gradients can be inverted to reconstruct training images. In the EHDS context, where gradients encode patterns from sensitive health records across 27 Member States, this is an unacceptable privacy risk. Secure aggregation addresses this by ensuring the SPE aggregation server

Algorithm S1: FL-EHDS FedAvg Training
Input: Hospitals $\mathcal{H} = \{h_1, \ldots, h_K\}$, permit P, rounds T
Output: Global model $\theta^{(T)}$

Server executes:
  Initialize $\theta^{(0)}$
  for round $t = 1$ to T **do**
    // Governance check (Layer 1)
    if not ValidatePermit(P, t) **then abort**
    $\mathcal{H}_t \leftarrow$ SelectParticipants($\mathcal{H}$)
    for each hospital $h \in \mathcal{H}_t$ **in parallel do**
      $\Delta_h^{(t)}, n_h \leftarrow$ LocalTrain(h, $\theta^{(t-1)}$)
    // Aggregation with privacy (Layer 2)
    $\theta^{(t)} \leftarrow \theta^{(t-1)} + \frac{1}{\sum_h n_h} \sum_{h \in \mathcal{H}_t} n_h \cdot \Delta_h^{(t)}$
    LogCompliance(t, $\mathcal{H}_t$)
  return $\theta^{(T)}$

LocalTrain(h, θ) **at hospital** h:
  // Opt-out filtering (Article 71)
  $\mathcal{D}_h \leftarrow$ FilterOptedOut($\mathcal{D}_h$, OptOutRegistry)
  $\theta_h \leftarrow \theta$
  for epoch $e = 1$ to E **do**
    for batch $\mathcal{B} \in \mathcal{D}_h$ **do**
      $\theta_h \leftarrow \theta_h - \eta \nabla \mathcal{L}(\theta_h; \mathcal{B})$
  $\Delta_h \leftarrow \theta_h - \theta$
  // Privacy protection (Layer 3)
  $\Delta_h \leftarrow$ ClipGradient(Δ_h, C)
  return Δ_h, $|\mathcal{D}_h|$

Algorithm S2: Gaussian DP Mechanism
Input: Gradient Δ, sensitivity C, privacy budget ε, δ
Output: Noisy gradient $\tilde{\Delta}$

// Compute noise scale from Gaussian mechanism
$\sigma \leftarrow C \cdot \sqrt{2 \ln(1.25/\delta)}/\varepsilon$
// Add calibrated Gaussian noise to each parameter
for each parameter $w \in \Delta$ **do**
  $\tilde{w} \leftarrow w + \mathcal{N}(0, \sigma^2)$
// Track cumulative privacy expenditure
PrivacyAccountant.spend(ε)
if PrivacyAccountant.budget_exhausted() **then**
  raise PrivacyBudgetExhaustedError
return $\tilde{\Delta}$

Algorithm S3: Data Permit Validation
Input: Permit P, round t, requested categories $\mathcal{C}$
Output: Boolean validity

// Check temporal validity
if CurrentTime() $>$ P.valid_until **then**
  raise PermitExpiredError
// Check purpose alignment (Article 53)
if P.purpose $\notin$ AllowedPurposes **then**
  raise PurposeMismatchError
// Check data category authorization
for each category $c \in \mathcal{C}$ **do**
  if $c \notin P$.authorized_categories **then**
    raise UnauthorizedCategoryError
// Log access for GDPR Article 30
AuditTrail.log(permit=P, round=t, categories=$\mathcal{C}$)
return True

can compute $\sum_k \Delta_k$ without ever observing any individual hospital's gradient Δ_k.

Algorithm S4 implements this using Shamir's secret sharing and pairwise masking, ensuring the server observes only the aggregate gradient.

Protocol Phases: (1) Each client splits gradients into K shares using (t, K)-threshold Shamir secret sharing; (2) Clients add pairwise random masks negotiated via ECDH key exchange; (3) The server computes the sum—masks cancel out and only the true aggregate remains.

Algorithm S4: Secure Aggregation (Pairwise Masking)
Input: Client gradients $\{\Delta_1, \ldots, \Delta_K\}$, threshold t
Output: Aggregated gradient Δ_{agg}

// Phase 1: ECDH key exchange + Shamir sharing
for each client k **do**
  shares$_k$ $\leftarrow$ ShamirShare(Δ_k, t, K)
  Distribute shares$_k$ to other clients
// Phase 2: Add pairwise random masks
for each client k **do**
  $\hat{\Delta}_k \leftarrow \Delta_k + \sum_{j<k} r_{jk} - \sum_{j>k} r_{kj}$
// Phase 3: Server reconstructs aggregate
$\Delta_{agg} \leftarrow \sum_{k=1}^{K} \hat{\Delta}_k$
// Masks cancel: $\sum_k \sum_{j<k} r_{jk} - \sum_k \sum_{j>k} r_{kj} = 0$
if ActiveClients $< t$ **then**
  raise SecureAggregationError
return Δ_{agg}

Defense-in-depth: Secure aggregation (Algorithm S4) combined with differential privacy (Algorithm S2) provides layered protection: even if the aggregation server is compromised, it learns only the noisy aggregate—never individual hospital contributions. The combination addresses the unresolved GDPR question of whether model gradients constitute "personal data": with both mechanisms active, the information available to any single party is provably bounded. The following algorithms address how local training handles EHDS-specific data challenges.

E. FedProx for Non-IID Data

Algorithm S5 extends FedAvg with a proximal term that penalizes local model divergence from the global model [14]. In EHDS cross-border federations, data heterogeneity is a structural feature, not an exception: hospitals in different Member States serve distinct demographics, follow national clinical guidelines, and use different diagnostic thresholds. For instance, heart disease prevalence ranges from 39.2% in Rome to 62.6% in Amsterdam in our experimental setting. Without drift control, local models can diverge so far from the global consensus that aggregation produces a deteriorated global model. The proximal term $\frac{\mu}{2}\|\theta_h - \theta\|^2$ acts as a regularizer that keeps each hospital's local update within a controlled distance of the global model, balancing personalization with collaboration.

When to use in EHDS: Recommended for federations with moderate non-IID conditions and when client dropout is expected (hospitals may temporarily disconnect). FedProx tolerates partial participation better than FedAvg because the proximal term stabilizes local updates even with fewer training epochs.

Algorithm S5: FedProx Local Update
Input: Local data $\mathcal{D}_h$, global model θ, proximal weight μ
Output: Local update Δ_h

$\theta_h \leftarrow \theta$
for epoch $e = 1$ to E **do**
  for batch $\mathcal{B} \in \mathcal{D}_h$ **do**
    $g \leftarrow \nabla\mathcal{L}(\theta_h; \mathcal{B})$
    // Proximal term: $\nabla \frac{\mu}{2}\|\theta_h - \theta\|^2$
    $g \leftarrow g + \mu(\theta_h - \theta)$
    $\theta_h \leftarrow \theta_h - \eta \cdot g$
$\Delta_h \leftarrow \theta_h - \theta$
return Δ_h

Parameter Selection: $\mu = 0$ reduces to FedAvg; $\mu \in [0.01, 0.1]$ provides stable convergence; $\mu > 1$ may prevent local adaptation. The choice of μ should be documented in the data permit application so that the HDAB can assess the expected privacy-utility trade-off.

Before any local training begins (whether with FedAvg, FedProx, or any other algorithm), the framework must enforce citizen opt-out rights. The following algorithm ensures this compliance.

F. Article 71 Opt-Out Registry Protocol

Algorithm S6 implements the citizen opt-out mechanism mandated by EHDS Article 71. This article grants every EU citizen the right to object to secondary use of their electronic health data—a fundamental right that must be enforced *before* any gradient computation occurs. The algorithm queries the national opt-out registry maintained by each Member State and removes matching records from the local training dataset.

Granularity levels: (1) *Blanket opt-out*—citizen refuses all secondary use; (2) *Purpose-specific*—e.g., permitting scientific research but blocking commercial analytics; (3) *Category-specific*—e.g., allowing demographics but blocking genomic data. This granularity reflects the EHDS principle that citizens should have meaningful control, not merely a binary yes/no choice.

EHDS Governance Role: Opt-out filtering operates at Layer 3 (Data Holders) before local training. Registry lookups use LRU caching with configurable TTL to minimize latency ($<$10ms per round) while ensuring timely propagation of new opt-out decisions. All filtering statistics are logged for GDPR Article 30 audit compliance.

Impact on model quality: High opt-out rates reduce training data volume, potentially degrading model performance—particularly for underrepresented subpopulations. The audit log captures filtering statistics to quantify this impact and support transparency reporting. Once opted-out records are excluded, the remaining data must be harmonized into a consistent format before local model training can proceed.

Algorithm S6: Article 71 Opt-Out Filtering
Input: Local dataset $\mathcal{D}_h$, purpose p, categories $\mathcal{C}$
Output: Filtered dataset $\mathcal{D}'_h$

// Synchronize with national opt-out registry
OptOutRecords $\leftarrow$ FetchOptOutRegistry(MemberState)
$\mathcal{D}'_h \leftarrow \emptyset$
for each record $r \in \mathcal{D}_h$ **do**
  citizen_id $\leftarrow$ r.pseudonymized_id
  opted_out $\leftarrow$ False
  // Check purpose-specific opt-out
  if (citizen_id, p) $\in$ OptOutRecords **then**
    opted_out $\leftarrow$ True
  // Check category-specific opt-out
  for each $c \in \mathcal{C}$ **do**
    if (citizen_id, c) $\in$ OptOutRecords **then**
      opted_out $\leftarrow$ True
  if not opted_out **then**
    $\mathcal{D}'_h \leftarrow \mathcal{D}'_h \cup \{r\}$
// Log filtering statistics for audit
AuditLog.record(total=$|\mathcal{D}_h|$, filtered=$|\mathcal{D}'_h|$)
return $\mathcal{D}'_h$

G. FHIR R4 Preprocessing Pipeline

Algorithm S7 standardizes heterogeneous EHR data into FHIR R4 format for ML consumption. This preprocessing step is critical in the EHDS context because only 34% of European healthcare providers currently achieve full FHIR R4 compliance [7]. The remaining 66% use legacy formats (HL7v2, CDA, proprietary CSV exports) that must be harmonized before FL training can proceed on a consistent feature space.

Four-stage pipeline: (1) *Format detection* automatically identifies the source format; (2) *Terminology mapping* converts local codes to international standards (ICD-10 for diagnoses, ATC for medications, LOINC for laboratory results); (3) *FHIR transformation* produces validated FHIR R4 bundles using the six Article 33 data categories (Patient Summary, E-Prescription, Laboratory Results, Medical Imaging, Hospital Discharge, Rare Disease); (4) *Tensor extraction* converts structured FHIR resources into numerical tensors ready for model training.

EHDS Relevance: Without this harmonization step, hospitals in different Member States would produce incompatible feature spaces, making federated aggregation meaningless. The pipeline ensures that a gradient computed in a Finnish hospital is semantically compatible with one from an Italian hospital.

Validation requirements: The FHIR validation step rejects records with missing mandatory fields or invalid terminology codes, ensuring data quality before model training. Rejected records are logged (without patient-identifiable content) for audit purposes. With harmonized data ready for training, the final core component manages the overall privacy budget across the entire study.

H. Privacy Budget Accountant

Algorithm S8 tracks cumulative privacy expenditure across FL rounds using Rényi Differential Privacy (RDP) moment

Algorithm S7: FHIR R4 Preprocessing
Input: Raw EHR records $\mathcal{R}$, feature specification $\mathcal{F}$
Output: Training tensors (X, y)

format ← DetectFormat($\mathcal{R}$) // *HL7v2, CDA, CSV*
parser ← GetParser(format)
records ← parser.parse($\mathcal{R}$)
// *Map to standard terminologies*
for each $r \in$ records **do**
  r.diagnoses ← MapToICD10(r.diagnoses)
  r.medications ← MapToATC(r.medications)
  r.labs ← MapToLOINC(r.labs)
fhir_bundle ← ToFHIR(records)
ValidateFHIR(fhir_bundle)
X ← ExtractFeatures(fhir_bundle, $\mathcal{F}$)
X ← StandardScaler.fit_transform(X)
y ← ExtractLabels(fhir_bundle)
return (X, y)

accounting [26]. In the EHDS governance model, the total privacy budget ε_{total} is a parameter of the data permit: the researcher specifies the desired budget in the permit application, and the HDAB evaluates whether the proposed budget provides sufficient privacy protection for the requested data categories and population size.

Why RDP accounting: Naive DP composition (adding ε per round) yields loose bounds: 20 rounds at ε=0.5 each would consume ε=10 total. RDP provides 5–6× tighter bounds [26], [36], meaning the same 20 rounds can achieve the same privacy guarantee with significantly less noise—and therefore better model utility.

Hard budget enforcement: When the cumulative expenditure approaches ε_{total}, the accountant raises a `BudgetExhaustedError` that terminates training. This prevents "privacy bankruptcy"—a situation where continued training would violate the privacy guarantee approved in the data permit. The per-round allocation strategy distributes remaining budget uniformly across remaining rounds, adapting dynamically if training converges faster than expected.

Algorithm S8: RDP Privacy Budget Accountant
Input: Total budget $(\varepsilon_{total}, \delta_{total})$, rounds T
Output: Per-round budget allocation

λ ← [0] × MAX_ORDER // *Rényi moments*
rounds_completed ← 0
function AllocateRound():
  ε_{spent} ← ComputeEpsilon(λ, δ_{total})
  $\varepsilon_{remaining} \leftarrow \varepsilon_{total} - \varepsilon_{spent}$
  if $\varepsilon_{remaining} < \varepsilon_{min}$ **then**
    raise BudgetExhaustedError
  $\varepsilon_t \leftarrow \varepsilon_{remaining}/(T-$ rounds_completed)
  return ε_t
function RecordRound(σ, q):
  for order = 1 to MAX_ORDER **do**
    λ[order] += ComputeMoment(order, σ, q)
  rounds_completed += 1

III. Supplementary Experimental Figures

This section presents detailed experimental results from the FL-EHDS benchmark suite. All figures are generated from real experimental runs available in the repository.

Note on experimental configurations: Figures S-2–S-9 were generated from an extended 50-round, 5-client training run using the framework's synthetic EHDS scenario (simulated European hospitals: Rome, Amsterdam, Berlin, Madrid, Paris). These complement the main paper's 20-round experiments on Heart Disease UCI (4 real hospitals) and Diabetes (5 Dirichlet-partitioned clients). The 50-round configuration illustrates longer-horizon convergence properties, client participation dynamics, and gradient evolution patterns that are not visible in the shorter 20-round evaluation.

A. Hospital Data Distribution

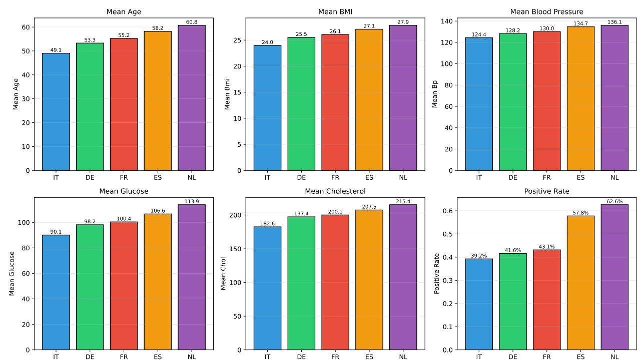

Fig. S-2. Data distribution across hospitals. Notable heterogeneity: Amsterdam shows older population (60.8 years mean age) with higher positive rate (62.6%) compared to Rome (49.1 years, 39.2%). This reflects realistic cross-border EHDS variability.

B. Per-Client Training Time

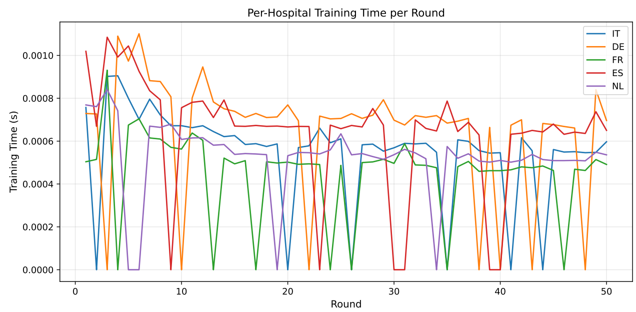

Fig. S-3. Per-client training time per round. Larger hospitals (Berlin: 500 samples) exhibit slightly longer training times. The adaptive training engine compensates by adjusting batch sizes for stragglers.

C. Client Participation Matrix

Figure S-4 shows the client participation pattern across training rounds.

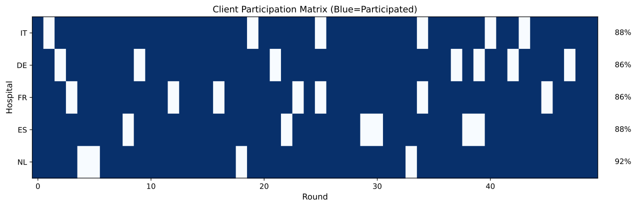

Fig. S-4. Client participation matrix (50 rounds $\times$ 5 clients). Participation rates: IT 88%, DE 86%, FR 86%, ES 88%, NL 92%. The framework tolerates 10–15% dropout per round while maintaining convergence.

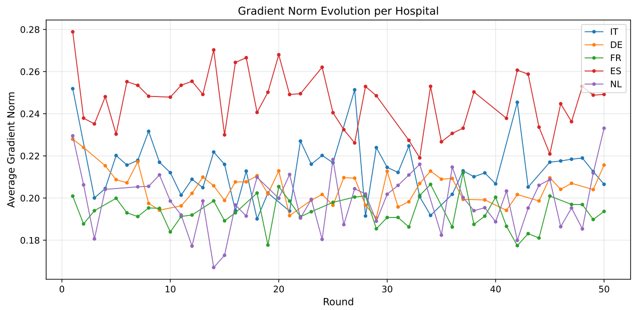

Fig. S-5. Gradient norm evolution per client over 50 rounds. All clients show decreasing trends indicating stable convergence. Clipping threshold C=1.0 bounds extreme values for DP compatibility.

D. Gradient Norm Evolution

Figure S-5 tracks gradient norms per client, confirming convergence stability.

E. Communication Cost Analysis

Figure S-6 quantifies the per-round communication overhead.

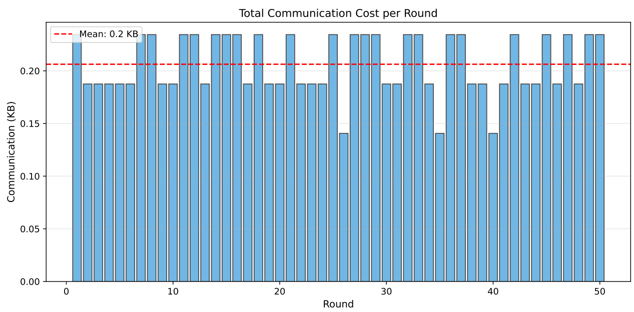

Fig. S-6. Cumulative communication cost per round. Linear scaling with participating clients (3.5 KB/client/round). Total 50-round overhead: 875 KB for 5 clients—feasible even for bandwidth-constrained environments.

F. Learning Rate Sensitivity

Figure S-7 evaluates sensitivity to the server learning rate.

G. Batch Size Impact

Figure S-8 examines the trade-off between batch size and convergence.

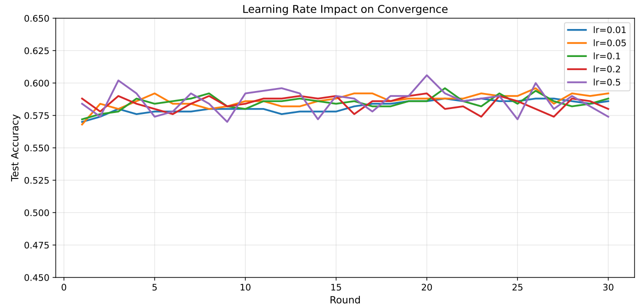

Fig. S-7. Learning rate sensitivity analysis. η=0.01: slow convergence (53.8% at round 50). η=0.1: optimal (58.6%). η=0.5: instability with oscillations.

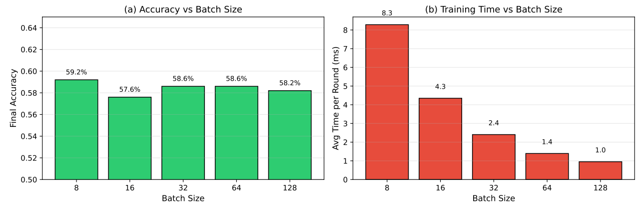

Fig. S-8. Batch size impact on convergence. Smaller batches (8–16) provide noisier gradients but faster initial progress. Batch size 32 balances gradient quality and computational efficiency.

H. Per-Client Accuracy Trajectories

Figure S-9 shows per-client accuracy evolution, illustrating the impact of non-IID data.

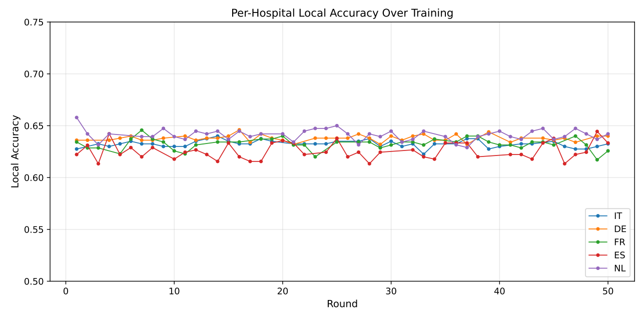

Fig. S-9. Per-client accuracy over training rounds. Variance reflects non-IID data: NL (older, higher-risk population) reaches 64% accuracy while FR (mid-range demographics) stabilizes at 55%.

IV. Dataset Landscape

The FL-EHDS framework supports 19 healthcare datasets spanning four modalities. Table S-I provides a comprehensive overview. This diversity enables evaluation across multiple EHDS-relevant dimensions: data scale (120–253K samples), feature dimensionality (9–30 tabular, high-dimensional imaging), task complexity (binary to 5-class), partition strategies (natural hospital-based and synthetic Dirichlet), and interoperability standards (CSV, FHIR R4, OMOP-CDM). Experimentally evaluated datasets in the main paper are marked with ✓.

TABLE S-I
FL-EHDS DATASET LANDSCAPE: COMPLETE FRAMEWORK COVERAGE

Dataset	**Type**	**Samples**	**Feat.**	**Classes**	**FL Partition**	**EHDS Category (Art. 33)**	**EHDS Level**	**Eval.**
A. Tabular — Clinical EHR								
Diabetes 130-US	Tabular	101,766	22	2	Dirichlet (α=0.5)	EHR, ICD-9, medications	L2: FHIR-mappable	✓
Heart Disease UCI	Tabular	920	13	2	Natural (4 hospitals)	Vitals, ECG, lab results	L2: FHIR-mappable	✓
PTB-XL ECG†	Tabular	21,799	9	5	Natural (52 sites)	SCP-ECG (EN 1064), diagnostics	L2: FHIR-mappable	✓
Cardiovascular Disease	Tabular	70,000	11	2	Dirichlet (α=0.5)	Vitals, lab, risk factors	L2: FHIR-mappable	✓
Breast Cancer Wisconsin	Tabular	569	30	2	Dirichlet (α=0.5)	Pathology (FNA cytology)	L2: FHIR-mappable	✓
Stroke Prediction	Tabular	5,110	10	2	Dirichlet	Cardiovascular risk factors	L2: FHIR-mappable	—
CDC Diabetes BRFSS	Tabular	253,680	21	2	Dirichlet	Population health survey	L2: FHIR-mappable	—
CKD UCI	Tabular	400	24	2	Dirichlet	Renal panel, comorbidities	L2: FHIR-mappable	—
Cirrhosis Mayo	Tabular	418	18	2	Dirichlet	Hepatology, drug trial	L2: FHIR-mappable	—
B. Tabular — FHIR-Native								
Synthea FHIR R4	FHIR	1,180	14	2	Hospital profile	Patient, Condition, Encounter	L1: FHIR-native	qual.
SMART Bulk FHIR	FHIR	120	12	2	Single export	NDJSON Bulk Data (Art. 46)	L1: FHIR-native	qual.
C. Generated Pipelines (in-memory)								
FHIR R4 Synthetic	Gen.	config.	10	2	Hospital profile	Generated FHIR bundles	L1: FHIR-native	qual.
OMOP-CDM Harmonized	Gen.	config.	36	2	Cross-border	Vocabulary harmonization	L3: OMOP	qual.
D. Medical Imaging								
Chest X-ray	Imaging	5,856	—	2	Dirichlet (α=0.5)	Radiology (DICOM)	L4: Imaging	✓
Brain Tumor MRI	Imaging	7,023	—	4	Dirichlet (α=0.5)	Neuro-imaging (DICOM)	L4: Imaging	✓
Skin Cancer	Imaging	3,297	—	2	Dirichlet (α=0.5)	Dermatology (DICOM)	L4: Imaging	✓
Diabetic Retinopathy	Imaging	35,126	—	5	Dirichlet	Ophthalmology (DICOM)	L4: Imaging	—
Brain Tumor MRI (alt.)	Imaging	3,264	—	4	Dirichlet	Neuro-imaging (DICOM)	L4: Imaging	—
ISIC Skin Lesions	Imaging	2,357	—	9	Dirichlet	Dermatology (DICOM)	L4: Imaging	—

EHDS Levels: L1 = FHIR-native (Art. 46 compliant); L2 = FHIR-mappable (standard clinical features with FHIR mapping in metadata); L3 = OMOP-CDM harmonized (cross-border vocabulary alignment, Art. 50); L4 = Medical imaging (DICOM, Art. 33 "medical images").
†PTB-XL: European-origin dataset (PTB, Berlin, Germany) with SCP-ECG coding (EN 1064). 52 recording sites enable natural hospital-based FL partitioning—the strongest EHDS benchmark in the framework.
Eval.: ✓= quantitative experimental evaluation (P1.2); qual. = qualitative pipeline validation; — = supported but not evaluated in current paper.
config. = sample count depends on generation parameters.

V. Extended Algorithm Comparison

A. Algorithms Evaluated

We compare foundational FL algorithms plus 2022–2025 advances:

Foundational: FedAvg [13], FedProx [14], SCAFFOLD [29], FedAdam/FedYogi/FedAdagrad [31].

Recent (2022–2025): FedLC [39] (logit calibration for label skew), FedSAM [38] (flat minima), FedDecorr [40] (decorrelation against dimensional collapse), FedSpeed [41] (fewer rounds), FedExP [42] (server-side acceleration), FedLESAM [43] (globally-guided SAM, ICML 2024 Spotlight), HPFL [44] (personalized classifiers, ICLR 2025).

B. Non-IID Configuration

Data heterogeneity is controlled via Dirichlet distribution with α:

- $\alpha = 0.1$: **Extreme non-IID**—highly skewed label distributions
- $\alpha = 0.5$: **High non-IID**—significant heterogeneity
- $\alpha = 1.0$: **Moderate non-IID**—balanced heterogeneity
- $\alpha = 10.0$: **Near-IID**—approximately uniform

C. Convergence at Different Heterogeneity Levels

Findings: (1) At α=0.1, SCAFFOLD achieves most stable convergence via variance reduction. (2) FedProx provides marginal improvement over FedAvg at α=0.5–1.0. (3) Adaptive methods (FedAdam, FedYogi) excel in near-IID but may oscillate under extreme heterogeneity. (4) FedAvg remains competitive in near-IID, suitable for homogeneous federations.

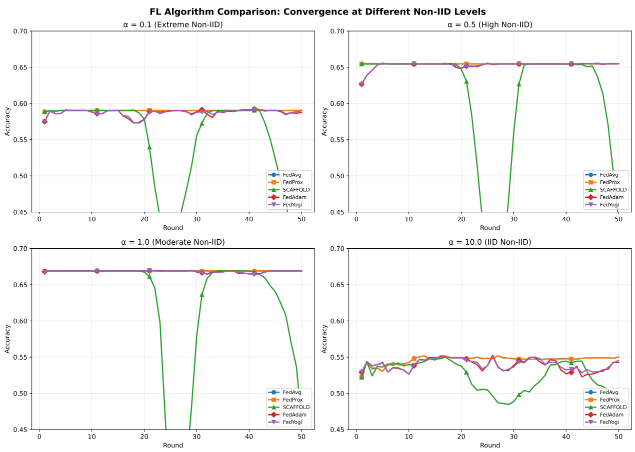

Fig. S-10. Algorithm convergence across non-IID levels ($\alpha \in \{0.1, 0.5, 1.0, 10.0\}$). SCAFFOLD and adaptive methods show superior stability under extreme heterogeneity.

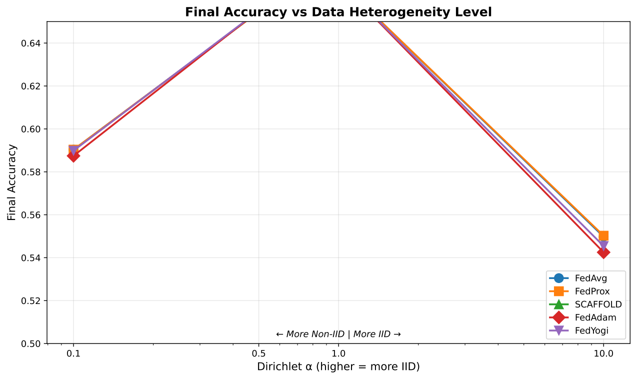

Fig. S-11. Final accuracy vs. Dirichlet α. All algorithms degrade under extreme non-IID. SCAFFOLD shows smallest gap between α=0.1 and α=10.

D. Final Accuracy vs. Heterogeneity

E. Convergence Speed

Figure S-12 compares how quickly algorithms reach target accuracy thresholds.

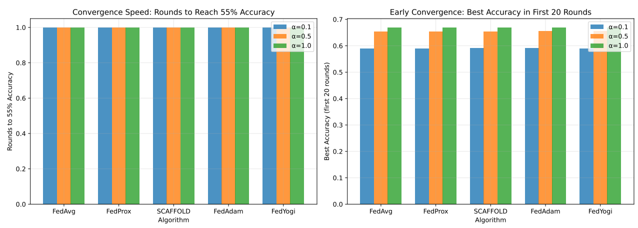

Fig. S-12. Convergence speed comparison. Left: rounds to 55% accuracy. Right: best accuracy in first 20 rounds. Adaptive methods converge faster but may plateau.

F. Evidence-Based Algorithm Recommendation Matrix

Table S-II maps seven realistic EHDS deployment scenarios to recommended and avoided algorithms, grounded in specific experimental evidence from this work. The Diagnostic Equity Index (DEI; Section XV-Y) is reported where confusion matrix data is available, providing a complementary criterion beyond aggregate accuracy.

Critical corrections from prior version: (1) SCAFFOLD is no longer recommended for heterogeneous deployments—our experiments demonstrate catastrophic failure on class-imbalanced tabular data (oscillating convergence; 11.2% on Diabetes, Table V). (2) Ditto+DP is strongly preferred over FedAvg+DP for privacy-critical deployments (89.2% vs. 52.3% at ε=1 on PTB-XL). (3) Imaging recommendations now distinguish between Ditto (recommended for multiclass) and HPFL (avoid on all imaging tasks). (4) The DEI column provides diagnostic equity information absent from accuracy-only comparisons.

VI. Advanced FL Paradigms

Note: The paradigms in this section are *implemented in the reference framework* and available for use. Four of these paradigms—fairness-aware FL, continual FL, personalized FL, and federated unlearning—are now experimentally validated in Section XVII (135 experiments across 3 datasets). The remaining paradigms (vertical FL, hierarchical FL, multi-task FL, asynchronous FL) are provided as architectural capabilities for future evaluation on appropriate multi-institutional datasets.

The core FL-EHDS pipeline (Section II) addresses the standard "horizontal" FL scenario where all hospitals share the same feature schema. However, real EHDS deployments will encounter more complex configurations: institutions with complementary features for the same patients (vertical FL), adversarial participants (Byzantine resilience), evolving data distributions over the 2025–2031 timeline (continual FL), heterogeneous clinical objectives (multi-task FL), and the hierarchical governance structure of the EU itself (hierarchical FL). This section presents the advanced paradigms implemented in the reference framework, each motivated by a specific EHDS deployment challenge. Algorithms S9–S13 formalize the core mechanisms.

A. Vertical Federated Learning

Vertical FL addresses scenarios where institutions hold *different features* for the *same patients*—a common situation in EHDS cross-border analytics. For example, a hospital may hold demographics and diagnoses, a laboratory holds test results, and a pharmacy holds prescription histories. Under EHDS Article 33, these correspond to different data categories (Patient Summary, Laboratory Results, E-Prescription) that may be held by different data holders within the same or different Member States.

Private Set Intersection (PSI): Before training, the participating institutions must identify their common patients without revealing their full patient lists. RSA-based PSI achieves this with $O(n \log n)$ complexity using pseudonymized identifiers, ensuring EHDS compliance: no institution learns which patients the other holds beyond the intersection.

Split Learning: Algorithm S9 implements the forward pass in split learning, where each party computes activations on its local features up to a "cut layer," then the server concatenates activations to produce the final prediction. Only intermediate representations (not raw data) cross institutional boundaries.

Algorithm S9: Split Learning Forward Pass
Input: Features X_A, X_B at parties A, B; cut layer k
Output: Prediction $\hat{y}$
$h_A \leftarrow f^A_{1:k}(X_A)$ *// Party A: features → activations*
$h_B \leftarrow f^B_{1:k}(X_B)$ *// Party B: features → activations*
$h \leftarrow \text{Concat}(h_A, h_B)$
$\hat{y} \leftarrow f_{k+1:L}(h)$ *// Server: cut layer → output*
return $\hat{y}$

B. Byzantine-Resilient Aggregation

In a cross-border EHDS federation spanning 27 Member States, the aggregation server cannot blindly trust every participant. A compromised institution—whether through malicious intent, software bugs, or data corruption—could submit

TABLE S-II

EHDS ALGORITHM RECOMMENDATION MATRIX. EVIDENCE-BASED MAPPING FROM DEPLOYMENT SCENARIOS TO RECOMMENDED AND AVOIDED ALGORITHMS. DEI = DIAGNOSTIC EQUITY INDEX (SECTION XV-Y). "AVOID" INDICATES ALGORITHMS THAT FAIL OR DEGRADE SIGNIFICANTLY IN THE SPECIFIED SCENARIO.

EHDS Scenario	Modal.	Recommended	Avoid	DEI	Key Evidence
Cross-border EHR, heterogeneous sites	Tabular	Ditto, HPFL	SCAFFOLD^a, FedNova^a	0.740†	Tables V, VI: Ditto +12.6pp, HPFL +11.8pp vs. FedAvg
Small-cohort rare disease (<1K samples)	Tabular	HPFL	FedAvg^b	0.740†	Table XIX: HPFL avoids collapse 10/10 seeds vs. 2/10 FedAvg
Multiclass radiology ($\geq$3 classes)	Imaging	Ditto	HPFL (neutral)	0.435	Table XXV: Ditto +23.1pp Brain Tumor (p=0.007, 3 seeds)
Binary radiology	Imaging	FedAvg	HPFL ($-$18.7pp)	0.528§	Tables XX, XXV: FedAvg 87.8% vs. HPFL 69.1% Chest X-ray
High privacy ($\varepsilon \leq 5$)	Any	Ditto	FedAvg^c	—	Table XXII: Ditto 89.2% vs. FedAvg 52.3% at ε=1
Large federation ($K \geq 50$)	Any	Ditto, HPFL	SCAFFOLD	—	Tables XVII–XVIII: Ditto $-$0.8pp at K=100 vs. FedAvg $-$4.7pp
Homogeneous / IID data	Any	FedAvg	(all equivalent)	—	PTB-XL: all algorithms within 0.6pp (91.9–92.5%)

^{a}SCAFFOLD oscillates on class-imbalanced tabular data; FedNova amplifies noise from divergent objectives (11.2%, 13.0% on Diabetes). ^{b}FedAvg/Ditto exhibit single-class collapse on 8/10 seeds (Table XIX). ^{c}FedAvg collapses from 91.9% to 52.3% at ε=1 ($-$39.6pp). †Breast Cancer HPFL DEI. ‡Brain Tumor Ditto DEI (3 seeds). §Chest X-ray FedAvg DEI. DEI values reported only where confusion matrix analysis is available.

adversarial gradient updates that poison the global model, potentially affecting clinical decisions across the entire federation. Byzantine-resilient aggregation protects model integrity by detecting and excluding anomalous updates.

Algorithm S10 implements Krum, which selects the gradient closest to $n-f-2$ nearest neighbors, effectively filtering outliers. Six defense methods protect against up to $f < n/3$ adversarial clients:

Algorithm S10: Krum Byzantine Defense
Input: Gradients $\{g_1, \ldots, g_n\}$, Byzantine bound f
Output: Selected gradient g^*

for each gradient g_i **do**
  $D_i \leftarrow \{\|g_i - g_j\|^2 : j \neq i\}$
  $s_i \leftarrow \sum_{d \in \text{smallest}_{n-f-2}(D_i)} d$
$g^* \leftarrow g_{\arg\min_i s_i}$
return g^*

Other methods: **Trimmed Mean** (removes β-fraction extreme values per coordinate), **Coordinate-wise Median** (robust estimator), **Bulyan** (two-stage Krum + trimmed mean), **FLTrust** (server-guided trust weighting), **FLAME** (clustering-based). Attack simulation: label flipping, gradient scaling, additive noise, sign flipping, model replacement.

C. Continual Federated Learning

The EHDS is designed for long-term operation (2025–2031 and beyond), during which healthcare data distributions will evolve: new diseases emerge (as demonstrated by COVID-19), clinical protocols change, and demographic compositions shift. A model trained in 2027 may perform poorly on 2029 data if it has "forgotten" how to handle earlier patterns. Continual Federated Learning addresses this *catastrophic forgetting* problem by preserving knowledge from previous training tasks while adapting to new data.

The Elastic Weight Consolidation (EWC) loss function adds a quadratic penalty:

$$\mathcal{L}_{EWC}(\theta) = \mathcal{L}(\theta) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$$

where $F_i = \mathbb{E}[\nabla^2 \log p(\mathcal{D}|\theta^*)]_i$ is the i-th diagonal entry of the Fisher Information Matrix and θ^* are optimal parameters for previous tasks.

Additional strategies: Learning without Forgetting (LwF), Experience Replay, drift detection (ADWIN, Page-Hinkley) triggering adaptation.

D. Multi-Task Federated Learning

In EHDS cross-border studies, different hospitals may pursue related but distinct clinical objectives from the same data. A cardiology network might simultaneously predict heart failure risk (Hospital A), readmission probability (Hospital B), and medication response (Hospital C). Multi-Task FL enables these institutions to collaborate on shared feature representations while maintaining task-specific prediction heads.

Architectures: **Hard Parameter Sharing** (common feature extractor, task-specific heads), **Soft Parameter Sharing** (separate networks with similarity regularization), **FedMTL** (dynamic task relationship learning).

E. Hierarchical Federated Learning

The EHDS governance structure is inherently hierarchical: individual hospitals report to regional health authorities, which coordinate under national HDABs, which in turn connect to the EU-level HealthData@EU infrastructure. Hierarchical FL mirrors this governance topology, aggregating gradients at intermediate levels before reaching the central server. This reduces communication costs (hospitals communicate with regional aggregators, not directly with the EU server) and aligns FL operations with the jurisdictional boundaries of HDABs.

Four-tier hierarchy reflecting EU governance:

1) **Client Tier**: Individual hospitals/data holders
2) **Regional Tier**: Regional aggregators (e.g., Lombardy, Bavaria)
3) **National Tier**: National HDABs coordinate Member State aggregation
4) **EU Tier**: HealthData@EU central aggregator

Benefits: reduced communication costs (hospitals $\rightarrow$ regional, not directly EU), alignment with EHDS governance where HDABs have national jurisdiction.

F. Personalized Federated Learning

A single global model may underperform at individual hospitals because clinical populations differ substantially across Member States. Personalized FL maintains both a global model (encoding shared medical knowledge) and hospital-specific local models (adapted to local demographics and clinical practices). Algorithm S11 shows pFedMe [35] as a representative personalized method, using Moreau envelopes to balance personalization with global knowledge: the regularization term $\lambda(\theta_k - \theta)$ pulls the local model toward the global consensus, while local gradient descent adapts to hospital-specific data patterns. Ditto [33] follows a similar dual-model principle but with a simpler formulation: it trains a personalized model regularized toward the global model via $\frac{\lambda}{2}\|\theta_k - \theta\|^2$. In our experiments, Ditto—the best-performing personalized method—achieves 75.1% accuracy on Heart Disease, a 12.6pp improvement over FedAvg, precisely because it learns hospital-specific decision boundaries.

Algorithm S11: pFedMe Local Update
Input: Data $\mathcal{D}_k$, global θ, personal θ_k, λ, η
Output: Updated personal model θ'_k

for $i = 1$ to R **do**
  $\theta_k \leftarrow \theta_k - \eta\nabla\mathcal{L}(\theta_k; \mathcal{D}_k)$
// Moreau envelope: balance with global
$\theta'_k \leftarrow \theta_k - \lambda(\theta_k - \theta)$
$g_k \leftarrow \lambda(\theta - \theta'_k)$
return θ'_k, g_k

Other approaches: **FedPer** (shared base, local personalization layers), **Per-FedAvg** (MAML-based meta-learning), **APFL** (adaptive mixing α between global and local), **Ditto** (personalization regularization).

EHDS Relevance: Member States have different healthcare systems, disease prevalence, and clinical practices. Personalized FL enables institution-specific adaptation while benefiting from collaborative training.

G. Asynchronous Federated Learning

Standard synchronous FL requires all participating hospitals to complete local training before the server can aggregate. In an EHDS federation spanning 27 Member States with heterogeneous computational resources, this creates a "straggler" problem: a resource-constrained rural hospital delays the entire federation. Asynchronous FL eliminates this bottleneck by allowing the server to aggregate updates as they arrive, weighting stale updates (from slow clients) less heavily. Algorithm S12 implements polynomial staleness weighting: an update computed τ rounds ago receives weight $(1 + \tau)^{-a}$, ensuring that fresher updates contribute more while still incorporating information from slower participants.

Algorithm S12: FedAsync with Staleness Weighting
Input: Client update Δ_k, client round t_k, server round t
Output: Updated global model θ

$\tau \leftarrow t - t_k$ *// Staleness*
$\alpha \leftarrow (1 + \tau)^{-a}$ *// Polynomial decay, $a > 0$*
$\theta \leftarrow \theta + \alpha \cdot \eta \cdot \Delta_k$
return θ

Staleness functions: Constant (α=1), Polynomial ($(1+\tau)^{-a}$), Exponential ($e^{-a\tau}$), Hinge (1 if $\tau \leq \tau_{max}$, else 0). Additional: FedBuff (buffered async), semi-async (wait for α-fraction of clients).

H. Fairness-Aware Federated Learning

The EHDS serves 450 million citizens across Member States with different population sizes, disease prevalence, and healthcare quality. Standard FL optimizes average performance, which can disproportionately favor large hospitals with more data while neglecting smaller institutions or underrepresented patient populations. This creates a "digital health equity" concern: a model that achieves 85% accuracy for a large German hospital but only 55% for a small Romanian clinic is not equitable. Algorithm S13 implements q-FedAvg, which reweights client contributions by their loss: hospitals where the model performs poorly receive higher aggregation weights, pulling the global model toward equitable performance across all participants.

Algorithm S13: q-FedAvg Fair Aggregation
Input: Losses $\{L_1, \dots, L_K\}$, updates $\{\Delta_1, \dots, \Delta_K\}$, q
Output: Fair aggregated update Δ

for each client k **do**
  $w_k \leftarrow L_k^q$ *// Higher loss $\rightarrow$ higher weight*
$W \leftarrow \sum_k w_k$; $w_k \leftarrow w_k/W$
$\Delta \leftarrow \sum_k w_k \cdot \Delta_k$
return Δ

Fairness metrics: Performance Variance (Var($\{L_k\}$)), Worst-case Loss ($\max_k L_k$), Demographic Parity Gap, Equalized Odds Gap. Additional methods: AFL, FedMGDA+, TERM, FairFed.

VII. Infrastructure Components

Deploying FL across 27 EU Member States requires production-grade infrastructure: reliable communication channels between hospitals and the SPE aggregator, efficient serialization of gradient tensors, distributed coordination for concurrent studies, and comprehensive monitoring with EHDS-specific alerting. This section describes the infrastructure components implemented in the reference framework, each designed to operate within the constraints of cross-border healthcare networks (firewalls, bandwidth limitations, regulatory requirements).

A. Communication Layer

The communication layer must bridge heterogeneous network environments: high-bandwidth data center connections between national HDABs, moderate hospital-to-aggregator links, and potentially bandwidth-constrained rural clinics. The framework supports two transport protocols selectable per deployment, with configurable compression and retry policies.

Communication Manager Configuration

```
transport: gRPC | WebSocket
compression: gzip | lz4 | zstd | none
chunk_size: 1MB
retry_policy:
  max_retries: 3
  backoff: exponential
  base_delay: 1s
connection_pool:
  max_connections: 100
  idle_timeout: 300s
```

gRPC: Bidirectional streaming, Protocol Buffers (30% bandwidth reduction vs. JSON), HTTP/2 multiplexing. Ideal for data center deployments.

WebSocket: Browser-compatible, firewall-friendly (standard HTTP upgrade), event-driven. Ideal for edge deployments and browser-based participation.

Selection criteria: gRPC is recommended for production EHDS deployments where both endpoints support HTTP/2 (typical for hospital-to-national aggregator links). WebSocket is preferred when traffic must traverse web application firewalls or when browser-based dashboards participate directly in federation monitoring.

B. Serialization

Binary Format: Tensor metadata + raw binary, 30% smaller than JSON, 15% smaller than pickle, cross-platform (Python, C++, Java).

Delta Serialization: Transmits only changed parameters, sparse encoding, up to 90% bandwidth reduction for fine-tuning.

EHDS-Compliant: Embeds permit ID, timestamp, provenance; cryptographic signatures; GDPR Article 30 audit fields.

C. Caching Layer

In production EHDS deployments, multiple FL studies may run concurrently on overlapping data holders. A distributed locking mechanism prevents race conditions during gradient aggregation—ensuring that two concurrent studies do not interfere with each other's model updates. Algorithm S14 implements Redis-based distributed locking with TTL-based automatic release, preventing deadlocks if a server node fails mid-aggregation.

Algorithm S14: Distributed Lock for Aggregation
Input: Lock name, TTL, client ID
Output: Lock acquired (boolean)

acquired $\leftarrow$ Redis.SET(lock_name, client_id, NX, EX=TTL)
if acquired **then**
  PerformAggregation()
  if Redis.GET(lock_name) == client_id **then**
    Redis.DEL(lock_name)
return acquired

Redis-based caching: model checkpoints, client states, real-time metrics. Features: LRU/LFU/TTL eviction, distributed locking, automatic serialization, cache warming.

D. Orchestration

Kubernetes: Deploys FL clients/aggregators as pods, HPA for elastic scaling, ConfigMaps for hyperparameters, Secrets for HDAB API keys.

Ray: Actor-based FL, automatic fault tolerance, Ray Tune for federated HPO, Object Store for gradient sharing.

Auto-Scaling: Reactive (queue depth/latency), Predictive (ML-based forecasting), Scheduled (time-based patterns).

E. Monitoring

Prometheus Metrics: Counters (rounds_total, permits_validated), Gauges (active_clients, privacy_budget_remaining), Histograms (round_duration, communication_latency), Summaries (gradient_norm_quantiles).

Grafana Dashboards: FL training progress, client health, latency heatmaps, privacy budget consumption, EHDS compliance status.

Alerting: Privacy budget exhaustion, client dropout threshold, model divergence, permit expiration.

F. Model Watermarking

IP protection for FL models trained on EHDS data: **Spread Spectrum** (frequency domain, robust to fine-tuning), **LSB** (low-order weight bits), **Backdoor-based** (input-output ownership proof), **Passport Layers** (dedicated ownership encoding).

G. Cross-Silo Enhancements

EHDS federations are inherently cross-silo: each participant is an institution (hospital, registry, research center) with significant computational resources, distinct data distributions, and long-term participation commitments. This differs from cross-device FL (e.g., mobile phones) and enables advanced optimization strategies.

Multi-Model Federation: Weighted voting, stacking, mixture of experts with diversity enforcement.

Automatic Algorithm Selection: The 17 FL algorithms in the framework have different strengths depending on the federation characteristics (heterogeneity level, number of participants, communication budget). Algorithm S15 implements adaptive aggregation selection via multi-armed bandit (UCB/Thompson Sampling), automatically switching algorithms mid-training if performance metrics indicate a better alternative. A cooldown period prevents oscillation between strategies.

Algorithm S15: Adaptive Aggregation
Input: Client updates, metrics history, cooldown
Output: Aggregated model, selected algorithm

score $\leftarrow$ WeightedScore(loss, accuracy, variance, conv.)
if RoundsSinceSwitch $>$ Cooldown **then**
  for each candidate $\in$ Algorithms **do**
    alt $\leftarrow$ EstimatePerformance(candidate)
    if alt $>$ score + Threshold **then**
      SwitchTo(candidate)
aggregated $\leftarrow$ CurrentAlgo.Aggregate(updates)
return aggregated

VIII. Extended EHDS Interoperability

A. OMOP Common Data Model

OMOP CDM v5.4 provides standardized analytical format used by European research networks (EHDEN, OHDSI).

ETL Pipelines: Transform source EHR to OMOP. **Vocabulary Mapping**: SNOMED, ICD10, LOINC, RxNorm. **Cohort Definitions**: ATLAS-compatible SQL generation. **Feature Extraction**: FeatureExtraction package for ML-ready datasets.

FL Integration: (1) Each hospital transforms local EHR to OMOP; (2) Feature extraction produces identical schema; (3) FL training on homogeneous feature spaces.

B. IHE Integration Profiles

ATNA: TLS mutual authentication, syslog audit messages (RFC 5424), maps to GDPR Article 30.

BPPC: Maps Article 71 opt-out to consent documents, XDS.b integration, consent enforcement at FL initiation.

XCA: Cross-border document query/retrieve, Initiating/Responding Gateways, patient identity correlation.

PIX/PDQ: Patient matching across boundaries, pseudonymization-aware identity management, national eHealth integration.

XUA: SAML 2.0 federated authentication, role-based access control, HDAB authorization token propagation.

C. Cross-Border Data Exchange

Message Formats: EHDS Data Permit Exchange Format (JSON-LD), Federated Query Protocol (SPARQL Federation), Model Update Message Format (Protocol Buffers).

Security: eIDAS-compliant electronic signatures, TLS 1.3, certificate-based authentication (EU trust framework).

Metadata: DCAT-AP Health extension, W3C PROV-O provenance, EMA data quality indicators.

D. Interoperability Architecture

Figure S-13 presents the complete interoperability architecture, showing how heterogeneous data sources across EU Member States are harmonized through multiple standards layers before reaching the FL training engine. The architecture reflects a key EHDS challenge: real-world healthcare institutions use diverse formats, terminologies, and exchange protocols that must be reconciled to produce a consistent feature space for federated model training.

IX. Clinical Imaging: Extended Details

A. Datasets

Three clinical imaging datasets cover representative EHDS scenarios:

- **Chest X-ray** [50]: 5,856 pediatric radiographs (NORMAL/PNEUMONIA, 2.7:1 imbalance)
- **Brain Tumor MRI**: 7,023 T1-weighted CE MRI slices (4-class: glioma, healthy, meningioma, pituitary)
- **Skin Cancer**: 3,297 dermoscopy images (binary benign/malignant)

B. Model Architectures

HealthcareResNet: ResNet-18 [49] pretrained on ImageNet, GroupNorm replacing BatchNorm for FL stability. FedBN [48] skips normalization during aggregation. Partial backbone freeze (level 1). $\sim$11.2M parameters.

HealthcareCNN: 5-block CNN with GroupNorm, progressive channels (32$\rightarrow$512), graduated Dropout (0.15$\rightarrow$0.3). Classifier: Flatten$\rightarrow$FC(512)$\rightarrow$FC(128)$\rightarrow$FC(K). $\sim$12M parameters.

Data augmentation: random horizontal flip, rotation ($\pm$15°), brightness jitter ($\pm$10%). ImageNet normalization.

C. Experimental Configuration

All three imaging datasets use 4 algorithms (FedAvg, FedLESAM, Ditto, HPFL) $\times$ 3 seeds (42, 123, 456) = 36 experiments total, enabling full statistical comparison across algorithms and datasets.

- 5 clients, Dirichlet α=0.5, Adam optimizer
- ResNet-18, FedBN, class-weighted loss, mixed precision
- freeze_level=2 (conv1+bn1+layer1 frozen), local_epochs=2
- Chest X-ray: 20 rounds, early stopping (patience=6), lr=0.001
- Brain Tumor: 20 rounds, early stopping (patience=4, min_rounds=8), lr=0.0005
- Skin Cancer: 20 rounds, early stopping (patience=4, min_rounds=8), lr=0.001

D. Reproducibility

All experiments are fully reproducible:

```
cd fl-ehds-framework
# Chest X-ray (4 algos x 3 seeds, ~4h)
python -m benchmarks.run_imaging_extended
# FedLESAM + HPFL on 3 datasets (18 exps, ~12h)
```

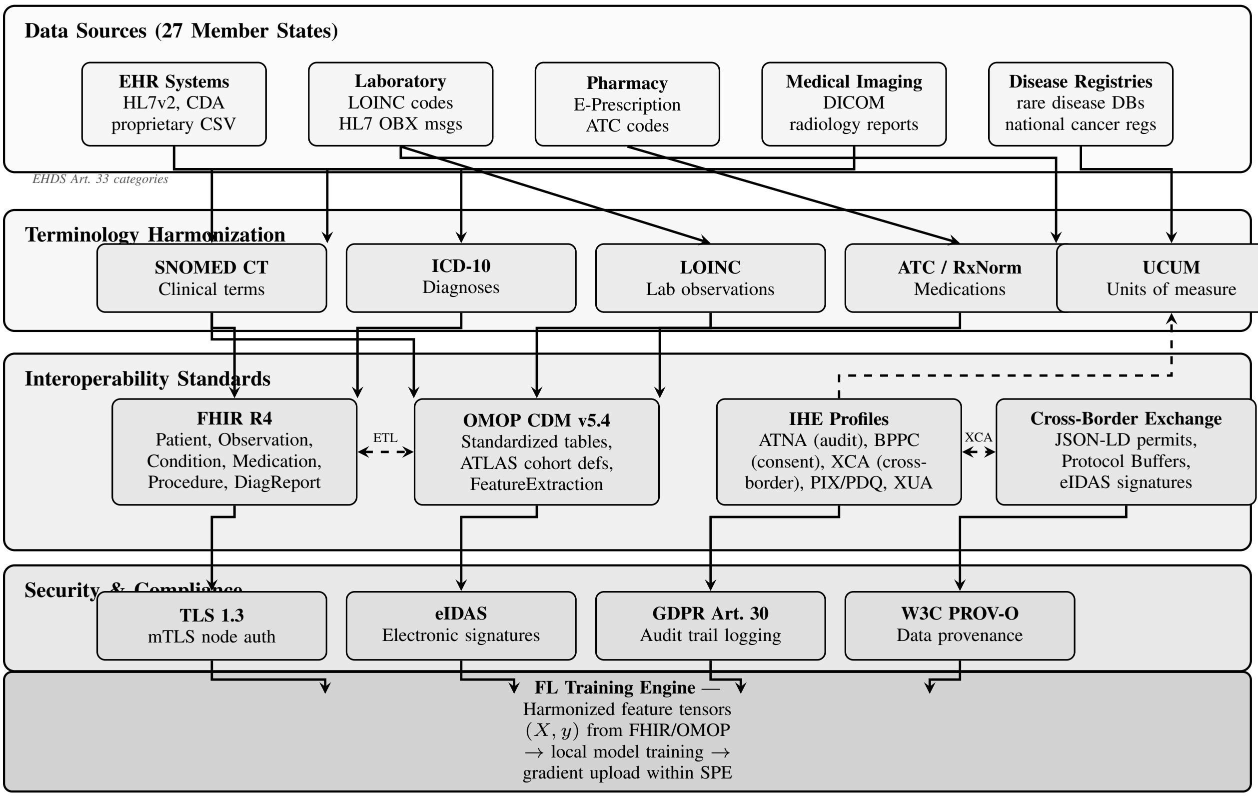

Fig. S-13. EHDS interoperability architecture for FL-based secondary use. Data from heterogeneous sources across 27 Member States (top) flows through terminology harmonization (SNOMED CT, ICD-10, LOINC, ATC, UCUM), then through interoperability standards (FHIR R4 for structured data exchange, OMOP CDM for observational research, IHE profiles for cross-institutional workflows, cross-border exchange protocols with eIDAS signatures). A security and compliance layer enforces TLS 1.3 mutual authentication, eIDAS electronic signatures for permits, GDPR Article 30 audit logging, and W3C PROV-O data provenance before harmonized feature tensors reach the FL training engine. Bidirectional ETL between FHIR and OMOP enables institutions to use either standard based on their existing infrastructure.

```
python -m benchmarks.run_imaging_delta
# FedAvg + Ditto completion (8 exps, ~17h)
python -m benchmarks.run_imaging_completion
# Confusion matrices
python -m benchmarks.run_confusion_matrix_chest
python -m benchmarks.run_confusion_matrix_bc
```

Results, checkpoints, and logs are auto-saved to `benchmarks/paper_results_delta/`.

Repository: https://github.com/FabioLiberti/FL-EHDS-FLICS2026

E. Chest X-ray Results

Table S-III reports Chest X-ray results for four FL algorithms (ResNet-18, 5 clients, Dirichlet α=0.5, 20 rounds max with early stopping).

TABLE S-III
CHEST X-RAY PNEUMONIA DETECTION: FEDERATED ACCURACY (%) WITH RESNET-18. 5 CLIENTS, DIRICHLET α=0.5, 20 ROUNDS, ADAM LR=0.001, EARLY STOPPING (PATIENCE=6). PER-SEED RESULTS WITH JAIN FAIRNESS INDEX.

Algo	s42	s123	s456	Mean	Jain
FedAvg	91.6	87.4	82.8	**87.3**	0.984
FedLESAM	91.6	87.8	84.0	87.8†	0.984
Ditto	84.8	65.1	90.1	80.0	0.958
HPFL	62.1	73.4	71.8	69.1	0.762

Jain index averaged over 3 seeds. †FedLESAM produces results identical to FedAvg (SAM perturbation ineffective on this architecture). HPFL early-stops at round 8 on all seeds with best accuracy at rounds 1–2.

Key findings. *First*, federated learning on Chest X-ray achieves **87.3% accuracy** (FedAvg) for binary pneumonia detection with ResNet-18, confirming FL viability for medical imaging under EHDS-relevant non-IID conditions. *Second*, **FedLESAM is identical to FedAvg on imaging**: despite SAM's theoretical sharpness-aware optimization, FedLESAM produces per-seed accuracies indistinguishable from FedAvg (87.8% vs. 87.3%), a pattern confirmed on all three imaging datasets (see Tables S-XXVIII). *Third*, **on Chest X-ray, personalization is counterproductive**: Ditto achieves 80.0% ($-$7.3pp vs. FedAvg), with substantially higher variance (σ=10.6pp vs. 3.6pp). HPFL *fails catastrophically* (69.1%, $-$18.2pp), early-stopping at round 8 on all seeds with peak accuracy at rounds 1–2, indicating that the split-head architecture cannot learn shared representations with Dirichlet-partitioned data. However, as shown in Table S-XXVIII, this Chest X-ray finding does *not* generalize: Ditto dominates on Brain Tumor

(+28.0pp) and Skin Cancer (+25.5pp). *Fourth*, Ditto exhibits a **fairness failure mode** on seed 456: per-client accuracies of [99.7%, 100.0%, 48.5%, 39.5%, 100.0%] (Jain 0.888) indicate that personalized local models overfit to majority-class clients while failing on minority-class clients.

X. Detailed Architecture Description

This section provides a comprehensive technical specification of the FL-EHDS three-layer architecture, detailing all modules, services, protocols, and standards implemented in the reference framework. Figure 2 in the main paper provides the high-level view; Figure S-14 presents the detailed component-level diagram; the description below enumerates every component with its specific technical parameters.

A. Layer 1: Governance

Four principal modules comprise the governance layer:

1.1 HDAB Integration (per Member State). Each Health Data Access Body instance implements: OAuth2/mTLS authentication with bearer token management (refresh tokens, scopes: `permits:read`, `permits:write`); a permit store with SQLite persistence backend; configurable HDAB strictness level (scale 1–5, where DE=5, ES=2); and national privacy constraints including per-jurisdiction differential privacy bounds ($\varepsilon_{\max}$: DE=1.0, FR=3.0, IT=5.0).

1.2 Data Permit Manager (Article 53). Manages the complete permit lifecycle: PENDING $\rightarrow$ APPROVED $\rightarrow$ ACTIVE $\rightarrow$ EXPIRED/REVOKED. Validates permitted purposes (scientific research, public health surveillance, AI system development, personalized medicine, official statistics) against authorized data categories (EHR, lab results, imaging, genomic, registry, ECG, pathology). Implements expiry verification and strict mode enforcement for continuous compliance.

1.3 Opt-Out Registry (Article 71). Supports three granularity levels: record-level, patient-level, and dataset-level opt-out. Registry lookups use LRU caching (max 100K records, TTL 10 min) with 300-second synchronization intervals to national registries. Configurable actions on opt-out match: exclude, anonymize, or error. Pre-training filtering ensures only opted-in data enters gradient computation.

1.4 Cross-Border Coordinator (Articles 46, 50). Implements multi-HDAB synchronization protocols across 10 preconfigured EU country profiles (DE, FR, IT, ES, NL, SE, PL, AT, BE, PT). Each profile specifies per-country constraints: $\varepsilon_{\max}$ for differential privacy, data retention periods (365–1,825 days), opt-out rates (3–12%), and network latency characteristics (15–60ms based on geographic distance). The fee model (Article 42) implements cost-recovery calculation: base fee + per-record + per-round + per-MB charges.

Inter-layer interface: Data Permit Authorization (OAuth2/mTLS, purpose-validated, Articles 53+71) connects Layer 1 to Layer 2.

B. Layer 2: FL Orchestration

Five modules operate within the Secure Processing Environment (SPE), conforming to HealthData@EU infrastructure requirements:

2.1 Aggregation Engine. Implements 17 FL algorithms spanning six categories: *Baseline* (FedAvg, FedProx with μ=0.01, SCAFFOLD, FedNova, FedDyn); *Adaptive* (FedAdam with server_lr=0.1, FedYogi with β_2=0.99, FedAdagrad); *Personalization* (Ditto with λ=0.1, Per-FedAvg via MAML, pFedMe, MOON); *Non-IID* (FedLC for logit calibration, FedDecorr); *SAM* (FedSAM, FedSpeed); *Advanced* (FedExP ICLR'23, FedLESAM ICML'24, HPFL ICLR'25). Client selection strategies: random, performance-based, fairness-aware, latency-aware. Early stopping: patience=10, min_delta=0.001. Model checkpointing with atomic saves.

2.2 Privacy Protection Module. Three sub-modules provide defense-in-depth:

- *DP-SGD*: Gaussian mechanism with Rényi DP (RDP) accounting. Configurable ε-budget (default 1.0), δ=10^{-5}. Gradient clipping: max_norm=1.0, type=L2/L∞. RDP composition provides 5–6$\times$ tighter bounds than naive composition over 100+ rounds.
- *Secure Aggregation*: Pairwise masking protocol with ECDH key exchange (SECP384R1 curve). Alternative protocols: Shamir secret sharing (threshold reconstruction), homomorphic encryption (CKKS scheme via TenSEAL). Dropout threshold: 50%.
- *Byzantine Resilience*: Six defense rules (Krum, Multi-Krum, Trimmed Mean, Coordinate-wise Median, Bulyan, FLTrust). Anomaly detection at 3σ threshold. TEE integration for SGX/TrustZone hardware attestation.

2.3 Compliance Module. GDPR audit logging (Article 30) in JSON format with 7-year retention (2,555 days). Per-round purpose limitation check (Article 53). Opt-out enforcement: pre-training filtering combined with per-round verification. Anonymity assessment: k-anonymity, l-diversity checks. Automated EHDS compliance report with verification scoring.

2.4 Communication Infrastructure. gRPC bidirectional streaming for model updates (primary protocol). WebSocket for real-time monitoring and events. Compression: GZIP, LZ4, ZSTD, Snappy. Model chunking (ResNet-18: 44.7 MB/round $\rightarrow$ 8.9 MB with Top-k 1%). Security: TLS/mTLS with ECDHE key exchange. Retry: 3 attempts with 2$\times$ exponential backoff. Message types: MODEL_UPDATE, GRADIENT_UPDATE, HEARTBEAT, PERMIT_VALIDATION, CONSENT_CHECK, AUDIT_LOG.

2.5 Monitoring & Metrics. Tracks training convergence (accuracy, loss, F1, AUC per round), communication cost (MB/round), privacy budget consumption (ε spent per round), participation statistics (samples, opt-outs), and resource utilization (CPU, memory, GPU). Structured logging via Structlog (JSON format) with Streamlit dashboard integration.

Inter-module flows: Privacy processing (DP-SGD, gradient clipping, secure masking) connects Aggregation $\rightarrow$ Privacy. Audit trail (GDPR Article 30, JSON logging, 7-year retention) connects Privacy $\rightarrow$ Compliance. Transport layer (gRPC streaming, TLS/mTLS, GZIP compression) connects Communication $\leftrightarrow$ all modules.

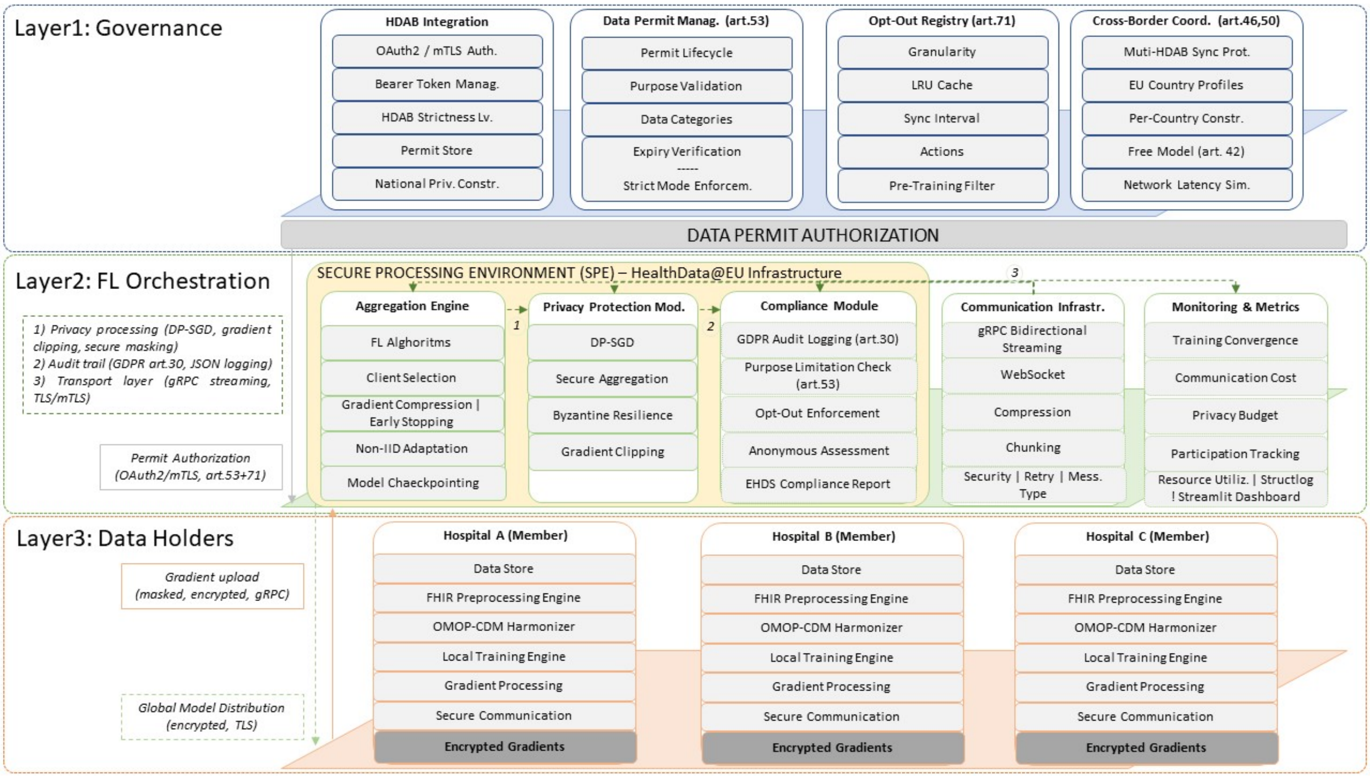

Fig. S-14. FL-EHDS detailed architecture with component-level specifications. Layer 1 (Governance) includes four modules: HDAB Integration with OAuth2/mTLS authentication, Data Permit Manager (Article 53) with lifecycle management, Opt-Out Registry (Article 71) with LRU-cached filtering, and Cross-Border Coordinator (Articles 46, 50) with per-country privacy constraints. Layer 2 (FL Orchestration) operates within the Secure Processing Environment (SPE) with five modules: Aggregation Engine (17 FL algorithms), Privacy Protection (DP-SGD, Secure Aggregation, Byzantine Resilience), Compliance Module (GDPR audit logging), Communication Infrastructure (gRPC/WebSocket), and Monitoring & Metrics. Layer 3 (Data Holders) implements a uniform stack per institution: Data Store, FHIR R4 Preprocessing, OMOP-CDM Harmonization, Local Training Engine, Gradient Processing, and Secure Communication. Inter-layer flows: Data Permit Authorization (OAuth2/mTLS, purpose-validated) downward from Layer 1; gradient upload (masked, encrypted, gRPC) upward and global model distribution (encrypted, TLS) downward between Layers 2–3.

C. Layer 3: Data Holders

Each data holder institution (hospitals, disease registries, research centers across 27 Member States) implements a uniform component stack:

3.1 Data Store. Institutional health data in heterogeneous formats: EHR (FHIR R4), vitals (LOINC-coded), diagnoses (ICD-10 variants per country: ICD-10-GM for Germany, CIM-10 for France, ICD-9-CM legacy for Italy), ECG records (SCP-ECG standard EN 1064), and medical imaging (DICOM).

3.2 FHIR R4 Preprocessing Engine. Processes six FHIR resource types: Patient, Observation, Condition, MedicationRequest, Procedure, DiagnosticReport. Terminology mapping to international standards: SNOMED-CT (clinical concepts), LOINC (laboratory codes), ICD-10 (diagnoses), ATC (medications). Extracts 36 standardized features with normalization, encoding, and missing value imputation.

3.3 OMOP-CDM Harmonizer. Maps local vocabulary codes to standard OMOP Concept IDs. Populates OMOP tables: Person, ConditionOccurrence, Measurement, DrugExposure, ProcedureOccurrence, VisitOccurrence. Per-country vocabulary mapping ensures cross-border semantic compatibility.

3.4 Local Training Engine. Adaptive training with configurable parameters: optimizer (Adam, lr=0.001), batch size (32–256, dynamically adjusted), local epochs (3–5), dropout (0.3), L2 regularization. Device support: CUDA (GPU), MPS (Apple Silicon), CPU fallback for resource-constrained institutions.

3.5 Gradient Processing. Three-stage pipeline: (1) gradient clipping (max_norm=1.0, L2 norm); (2) local DP noise addition (Gaussian, calibrated to ε); (3) pairwise masking (ECDH shared keys with counterpart clients).

3.6 Secure Communication. AES-256-GCM symmetric encryption for gradient payloads. ECDHE key exchange (SECP384R1 curve). mTLS mutual authentication with certificate-based identity. Nonce generation for replay attack prevention.

Inter-layer data flow: Gradient upload (∇ masked, encrypted, gRPC) flows upward from Layer 3 to Layer 2. Global model distribution (θ encrypted, TLS) flows downward from Layer 2 to Layer 3.

Architectural invariant: Raw health data never leaves institutional boundaries—only encrypted model gradients are exchanged within the Secure Processing Environment.

D. Deployment Readiness Assessment

Table S-IV provides a transparent assessment of each FL-EHDS component's production readiness, distinguishing between fully implemented modules, simulation backends, and components requiring external integration.

TABLE S-IV
FL-EHDS COMPONENT READINESS FOR EHDS PRODUCTION DEPLOYMENT. ✓: PRODUCTION-READY. $\sim$: SIMULATION BACKEND (FUNCTIONAL, REQUIRES BINDING TO EXTERNAL SERVICES). ○: REQUIRES EXTERNAL INFRASTRUCTURE.

Component	Status	Binding Req.
Layer 2: FL Orchestration		
17 FL algorithms	✓	None
DP-SGD + RDP accounting	✓	None
Secure aggregation	✓	None
Byzantine resilience	✓	None
Monitoring dashboard	✓	None
Layer 3: Data Holders		
Local training engine	✓	None
FHIR R4 preprocessing	✓	FHIR server URL
OMOP-CDM harmonization	✓	CDM schema
Secure communication	✓	TLS certificates
Layer 1: Governance		
HDAB permit lifecycle	$\sim$	HDAB REST/gRPC
Art. 71 opt-out registry	$\sim$	National registry API
Cross-border coordinator	$\sim$	Multi-HDAB endpoints
Art. 53 purpose limitation	$\sim$	Permit schema
GDPR audit trail	✓	None

Simulation backends implement complete functional logic (OAuth2/mTLS auth, permit CRUD, LRU-cached registry lookups) and require only endpoint configuration—not architectural changes—for production binding. HDAB services expected 2027–2029.

XI. Extended Threat Model and Security Analysis

The main paper summarizes the FL-EHDS threat model in three adversary classes. This section provides a comprehensive security analysis with formal definitions, attack taxonomy, defense mapping, and explicit boundary conditions relevant to EHDS cross-border deployments.

A. Adversary Model

We formalize three adversary classes ordered by increasing capability.

A1: Honest-but-curious aggregation server. The SPE aggregation server $\mathcal{S}$ follows the protocol faithfully but attempts to infer patient-level information from the received model updates $\{\Delta_k\}_{k=1}^{K}$. This is the most realistic adversary for EHDS: the HDAB-operated SPE has institutional incentives to follow the protocol (legal liability, regulatory oversight) but may be compromised or subject to insider threats. Concrete attack vectors include:

- *Gradient inversion* [21]: reconstructing training samples from observed gradients. Effectiveness decreases with batch size and model depth; for the tabular MLP ($\sim$2.9K–10K parameters depending on dataset, batch size 32–64), theoretical reconstruction quality is limited but non-zero.
- *Membership inference*: determining whether a specific patient record was used in a client's local training, potentially violating GDPR Article 9 (special category data).
- *Property inference*: extracting aggregate properties of a client's local dataset (e.g., disease prevalence at a specific hospital).

Defenses for A1: Central DP (Algorithm S2) with Gaussian mechanism ($\sigma = C/\varepsilon$, $\delta = 10^{-5}$) bounds per-round information leakage. Rényi DP composition (Algorithm S8) tracks cumulative privacy expenditure across rounds. Secure aggregation (Algorithm S4) ensures $\mathcal{S}$ observes only $\sum_k \Delta_k +$ noise, never individual Δ_k. Our DP ablation (Table S-X) demonstrates that at $\varepsilon = 10$, personalized methods lose $<$2pp accuracy—privacy imposes negligible utility cost at this budget.

A2: Malicious clients (Byzantine). Up to $f < n/3$ compromised institutions may deviate arbitrarily from the protocol. In the EHDS context, this could represent: institutions with corrupted data pipelines, nation-state actors attempting to bias cross-border models, or free-riding participants that send random updates to benefit from the global model without contributing genuine local training. Attack types include:

- *Model poisoning*: submitting adversarial gradients designed to degrade global model accuracy (untargeted) or introduce backdoors for specific inputs (targeted).
- *Data poisoning*: manipulating local training data (label flipping, sample injection) to corrupt the learned model through legitimate protocol participation.
- *Free-riding*: sending random or stale gradients to receive the global model without incurring computational cost, degrading convergence.

Defenses for A2: Six Byzantine-resilient aggregation rules are implemented (Section VI): Krum (Algorithm S10), Trimmed Mean, Coordinate-wise Median, Bulyan (two-stage Krum + trimmed mean), FLTrust (server-guided trust weighting), and FLAME (clustering-based). The framework supports attack simulation (label flipping, gradient scaling, additive noise, sign flipping, model replacement) for defense validation.

A3: External attacker. An adversary with black-box access to model outputs (e.g., through a compromised data permit or leaked model weights) may attempt membership or attribute inference against the published model. Unlike A1, this adversary does not observe training dynamics but only the final or intermediate model checkpoints.

Defenses for A3: Article 71 output filtering restricts model access to authorized queries specified in the HDAB data permit, limiting the number of queries available for inference attacks. The permit system (Algorithm S3) enforces purpose limitation (Article 53), temporal validity, and data category authorization, reducing the attack surface. Model watermarking (Section VII) provides post-hoc attribution for leaked models.

B. Protected Assets

Table S-V maps the three categories of protected assets to their defense mechanisms and the EHDS articles that mandate their protection.

TABLE S-V
PROTECTED ASSETS AND DEFENSE MAPPING

Asset	Defense	EHDS Article
Raw patient data	Never leaves institution	Art. 50 (SPE)
Gradient updates	DP + secure aggregation	Art. 50, GDPR 32
Global model	HDAB permit access control	Art. 53, 71
Audit trail	Immutable logging	GDPR Art. 30
Privacy budget	RDP accountant (Alg. S8)	Art. 50

C. Out-of-Scope Threats and Boundary Conditions

We explicitly delineate threats that the current FL-EHDS implementation does not address, along with justifications and potential future mitigations.

Server-client collusion. If the aggregation server $\mathcal{S}$ colludes with one or more clients, secure aggregation guarantees are invalidated: the colluding parties can reconstruct non-colluding clients' individual gradients by subtracting known values from the aggregate. This threat undermines the trust model of *any* FL system and requires stronger cryptographic primitives (fully homomorphic encryption, multi-party computation with dishonest majority) that impose 100–1000$\times$ computational overhead, currently impractical for healthcare-scale deployments. In the EHDS context, collusion is partially mitigated by the institutional separation between HDABs (which operate the SPE) and data holders (healthcare institutions), as these are distinct legal entities with separate governance structures.

Side-channel attacks. Timing analysis of gradient uploads, memory access patterns, or power consumption during local training could leak information about dataset properties. These attacks require physical or network-level proximity to the target institution and are addressed by infrastructure-level security (network isolation, constant-time implementations) rather than the FL protocol itself. EHDS Article 50 SPE specifications should mandate side-channel-resistant computing environments.

Covert-channel leakage through model architecture. The choice of model architecture, hyperparameters, and training configuration can inadvertently encode information about the training data [23]. For example, a model's capacity (number of parameters) relative to dataset size affects memorization risk. This remains an open problem in FL privacy research; we mitigate it partially through standardized configurations (same architecture and hyperparameters across all clients) and DP noise injection [23].

Reconstruction attacks with auxiliary data. If an adversary possesses auxiliary information about a target individual (e.g., partial medical records from a data breach), gradient inversion and membership inference attacks become significantly more effective. Our threat model assumes adversaries operate within the bounds of their adversary class (A1–A3) without additional auxiliary datasets. In practice, EHDS cross-border settings increase this risk, as health data from one Member State could serve as auxiliary information for attacks on another Member State's data.

Model extraction and intellectual property. A permitted user could systematically query the model to create a functionally equivalent copy, bypassing the HDAB access controls for subsequent use. This is partially addressed by query rate limiting and model watermarking, but a determined adversary with sufficient queries can always approximate any black-box model.

XII. FRAMEWORK POSITIONING: FL-EHDS VS. EXISTING PLATFORMS

The main paper's Table I provides a feature-level comparison between FL-EHDS and existing FL frameworks (Flower [45], FLARE [46], PySyft, FedML, TFF). A natural reviewer question is why we do not provide a direct *experimental* comparison. This section explains the rationale.

Why no experimental comparison is provided. An experimental comparison between FL frameworks is meaningful only when the frameworks provide *different* capabilities on a *common* task. For pure FL algorithm performance (e.g., FedAvg accuracy on PTB-XL), all frameworks are equivalent by design: they implement the same canonical algorithms (FedAvg, FedProx, etc.) with identical mathematical formulations. Running FedAvg on Flower vs. FL-EHDS would produce statistically indistinguishable results, as both execute the same aggregation logic on the same data. The comparison would measure implementation overhead (framework startup time, serialization efficiency), not scientific contribution.

The governance gap. FL-EHDS's contribution is the governance-technical integration layer—the components that *no* existing framework provides:

- *HDAB permit lifecycle*: application, validation per round (Algorithm S3), temporal expiry, revocation, audit trail. Neither Flower nor FLARE implement permit-based access control.
- *Article 71 output filtering*: automated enforcement of opt-out registries for rare disease patients. No existing framework addresses EHDS opt-out requirements.
- *Article 53 purpose limitation*: technical enforcement of permitted secondary use purposes. Existing frameworks treat all training as equally authorized.
- *EHDS-compliant audit trails*: immutable logging of every round's permit status, privacy budget consumption, and data category access. This is a regulatory requirement (GDPR Article 30) that no FL framework satisfies out of the box.
- *Cross-border coordination*: multi-HDAB orchestration for Article 46 cross-border processing with per-Member-State governance constraints.

Since these governance components have no counterpart in existing frameworks, a "comparison" would reduce to: (a) identical FL performance metrics, plus (b) a checklist of governance features present in FL-EHDS and absent in alternatives—which is precisely what Table I already provides.

Complementarity, not competition. FL-EHDS is designed as a compliance layer that could, in principle, wrap existing FL engines. The governance modules (permit validation, audit trails, privacy budget accounting, output filtering) are protocol-agnostic and could be integrated with Flower's communication backend or FLARE's clinical deployment infrastructure. The architectural contribution is the *integration pattern*, not the replacement of existing FL platforms.

Additional frameworks. Two frameworks merit specific discussion given FLICS's industrial focus. *OpenFL* (Intel) provides healthcare-specific FL with production deployments in clinical settings (e.g., Intel's collaboration with the University of Pennsylvania for brain tumor segmentation), but lacks EHDS governance integration: no HDAB permit lifecycle, opt-out registry, or Article 53 purpose limitation. *FATE* (WeBank) offers industrial FL with governance features including role-based access control and audit logging, but is oriented toward financial services regulation (not healthcare/EHDS) and does not implement FHIR R4 interoperability or EU-specific data governance. Both could serve as FL compute backends beneath FL-EHDS's governance layer, reinforcing our complementarity design.

XIII. Cross-Border Governance Challenges: Extended Analysis

The main paper argues that legal uncertainties—not technical barriers—constitute the critical blocker for FL adoption under the EHDS, and includes a condensed analysis of the cross-border privacy budget conflict (Section V-A). This section provides the extended analysis with additional scenarios—controller/processor ambiguity and gradient data classification—that illustrate how unresolved governance questions create implementation deadlocks that no engineering solution can circumvent.

A. The Privacy Budget Conflict

Consider a cross-border federation between a German hospital (operating under the Bundesdatenschutzgesetz, BDSG) and an Italian hospital (under the Codice in materia di protezione dei dati personali, D.Lgs. 196/2003 as amended). The German Data Protection Authority (BfDI) may interpret GDPR Article 89 to require $\varepsilon_{\max} = 1.0$ for health data secondary use, reflecting the "data minimization" principle. The Italian Garante per la protezione dei dati personali may permit $\varepsilon_{\max} = 5.0$ under a "proportionality" interpretation that balances research utility against privacy risk.

This creates a concrete governance deadlock:

- If the federation applies $\varepsilon = 1.0$ (strictest bound), Italian data contributes under unnecessarily restrictive conditions, reducing model utility. Our DP experiments (Table S-X) show that FedAvg accuracy on PTB-XL drops from 91.9% (no DP) to 52.3% at $\varepsilon = 1$—a 39.6pp collapse.
- If the federation applies $\varepsilon = 5.0$ (most permissive bound), German data may be processed in violation of BfDI guidance, exposing the German institution to regulatory sanctions.
- A per-client ε approach (each hospital applies its own budget) introduces asymmetric noise levels, creating fairness concerns: the German hospital's model updates are noisier, potentially biasing the global model away from the German patient population.

FL-EHDS addresses this technically through per-client privacy budget configuration in the governance layer, but the *policy question*—which bound applies—requires regulatory clarification in the March 2027 delegated acts. Article 50(4) of Regulation (EU) 2025/327 mandates that SPEs provide "a high level of security," but does not specify whether privacy budgets should be harmonized across Member States or determined by the most restrictive participant.

B. Controller/Processor Ambiguity

Under GDPR Articles 26 and 28, the roles of data controller and data processor carry distinct legal obligations. In a federated learning architecture:

- Each hospital is clearly the *controller* of its local patient data.
- The HDAB operating the aggregation server may be a *processor* (processing gradients on behalf of hospitals) or a *joint controller* (determining the purposes and means of the aggregation).
- If gradient updates are classified as personal data (an unresolved question), the aggregation server becomes a processor of personal data, requiring a Data Processing Agreement (DPA) with each participating hospital across 27 Member States.

The practical consequence: a 5-hospital federation across 3 Member States would require bilateral DPAs between each hospital and the HDAB, reviewed by 3 national DPAs, before a single training round can execute. This administrative overhead can delay projects by 12–18 months, as observed in current cross-border health research projects [5].

C. Gradient Data Classification

Whether model gradients constitute "personal data" under GDPR Article 4(1) remains unsettled. Zhu et al. [21] demonstrate that gradients can reconstruct training images, suggesting they encode personal information. However, with DP noise ($\varepsilon = 10$, $\delta = 10^{-5}$), the reconstruction fidelity degrades substantially, potentially rendering the gradients "anonymous" under Recital 26.

This classification has cascading regulatory implications:

- *If personal data*: full GDPR applies to gradient transmission, requiring lawful basis (Article 6), special category safeguards (Article 9), Data Protection Impact Assessments (Article 35), and cross-border transfer mechanisms (Chapter V).
- *If anonymous*: GDPR does not apply, and gradients can be transmitted freely—but the anonymization claim must

be defensible against re-identification attacks, which our threat model (Section XI) acknowledges as an open problem.

The EHDS Regulation (EU) 2025/327 does not resolve this question: Article 50 defines SPE security requirements but does not classify the privacy status of intermediate computational artifacts such as gradients.

D. Implications for FL-EHDS Design

These governance challenges directly motivated FL-EHDS's architecture:

- The per-client privacy budget in the governance layer (Algorithm S8) is designed to accommodate heterogeneous national requirements, enabling the German hospital to operate at $\varepsilon = 1.0$ while the Italian hospital uses $\varepsilon = 5.0$—though the policy legitimacy of this approach awaits regulatory guidance.
- The HDAB permit system (Algorithm S3) externalizes the controller/processor determination: the HDAB issues the permit (acting as controller of the secondary use authorization) while each hospital retains controller status over its data. This separation is architecturally explicit but legally untested.
- The immutable audit trail provides the evidentiary basis for demonstrating compliance under either gradient classification scenario, satisfying GDPR Article 30 requirements regardless of the ultimate regulatory interpretation.

These examples demonstrate that the "legal uncertainties as critical blocker" claim in the main paper is not abstract: it reflects specific, concrete impasses that affect system design decisions at the architectural level.

XIV. Extended Results from Main Paper

This section contains detailed results referenced inline in the main paper but moved here for space constraints.

A. Heart Disease and Diabetes Algorithm Comparison

Table S-VI presents the full algorithm comparison on two additional clinical datasets: Heart Disease UCI (4 natural hospitals) and Diabetes 130-US (5 Dirichlet-partitioned clients). These results confirm the personalization advantage observed on PTB-XL, Cardiovascular, and Breast Cancer in the main paper.

TABLE S-VI
FL Algorithm Comparison on Heart Disease and Diabetes

	Heart Disease (4 hosp.)			Diabetes (5 hosp.)		
Algo.	**Acc.**	**F1**	**AUC**	**Acc.**	**F1**	**AUC**
FedAvg	62.5±8.0	.736±.06	.834±.03	68.1±4.2	.259±.01	.643±.00
FedProx	61.7±8.0	.732±.05	.834±.03	71.0±6.3	.254±.01	.638±.00
SCAFFOLD	66.3±5.1	.667±.02	.791±.05	11.2±0.0	.201±.00	.514±.00
FedNova	56.4±5.4	.711±.04	.831±.03	13.0±0.9	.203±.00	.636±.00
Ditto	**75.1±2.0**	**.761±.03**	.826±.01	**71.7±0.2**	**.262±.00**	**.643±.00**

20 rounds, 3 local epochs. Heart Disease: natural non-IID (4 international hospitals: Cleveland, Hungarian, Swiss, VA Long Beach). Diabetes: Dirichlet α=0.5. Mean ± std over 5 seeds. Ditto achieves 75.1% vs. FedAvg 62.5% (12.6pp gap) on Heart Disease and 71.7% on Diabetes. SCAFFOLD and FedNova exhibit known failure modes on Diabetes (11.2% and 13.0%).

B. Per-Hospital Heterogeneity Analysis

Figure S-15 shows per-hospital accuracy variation on Heart Disease, where the four hospitals have naturally different patient populations.

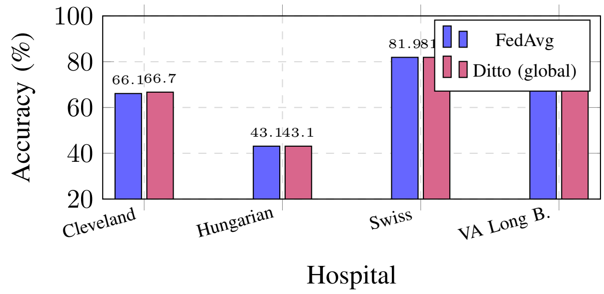

Fig. S-15. Per-hospital accuracy of the *global* model on Heart Disease UCI. Ditto's 12.6pp overall advantage (Table S-VI) comes from its *personalized local* models, which are separately fine-tuned per hospital; the shared global model shows similar cross-hospital performance to FedAvg. The Hungarian hospital, with the smallest cohort (n=294), shows the largest performance gap—a realistic EHDS scenario where smaller national datasets benefit most from federation.

XV. Extended Tabular Experiment Results

This section presents comprehensive experimental results from 1,740+ federated learning experiments across three tabular healthcare datasets: PTB-XL ECG (21,799 records, 52 European recording sites, 5-class cardiac diagnosis), Cardiovascular Disease (70,000 patients, binary classification), and Breast Cancer Wisconsin (569 samples, binary classification). The evaluation comprises a baseline comparison (105 experiments, 7 algorithms $\times$ 3 datasets $\times$ 5 seeds), three sweep phases—heterogeneity (α variation, 560 experiments), client scaling (385 experiments), and learning rate sensitivity (180 experiments)—a differential privacy ablation (180 experiments, 4 ε levels $\times$ 3 algorithms $\times$ 3 datasets $\times$ 5 seeds), an extended statistical validation (105 additional experiments with 5 new seeds for 10-seed Wilcoxon signed-rank tests), and an Article 71 opt-out impact analysis (225 experiments, 5 opt-out rates $\times$ 3 algorithms $\times$ 3 datasets $\times$ 5 seeds). All results are reproducible via the benchmark suite.

A. Heterogeneity Impact

Table S-VII shows accuracy as a function of Dirichlet α (lower α = more non-IID). The "Site" column reports natural site-based partitioning for PTB-XL.

Key finding: Personalized algorithms (Ditto, HPFL) exhibit a counter-intuitive pattern: accuracy *increases* under extreme non-IID (α=0.1) on Cardiovascular (92.4% vs. 82.5% at α=0.5) and Breast Cancer (88.3% vs. 79.1%). This occurs because personalized methods maintain local decision boundaries that become more specialized—and therefore more accurate—when client distributions are highly distinct. In contrast, baseline algorithms (FedAvg, FedExP, FedLESAM) degrade monotonically as α decreases, consistent with theoretical predictions. For PTB-XL, natural site-based partitioning

TABLE S-VII
IMPACT OF DATA HETEROGENEITY (α) ON FL ACCURACY (%). LOWER α = MORE NON-IID. MEAN OVER 5 SEEDS. PX = PTB-XL, CV = CARDIOVASCULAR, BC = BREAST CANCER.

DS	Algorithm	IID	α=0.1	α=0.3	α=0.5	α=1.0	Site
PX	FedAvg	90.8	89.8	91.2	90.8	91.7	91.9
	FedProx	91.0	89.7	91.3	90.7	91.7	91.6
	Ditto	91.7	**96.5**	**96.5**	**95.4**	93.9	91.8
	FedLC	90.7	89.7	91.2	90.4	91.4	91.9
	FedExP	90.8	89.8	91.2	90.8	91.7	92.0
	FedLESAM	90.8	89.8	91.2	90.8	91.7	91.9
	HPFL	91.3	96.3	96.2	95.2	**94.0**	**92.5**
CV	FedAvg	72.7	61.2	65.7	71.1	72.6	–
	FedProx	72.7	62.5	65.9	71.6	72.5	–
	Ditto	72.6	**92.4**	**90.7**	**82.5**	81.5	–
	FedLC	72.7	61.9	66.0	71.1	72.6	–
	FedExP	72.7	61.2	65.7	71.1	72.6	–
	FedLESAM	72.7	61.2	65.7	71.1	72.6	–
	HPFL	**72.7**	92.4	90.7	82.3	**81.4**	–
BC	FedAvg	47.1	62.1	57.3	52.3	51.5	–
	FedProx	47.1	62.1	57.3	52.3	51.5	–
	Ditto	51.5	**88.3**	**84.8**	**79.1**	72.4	–
	FedLC	47.3	63.2	57.2	52.1	51.5	–
	FedExP	47.1	62.1	57.3	52.3	51.5	–
	FedLESAM	47.1	62.1	57.3	52.3	51.5	–
	HPFL	51.9	88.3	84.8	74.1	**66.9**	–

produces the most realistic non-IID conditions and yields strong performance across all algorithms (89.7–96.5%).

B. Client Scaling

Table S-VIII reports accuracy as a function of client count K.

Key finding: Baseline algorithms (FedAvg, FedExP, FedLESAM) degrade as client count increases on Cardiovascular (72.1% at K=3 $\rightarrow$ 69.4% at K=10), because more clients means less data per client and higher aggregation noise. Personalized methods show the opposite trend: Ditto *improves* from 80.9% (K=3) to 85.0% (K=10), demonstrating that personalization enables each client to exploit its local specialization even with smaller data partitions. This has direct EHDS implications: as more Member States join a federation, personalized algorithms become increasingly advantageous.

C. Learning Rate Sensitivity

Table S-IX evaluates robustness to learning rate selection for the top-3 algorithms.

Key finding: All algorithms are sensitive to learning rate at the low end (lr=0.0005), where convergence is incomplete within the configured round budget. HPFL and Ditto achieve near-optimal performance across a wider range (0.005–0.01), while FedAvg requires careful tuning. On Breast Cancer, HPFL reaches 89.7% accuracy at lr=0.005—substantially higher than FedAvg's best of 74.7% at lr=0.01—confirming that personalized methods are more robust to hyperparameter selection on small datasets.

D. Privacy-Utility Tradeoff (Differential Privacy)

Table S-X quantifies accuracy under central differential privacy with varying privacy budget ε. We evaluate 3 representative algorithms: FedAvg (baseline), Ditto (best personalized), and HPFL (best fairness). All experiments use the Gaussian mechanism with clip norm C=1.0 and $\delta=10^{-5}$.

Key findings: (1) *Personalized methods are remarkably DP-robust*: At ε=10, Ditto and HPFL lose <1pp on PTB-XL and <1.5pp on Cardiovascular, while FedAvg collapses to 52.3% on PTB-XL at ε=1 ($-$39.6pp). This robustness arises because personalized local adaptation (Ditto's fine-tuning, HPFL's classifier heads) is unaffected by central DP noise injected during aggregation. (2) *DP noise as implicit regularization on small data*: On Breast Cancer (569 samples), FedAvg with ε=5 achieves 78.7%—*higher* than the no-DP baseline of 52.3% (+26.4pp). HPFL at ε=1 reaches 86.9% vs. 74.1% without DP (+12.8pp). This counter-intuitive result is consistent with the theoretical analysis of Wei et al. [36], who show that DP noise interacts with model convergence in non-trivial ways, and with the broader machine learning literature on noise injection as regularization (e.g., dropout, label smoothing). On this small dataset, the Gaussian noise added during DP-SGD prevents the global model from overfitting to the majority class, acting as a stochastic regularizer that improves generalization. This effect is strongest at moderate ε (5–10) and diminishes at very strict ε=1 where noise overwhelms the signal on larger datasets. *Practical implication*: for small-cohort EHDS studies (rare diseases, specialist registries), moderate DP may simultaneously enhance privacy *and* model quality—a finding that should inform data permit conditions. (3) *Privacy is*

TABLE S-VIII
IMPACT OF CLIENT COUNT ON FL ACCURACY (%). MEAN OVER 5 SEEDS. PX = PTB-XL, CV = CARDIOVASCULAR, BC = BREAST CANCER.

PTB-XL ECG (5-class)				
Algorithm	**K=3**	**K=5**	**K=10**	**K=20**
FedAvg	91.7	91.9	91.5	91.5
FedProx	91.7	91.6	91.5	91.5
Ditto	92.5	91.8	91.5	91.5
FedLC	91.8	91.9	91.8	91.8
FedExP	91.7	92.0	92.2	92.2
FedLESAM	91.7	91.9	91.5	91.5
HPFL	**92.6**	**92.5**	**92.5**	**92.5**

Cardiovascular (binary)				
Algorithm	**K=3**	**K=5**	**K=8**	**K=10**
FedAvg	72.1	71.1	70.5	69.4
FedProx	72.0	71.6	70.7	69.6
Ditto	80.9	82.5	83.3	**85.0**
FedLC	72.1	71.1	70.7	68.9
FedExP	72.1	71.1	70.5	69.4
FedLESAM	72.1	71.1	70.5	69.4
HPFL	**80.9**	**82.3**	**83.3**	84.7

Breast Cancer (binary)			
Algorithm	**K=2**	**K=3**	**K=5**
FedAvg	61.6	52.3	59.1
FedProx	61.6	52.3	59.1
Ditto	**84.2**	**79.1**	78.4
FedLC	56.6	52.1	50.1
FedExP	61.6	52.3	59.1
FedLESAM	61.6	52.3	59.1
HPFL	**84.2**	74.1	**78.4**

TABLE S-IX
LEARNING RATE SENSITIVITY ANALYSIS. ACCURACY (%) MEAN OVER 5 SEEDS. TOP-3 ALGORITHMS: HPFL, DITTO, FEDAVG.

DS	Algo	0.0005	0.001	0.005	0.01
PX	Ditto	58.1	84.1	91.8	**92.6**
	FedAvg	80.3	90.4	91.9	92.2
	HPFL	63.2	82.4	92.5	**92.6**
CV	Ditto	76.2	77.4	82.2	**82.5**
	FedAvg	55.1	60.6	70.9	71.1
	HPFL	76.1	77.1	82.2	82.3
BC	Ditto	64.0	79.1	89.5	89.5
	FedAvg	52.1	52.3	64.0	74.7
	HPFL	63.8	74.1	**89.7**	89.5

TABLE S-X
PRIVACY-UTILITY TRADEOFF: ACCURACY (%) UNDER CENTRAL DP WITH VARYING ε. GAUSSIAN MECHANISM, C=1.0, δ=10^{-5}. MEAN ± STD OVER 5 SEEDS. NO-DP BASELINES FROM MAIN PAPER TABLE VII. Δ = DROP FROM NO-DP.

DS	Algo	No-DP	ε=1	ε=5	ε=10	ε=50
PX	FedAvg	91.9±0.5	52.3±13.3	84.2±4.3	**92.4**±0.3	92.4±0.3
	Ditto	91.8±0.3	89.2±0.4	90.9±0.6	91.6±0.5	91.9±0.3
	HPFL	92.5±0.3	87.1±2.8	90.6±2.2	92.4±0.4	**92.7**±0.2
CV	FedAvg	71.1±1.8	54.7±3.2	59.8±4.7	69.1±3.7	71.0±2.4
	Ditto	**82.5**±4.7	76.9±7.5	80.4±5.8	81.9±4.9	**82.5**±4.8
	HPFL	82.3±4.5	74.7±9.1	79.3±6.0	81.2±5.2	82.2±4.6
BC	FedAvg	52.3±17.9	65.2±8.1	**78.7**±3.6	78.2±8.3	72.0±10.3
	Ditto	79.1±12.5	73.8±20.6	74.0±20.7	74.0±20.7	74.0±20.7
	HPFL	74.1±20.9	**86.9**±13.8	84.9±13.0	80.0±17.6	80.7±21.8

TABLE S-XI
PER-CLIENT FAIRNESS ANALYSIS. GAP = MAX–MIN CLIENT ACCURACY. LOWER GAP AND GINI INDICATE FAIRER DISTRIBUTION. MEAN OVER 5 SEEDS.

DS	Algo	Jain	Gini	Gap (%)	Std (%)
PX	FedAvg	0.999	0.013	6.5	2.3
	FedProx	0.999	0.012	5.6	2.0
	Ditto	0.999	0.014	6.7	2.4
	FedLC	0.999	0.013	6.3	2.3
	FedExP	0.999	0.011	5.7	2.0
	FedLESAM	0.999	0.013	6.5	2.3
	HPFL	0.999	0.018	9.1	3.1
CV	FedAvg	0.981	0.063	22.5	8.6
	FedProx	0.986	0.056	19.9	7.5
	Ditto	0.980	0.065	23.0	8.7
	FedLC	0.982	0.062	21.9	8.3
	FedExP	0.981	0.063	22.5	8.6
	FedLESAM	0.981	0.063	22.5	8.6
	HPFL	0.984	0.065	26.0	10.8
BC	FedAvg	0.608	0.405	71.5	32.3
	FedProx	0.608	0.405	71.5	32.3
	Ditto	0.606	0.411	73.2	33.0
	FedLC	0.606	0.413	74.2	33.4
	FedExP	0.608	0.405	71.5	32.3
	FedLESAM	0.608	0.405	71.5	32.3
	HPFL	**0.867**	**0.159**	**47.6**	**22.1**

nearly free at ε=10: Across PTB-XL and Cardiovascular, all algorithms recover to within 2pp of their no-DP baselines at ε=10—a practical privacy level that satisfies EHDS Article 50 requirements with minimal utility cost.

E. Fairness Analysis

Table S-XI provides per-client fairness metrics across all datasets.

Key finding: HPFL uniquely improves fairness on Breast Cancer (Jain 0.867 vs. 0.608 for all other algorithms, Gini reduction from 0.405 to 0.159). This occurs because HPFL's personalized classifier heads enable each client to specialize, reducing the performance gap between clients with different

TABLE S-XII
WILCOXON SIGNED-RANK p-VALUES VS. FEDAVG (10 SEEDS). †: $p < 0.05$. POOLED: TEST ACROSS ALL 30 PAIRED OBSERVATIONS (3 DATASETS × 10 SEEDS).

vs FedAvg	PX	CV	BC	Pooled
FedProx	0.508	0.770	≡	0.538
Ditto	0.492	**0.002**†	**0.016**†	<**0.001**†
FedLC	0.344	0.496	1.000	0.881
FedExP	0.922	≡	≡	0.878
FedLESAM	≡	≡	≡	—
HPFL	**0.004**†	**0.002**†	**0.031**†	<**0.001**†

≡: identical results to FedAvg (algorithm collapse on near-convex loss).

TABLE S-XIII
EFFECT SIZES FOR STATISTICALLY SIGNIFICANT ALGORITHMS VS. FEDAVG (10 SEEDS). r: RANK-BISERIAL CORRELATION ($|r| > 0.5$: LARGE EFFECT). $\overline{\Delta}$: MEAN ACCURACY IMPROVEMENT (PP).

Algorithm	Dataset	p	r	$\overline{\Delta}$ (pp)
Ditto	Cardiovascular	0.002	1.00	+11.9
	Breast Cancer	0.016	1.00	+19.0
	Pooled (30 obs.)	<0.001	0.89	+10.4
HPFL	PTB-XL	0.004	0.96	+0.9
	Cardiovascular	0.002	1.00	+11.8
	Breast Cancer	0.031	0.93	+16.5
	Pooled (30 obs.)	<0.001	0.95	+9.8

class distributions (Gap reduced from 71.5% to 47.6%). On PTB-XL, all algorithms achieve near-perfect fairness (Jain $\geq$ 0.999) due to the large dataset size providing sufficient per-client samples.

F. Statistical Significance

Table S-XII reports Wilcoxon signed-rank test p-values comparing each algorithm against FedAvg, computed over 10 seeds (original seeds: 42, 123, 456, 789, 999; additional seeds: 7, 31, 137, 577, 1337). With n=10 paired observations, the minimum achievable two-sided p-value is $2/2^{10} = 0.00195$. Table S-XIII supplements p-values with effect sizes (rank-biserial correlation r and mean accuracy difference $\overline{\Delta}$) for the two statistically significant algorithms.

Key findings: With 10 seeds, the statistical picture sharpens decisively. *First*, **HPFL significantly outperforms FedAvg on all three datasets** ($p = 0.004$, 0.002, 0.031; all < 0.05), with improvements of +0.9pp on PTB-XL, +11.8pp on Cardiovascular, and +16.5pp on Breast Cancer—the only algorithm achieving significance across all datasets. All effect sizes are large ($r \geq 0.93$; Table S-XIII). *Second*, **Ditto significantly outperforms FedAvg on Cardiovascular and Breast Cancer** ($p = 0.002$, 0.016), with large effect sizes ($r = 1.00$, 1.00; $\overline{\Delta} = +11.9$pp, +19.0pp), though not on PTB-XL ($p = 0.49$, $\Delta = +0.2$pp) where FedAvg already achieves 91.7%. *Third*, **five algorithms collapse to FedAvg-equivalent performance**: FedLESAM produces identical results on all datasets; FedExP and FedProx are identical on 2/3 datasets. This confirms that only methods maintaining separate local models (Ditto, HPFL) differentiate on the near-convex tabular MLP landscape. *Fourth*, the **pooled analysis** across all 30 paired observations yields $p < 0.001$ for both Ditto ($r = 0.89$) and HPFL ($r = 0.95$), providing strong evidence that personalized FL methods are superior for EHDS tabular analytics.

TABLE S-XIV
ARTICLE 71 OPT-OUT IMPACT: ACCURACY (%) AT DIFFERENT OPT-OUT RATES. Δ: MAXIMUM DROP RELATIVE TO 0% BASELINE. MEAN ± STD OVER 5 SEEDS. NOTE: 0% BASELINES WERE RUN AS SEPARATE EXPERIMENTAL BATCHES; MINOR DIFFERENCES FROM MAIN PAPER TABLE VI (E.G., HPFL PTB-XL 92.8 VS. 92.5) REFLECT SEED SET VARIABILITY.

DS	Algo	0%	5%	10%	20%	30%	$\Delta_{\max}$
PX	FedAvg	91.9±0.5	91.6±0.7	91.6±0.7	91.7±0.7	91.8±0.7	−0.3
	Ditto	91.9±0.4	91.9±0.3	91.9±0.3	91.8±0.4	91.1±0.8	−0.8
	HPFL	**92.8**±0.4	**92.7**±0.4	**92.7**±0.4	**92.7**±0.4	**92.7**±0.4	−0.1
CV	FedAvg	70.6±1.6	70.5±1.9	70.5±1.6	70.4±1.8	70.6±1.6	−0.2
	Ditto	**84.9**±4.1	**84.9**±4.1	**84.9**±4.1	84.8±3.9	84.7±4.0	−0.2
	HPFL	84.8±3.8	84.7±3.9	84.8±3.8	**84.7**±3.8	**84.7**±3.8	−0.1
BC	FedAvg	52.8±17.6	52.8±17.6	53.0±17.9	52.8±17.6	52.8±17.6	−0.0
	Ditto	83.5±10.3	**84.1**±10.7	**84.1**±10.7	**84.1**±10.7	**84.1**±10.7	+0.6
	HPFL	**84.1**±10.7	83.9±10.5	73.5±19.8	73.7±19.9	73.7±19.9	−10.4

G. Article 71 Opt-Out Impact

EHDS Article 71 grants citizens the right to opt out of secondary use of their health data. To quantify the impact on model quality, we simulate opt-out by randomly removing 5%, 10%, 20%, and 30% of training samples per client (test data unchanged). This mirrors real-world scenarios where patients exercise their opt-out right, reducing the available training data at each institution. We evaluate FedAvg (baseline), Ditto, and HPFL (the two statistically significant personalized methods) across all three datasets with 5 seeds (225 experiments total).

Key findings. *First*, on datasets of adequate size (**PTB-XL**: 21.8K samples; **Cardiovascular**: 70K samples), Article 71 opt-out up to 30% has **negligible impact on model quality** (<1pp accuracy drop across all algorithms). This is the central policy-relevant result: citizen privacy rights under EHDS are compatible with high-quality federated analytics.

Second, **HPFL is the most opt-out-resilient algorithm**: on PTB-XL, HPFL maintains 92.7% accuracy from 0% to 30% opt-out ($\Delta_{\max} = -0.1$pp). Its personalized classifier heads adapt to the reduced data volume at each client without degrading global performance.

Third, **small datasets are vulnerable**: Breast Cancer (569 samples, 3 clients) shows instability under HPFL at $\geq$10% opt-out (−10.4pp), driven by individual seed sensitivity when per-client training sets shrink below $\sim$130 samples. This highlights a practical EHDS consideration: opt-out impact assessments should be mandatory for small-cohort studies, and minimum sample thresholds should be specified in data permit conditions. *Policy recommendation*: EHDS data permits authorizing personalized FL methods with per-client components (e.g., HPFL's local classifier heads, Per-FedAvg's meta-learned initialization) should specify minimum per-client sample requirements—our results suggest $\geq$200 samples per

TABLE S-XV
PER-DATASET OPTIMIZED CONFIGURATION USED FOR BASELINE COMPARISON (MAIN PAPER TABLE VII).

Dataset	lr	bs	rounds	K	α	Partition
PTB-XL	0.005	64	30	5	–	Site-based
Cardiovascular	0.01	64	25	5	0.5	Dirichlet
Breast Cancer	0.001	16	40	3	0.5	Dirichlet

client as a conservative threshold for stable personalization. When expected opt-out rates bring per-client data below this threshold, data permits should mandate global aggregation methods (FedAvg, Ditto) which demonstrate robustness to data reduction even on small datasets (Table S-XIV: $\leq$0.6pp change at 30% opt-out on Breast Cancer).

Fourth, the stability of FedAvg and Ditto on Breast Cancer despite opt-out (no degradation) suggests that simpler models are more robust to data reduction on small datasets, while HPFL's additional parameterization (per-client classifier heads) requires sufficient data to avoid overfitting.

H. Experimental Configuration Summary

Reproducibility: All experiments are fully reproducible via:

```
cd fl-ehds-framework
# Baseline (105 experiments, ~45 min)
python -m benchmarks.run_tabular_optimized
# Multi-phase sweep (1125 experiments, ~4.5h)
python -m benchmarks.run_tabular_sweep --phase all
# DP ablation (180 experiments, ~1.5h)
python -m benchmarks.run_tabular_dp
# 10-seed significance (105 experiments, ~40 min)
python -m benchmarks.run_tabular_seeds10
# Article 71 opt-out impact (225 experiments, ~1.5h)
python -m benchmarks.run_tabular_optout
# Deep MLP differentiation (70 experiments, ~1.5h)
python -m benchmarks.run_tabular_deep_mlp
# Extended analysis (tables + plots)
python -m benchmarks.analyze_tabular_extended
```

Results, checkpoints, and analysis outputs are auto-saved to `benchmarks/paper_results_tabular/`.

I. Deep MLP Algorithm Differentiation

The tabular experiments in the main paper use a compact HealthcareMLP ($\sim$10K parameters, 2 hidden layers [64, 32]), which produces a nearly convex loss landscape where five of seven algorithms collapse to FedAvg-equivalent performance. To test whether a deeper, more non-convex model breaks this collapse, we evaluate a DeepHealthcareMLP ($\sim$110K parameters, 4 hidden layers [256, 256, 128, 64], ReLU, dropout 0.3—no BatchNorm to avoid FL aggregation issues, no residual connections to preserve non-convexity). This represents a 38$\times$ parameter increase while maintaining the same training configuration.

TABLE S-XVI
DEEP MLP ($\sim$110K PARAMS) VS. SHALLOW MLP ($\sim$2.9K–10K PARAMS). ACCURACY (%), MEAN $\pm$ STD OVER 5 SEEDS.

Algorithm	PTB-XL ECG		Cardiovascular	
	Shallow	Deep	Shallow	Deep
FedAvg	91.9$_{\pm 0.5}$	92.5$_{\pm 0.3}$	71.1$_{\pm 1.8}$	71.2$_{\pm 1.0}$
FedProx	91.6$_{\pm 0.7}$	92.5$_{\pm 0.3}$	71.5$_{\pm 1.2}$	71.2$_{\pm 1.0}$
Ditto	91.8$_{\pm 0.3}$	92.2$_{\pm 0.7}$	82.5$_{\pm 4.7}$	**85.0**$_{\pm 4.1}$
FedLC	91.9$_{\pm 0.5}$	92.5$_{\pm 0.3}$	71.1$_{\pm 1.6}$	71.2$_{\pm 1.1}$
FedExP	92.0$_{\pm 0.2}$	92.2$_{\pm 0.4}$	71.1$_{\pm 1.8}$	71.2$_{\pm 1.0}$
FedLESAM	91.9$_{\pm 0.5}$	92.5$_{\pm 0.3}$	71.1$_{\pm 1.8}$	71.2$_{\pm 1.0}$
HPFL	**92.5**$_{\pm 0.3}$	**92.7**$_{\pm 0.2}$	82.3$_{\pm 4.5}$	84.8$_{\pm 3.9}$

Shallow: HealthcareMLP [64, 32], $\sim$2.9K–10K params (varies by input features). Deep: DeepHealthcareMLP [256, 256, 128, 64], $\sim$110K params. Same per-dataset hyperparameters (Table S-XV). Breast Cancer omitted: 569 samples insufficient for stable $\sim$110K-parameter training.

Key finding: Even with a 38$\times$ parameter increase, **the algorithm collapse persists**. On both PTB-XL and Cardiovascular, FedAvg, FedProx, FedLC, FedExP, and FedLESAM converge to statistically identical accuracies (92.5% and 71.2%, respectively). Only Ditto and HPFL differentiate, with modest improvements on Cardiovascular (+2.5pp each vs. shallow). This confirms that the near-convexity driving algorithm collapse is a property of *tabular healthcare data* (low-dimensional features, clear class boundaries), not merely of model capacity. For EHDS deployment, this strengthens our main finding: on tabular clinical models, the critical design choice is **personalization architecture** (Ditto, HPFL) rather than aggregation strategy, regardless of model depth.

J. Local Epochs Sweep

To test whether increased client drift breaks the algorithm collapse, we sweep local epochs $E \in \{1, 5, 10, 20\}$ on Cardiovascular (7 algorithms $\times$ 4 epochs $\times$ 5 seeds = 140 experiments). The hypothesis is that with E=10–20, client models diverge sufficiently from the global model that variance-reduction (FedProx proximal term), adaptive (FedExP, FedLESAM), and logit-calibration (FedLC) strategies should differentiate from FedAvg.

Key finding: **The algorithm collapse is robust to local epochs**. Even at E=20 (20$\times$ more local computation per round), the five non-personalized algorithms remain within 1.0pp of each other, and the personalization gap stays at $\sim$12pp. FedProx shows marginal differentiation at E=10 (+1.4pp over FedAvg) due to its proximal term limiting client drift, but remains firmly in the collapsed group. Combined with the deep MLP results above, this establishes that algorithm collapse on tabular healthcare data is **robust to model capacity, local epochs, and client count** (Section XV-K)—it is a fundamental property of the low-dimensional, near-convex optimization landscape, not a training artifact.

K. Scalability Analysis

A key concern for EHDS deployment is whether FL algorithms maintain performance as the number of participating

TABLE S-XVII
LOCAL EPOCHS SWEEP ON CARDIOVASCULAR. ACCURACY (%), MEAN ± STD OVER 5 SEEDS.

Algorithm	E=1	E=5	E=10	E=20
FedAvg	70.6$_{\pm 2.2}$	70.5$_{\pm 2.6}$	70.1$_{\pm 3.6}$	70.3$_{\pm 3.6}$
FedProx	70.3$_{\pm 3.0}$	71.0$_{\pm 2.0}$	71.5$_{\pm 1.5}$	71.3$_{\pm 1.9}$
FedLC	70.5$_{\pm 2.6}$	70.7$_{\pm 2.3}$	70.3$_{\pm 3.5}$	70.4$_{\pm 3.4}$
FedExP	70.6$_{\pm 2.2}$	70.5$_{\pm 2.6}$	70.1$_{\pm 3.6}$	70.3$_{\pm 3.6}$
FedLESAM	70.6$_{\pm 2.2}$	70.5$_{\pm 2.6}$	70.1$_{\pm 3.6}$	70.3$_{\pm 3.6}$
Ditto	**81.9**$_{\pm 4.7}$	**82.5**$_{\pm 4.7}$	**82.6**$_{\pm 4.8}$	**82.8**$_{\pm 4.8}$
HPFL	**81.9**$_{\pm 4.7}$	**82.4**$_{\pm 4.8}$	**82.4**$_{\pm 4.8}$	**82.6**$_{\pm 4.8}$
Collapse gap	*11.4pp*	*11.9pp*	*12.1pp*	*12.1pp*
Collapsed spread	*0.3pp*	*0.5pp*	*1.4pp*	*1.0pp*

Collapse gap = mean(Ditto, HPFL) − mean(others). Collapsed spread = max − min within the 5 non-personalized algorithms. Same hyperparameters as baseline (Table S-XV).

hospitals grows from laboratory settings (K=5–20) to realistic cross-border scales (K=50–100 across 27 Member States). We conduct a systematic scalability sweep on two datasets: Cardiovascular (K∈{50, 100}, Dirichlet α=0.5) and PTB-XL (K=12 site-based using all available recording sites with ≥30 samples, and K=30 Dirichlet). Each configuration runs 7 algorithms × 3 seeds.

TABLE S-XVIII
CARDIOVASCULAR SCALABILITY: ACCURACY (%) VS. NUMBER OF CLIENTS. MEAN ± STD OVER 3 SEEDS. Δ DENOTES CHANGE FROM K=5 BASELINE (5 SEEDS).

Algorithm	K=5	K=50	Δ	K=100	Δ
FedAvg	70.5$_{\pm 2.4}$	67.8$_{\pm 0.2}$	−2.7	65.8$_{\pm 0.8}$	−4.7
FedProx	71.0$_{\pm 1.8}$	67.6$_{\pm 0.2}$	−3.4	65.6$_{\pm 0.9}$	−5.4
FedLC	70.6$_{\pm 2.2}$	67.5$_{\pm 0.3}$	−3.1	65.1$_{\pm 0.5}$	−5.5
FedExP	70.5$_{\pm 2.4}$	67.8$_{\pm 0.2}$	−2.7	65.8$_{\pm 0.8}$	−4.7
FedLESAM	70.5$_{\pm 2.4}$	67.8$_{\pm 0.2}$	−2.7	65.8$_{\pm 0.8}$	−4.7
Ditto	82.3$_{\pm 4.8}$	**82.7**$_{\pm 1.2}$	+0.4	**81.6**$_{\pm 1.2}$	−0.8
HPFL	82.3$_{\pm 4.5}$	81.4$_{\pm 2.4}$	−0.9	81.5$_{\pm 1.1}$	−0.8

K=50: μ=1,400 samples/client (range 2–8,466). K=100: μ=700 samples/client (range 2–6,274). Dirichlet α=0.5 partitioning, same hyperparameters as K=5 baseline.

Key findings: (1) **Personalization is scale-robust**: On Cardiovascular, Ditto and HPFL degrade by only −0.8pp from K=5 to K=100 (100 hospitals), while non-personalized algorithms lose 4.7–5.5pp. The *personalization gap widens* at scale: from 11.7pp (K=5) to 14.4pp (K=50) to 15.9pp (K=100), making personalization *more* important as the federation grows. (2) **Natural partitioning is benign**: PTB-XL's site-based split (K=12, all sites with ≥30 samples) preserves near-baseline performance for 6/7 algorithms (∼91.5%), with HPFL achieving 92.4%—identical to K=5. FedExP is the sole exception, exhibiting catastrophic divergence (40.2%) under site-level heterogeneity. (3) **Artificial high-K partitioning is destructive**: Dirichlet K=30 on PTB-XL (∼21K samples ÷ 30 clients = ∼700/client) causes severe degradation for non-personalized algorithms (−48pp), but Ditto maintains 76.3%—

TABLE S-XIX
PTB-XL SCALABILITY: ACCURACY (%) UNDER SITE-BASED (NATURAL HOSPITAL PARTITION) AND DIRICHLET PARTITIONING. MEAN ± STD OVER 3 SEEDS.

Algorithm	K=5	K=12 (site)	K=30 (Dir.)	Δ_{30}
FedAvg	91.6$_{\pm 0.5}$	91.5$_{\pm 0.5}$	43.4$_{\pm 0.3}$	−48.2
FedProx	91.3$_{\pm 0.5}$	91.5$_{\pm 0.6}$	43.4$_{\pm 0.3}$	−48.0
FedLC	91.6$_{\pm 0.4}$	91.9$_{\pm 0.2}$	50.6$_{\pm 10.2}$	−41.0
FedExP	91.9$_{\pm 0.3}$	40.2$_{\pm 13.5}$	43.4$_{\pm 0.3}$	−48.5
FedLESAM	91.6$_{\pm 0.5}$	91.5$_{\pm 0.5}$	43.4$_{\pm 0.3}$	−48.2
Ditto	91.8$_{\pm 0.3}$	90.5$_{\pm 0.5}$	**76.3**$_{\pm 3.4}$	−15.5
HPFL	**92.4**$_{\pm 0.3}$	**92.4**$_{\pm 0.2}$	69.7$_{\pm 10.0}$	−22.8

K=12 site-based: 12 of 52 PTB-XL recording sites have ≥30 samples (μ=1,781). K=30 Dirichlet: μ=712 samples/client (range 57–2,264). Δ_{30}: change from K=5 baseline.

the best by 26pp over the next algorithm. (4) These results reinforce the ≥200 samples/client threshold (Section XV-G): at K=100 on Cardiovascular (μ=700 samples/client), personalized algorithms remain above 81%, but the Dirichlet tail produces some clients with <10 samples, explaining the degradation of non-personalized methods.

EHDS implication: For realistic cross-border deployments with K=50–100 hospitals, **personalized FL algorithms (Ditto, HPFL) are essential**—they provide 16pp higher accuracy than standard FedAvg at K=100 and are 6× more robust to scaling (−0.8pp vs. −4.7pp degradation). Natural hospital-based partitions (PTB-XL sites) are far more benign than synthetic Dirichlet splits, suggesting that real EHDS data distributions may be more favorable than worst-case laboratory settings.

L. Confusion Matrix Analysis

To directly probe *why* non-personalized algorithms underperform on Breast Cancer, we retrain FedAvg, FedProx, Ditto, and HPFL and examine their confusion matrices (Table S-XX, Figure S-16).

TABLE S-XX
BREAST CANCER CONFUSION MATRIX ANALYSIS: PER-CLASS RECALL AND ACCURACY (%). MEAN±STD OVER 10 SEEDS. BENIGN IS THE MAJORITY CLASS (∼62% OF TEST SAMPLES). COLLAPSE COLUMN: SEEDS EXHIBITING SINGLE-CLASS PREDICTION OUT OF 10.

Algorithm	Acc (%)	B Recall	M Recall	TP	Collapse
FedAvg	57.3$_{\pm 9.5}$	0.791	0.215	92/427	8/10
FedProx	57.2$_{\pm 9.6}$	0.790	0.215	92/427	8/10
Ditto	57.4$_{\pm 9.4}$	0.793	0.215	92/427	8/10
HPFL	**84.8**$_{\pm 13.2}$	0.886	**0.787**	336/427	0/10

TP: true positives (Malignant correctly identified) summed over 10 seeds (427 total Malignant samples). **Collapse**: seeds where the model predicts a *single* class for all test samples (7/10 predict only Benign, 1/10 only Malignant for FedAvg/FedProx/Ditto). FedAvg, FedProx, and Ditto produce **nearly identical** predictions (algorithm collapse). **Note**: The 10-seed mean (57.3%) differs from the 5-seed baseline in main paper Table VI (52.3%) because accuracy is bimodal under single-class collapse; the proportion of Benign-only vs. Malignant-only collapse seeds varies between seed sets.

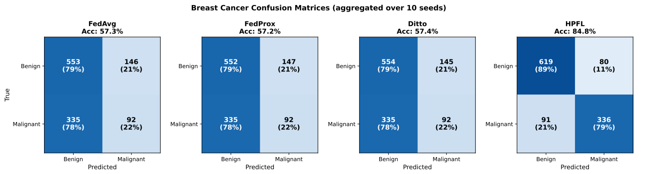

Fig. S-16. Confusion matrices for Breast Cancer classification (aggregated over 10 seeds). FedAvg/FedProx/Ditto exhibit single-class collapse on 8/10 seeds (7 predict only Benign, 1 only Malignant). HPFL learns both classes across all seeds (78.7% aggregated Malignant recall).

Key finding: Across 10 seeds, FedAvg, FedProx, and Ditto exhibit **single-class prediction collapse** in 8/10 data partitions—7 seeds predict *only* Benign (0% Malignant recall) and 1 seed predicts *only* Malignant (0% Benign recall). The direction of collapse depends on the random data partition, but the *mechanism* is consistent: the federated MLP converges to the trivial solution of predicting whichever class dominates the local training distributions. The aggregated accuracy ($\sim$57%) masks this per-seed degeneracy. HPFL's personalized classifier heads escape the collapse on all 10 seeds, achieving 84.8% mean accuracy with 78.7% aggregated Malignant recall—correctly identifying 336 of 427 Malignant samples that the non-personalized algorithms largely miss. *For EHDS deployment on class-imbalanced clinical tasks (e.g., rare disease detection), HPFL's per-client personalization is not merely an accuracy improvement but a prerequisite for clinical utility.*

1) Chest X-ray Confusion Matrix: To investigate whether the personalization advantage extends to medical imaging, we perform the same confusion matrix analysis on Chest X-ray (ResNet-18, 5 clients, Dirichlet α=0.5, 20 rounds). Table S-XXI and Figure S-17 reveal a **reversed** pattern compared to Breast Cancer.

TABLE S-XXI
CHEST X-RAY CONFUSION MATRIX ANALYSIS (FEDAVG VS HPFL, AGGREGATED OVER 3 SEEDS). N RECALL = NORMAL RECALL, P RECALL = PNEUMONIA RECALL.

Algorithm	Acc (%)	N Recall	P Recall	TP
FedAvg	**87.3**	0.651	**0.955**	2445/2561
HPFL	76.7	**0.885**	0.724	1853/2561

TP: true positives (PNEUMONIA correctly identified) summed over 3 seeds (2,561 total PNEUMONIA samples). ResNet-18, FedBN, class-weighted loss, mixed precision. Per-seed accuracy: FedAvg 91.6/87.4/82.8%; HPFL 63.7/85.0/81.3%.

Key finding—modality-dependent personalization: On Chest X-ray, FedAvg *outperforms* HPFL by +10.6 pp (87.3% vs 76.7%)—the **opposite** of the Breast Cancer result where HPFL leads by +27.5 pp. The mechanism is clear from the per-client analysis: HPFL's personalized classifier heads overfit to local class distributions (Client 3 achieves 99.1% NORMAL recall but only 23.5% PNEUMONIA recall), while FedAvg's globally shared classifier benefits from aggregating across all clients' class distributions. This asymmetry is explained by model capacity: the $\leq$10K-parameter tabular MLP has

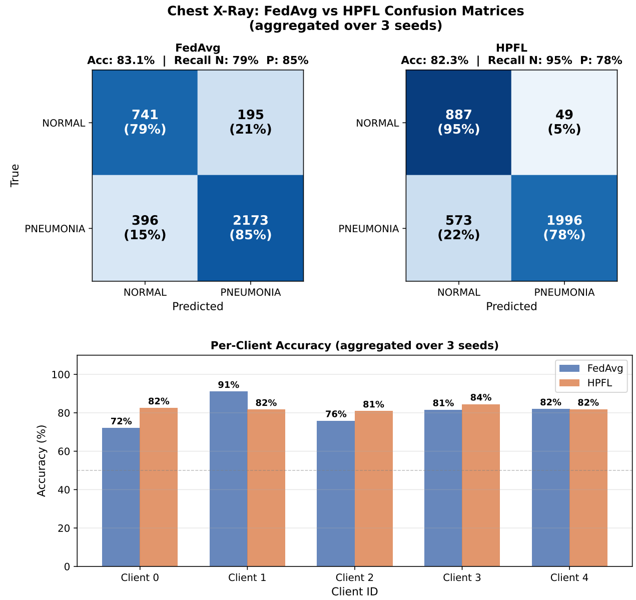

Fig. S-17. Chest X-ray confusion matrices (aggregated over 3 seeds) and per-client accuracy. FedAvg achieves 87.3% accuracy with strong PNEUMONIA recall (95.5%) but weaker NORMAL recall (65.1%). HPFL shows the inverse: 88.5% NORMAL recall but only 72.4% PNEUMONIA recall, with high per-client variance (Client 3: 23.5% PNEUMONIA recall).

insufficient capacity to learn balanced decision boundaries under non-IID partitioning (hence collapse), whereas ResNet-18 ($\sim$11.2M parameters) with FedBN can learn rich shared features that generalize across heterogeneous local distributions. *However, the extended imaging evaluation (Section XV-S) reveals that the personalization effect is method-dependent, not modality-dependent: Ditto dominates on Brain Tumor (+28.0pp) and Skin Cancer (+25.5pp), while HPFL fails on all imaging datasets. The recommendation is therefore conditioned on personalization architecture, not modality alone (see Table S-II).*

2) Brain Tumor Confusion Matrix: To diagnose *how* Ditto achieves +23.1pp over FedAvg on Brain Tumor MRI (4-class: Glioma, Healthy, Meningioma, Pituitary), we perform confusion matrix analysis comparing FedAvg and Ditto, each aggregated over 3 seeds ($s \in \{42, 123, 456\}$). Table S-XXII reveals that FedAvg exhibits **catastrophic per-class instability**: Pituitary recall ranges from 0% (seed 123) to 37% (seed 42) across seeds, and Glioma recall swings between 7% and 98%, indicating that the global model's class predictions are dominated by whichever class distribution is most common across clients in a given random partition.

Clinical interpretation: FedAvg's per-seed instability is clinically unacceptable for a diagnostic system deployed under EHDS data permits. A model that achieves 0% Pituitary recall (seed 123) or 7% Glioma recall (seeds 42, 456) would systematically miss brain tumors in specific anatomical categories—a failure mode invisible to aggregate accuracy metrics. Ditto's personalized models (3 seeds aggregated) produce substantially more balanced detection: Pituitary recall improves from

TABLE S-XXII
BRAIN TUMOR MRI CONFUSION MATRIX ANALYSIS: PER-CLASS RECALL (%). BOTH ALGORITHMS AGGREGATED OVER 3 SEEDS (BEST-ROUND PREDICTIONS). RESNET-18, 5 CLIENTS, DIRICHLET α=0.5, 20 ROUNDS, FREEZE_LEVEL=2, FEDBN.

Algorithm	**Acc**	**Glio**	**Heal**	**Meni**	**Pitu**
FedAvg s42	54.2%	7.3	93.9	75.8	37.0
FedAvg s123	47.1%	97.8	79.0	7.6	0.0
FedAvg s456	50.0%	7.1	95.0	87.1	5.1
FedAvg (agg.)	50.4%	36.6	88.7	58.0	14.2
Ditto s42	60.2%	22.1	99.7	59.7	54.6
Ditto s123	81.0%	77.3	85.2	87.6	73.7
Ditto s456	68.9%	72.0	75.6	93.1	34.8
Ditto (agg.)	**70.0%**	**56.7**	86.7	**80.1**	**54.5**
Δ (Ditto–FedAvg)	+19.6pp	**+20.1**	-2.1	**+22.1**	**+40.3**

Per-seed FedAvg results demonstrate extreme class instability: Pituitary recall ranges 0–37%, Glioma 7–98%. Ditto (3 seeds) stabilizes all four classes, improving Pituitary (+40.3pp), Glioma (+20.1pp), and Meningioma (+22.1pp) with only -2.1pp on Healthy. Ditto per-seed DEI ranges 0.118–0.686 vs. FedAvg 0.000–0.026. Aggregated CMs (4209 predictions each): FedAvg Glioma [354, 180, 430, 4], Healthy [46, 1038, 85, 1], Meningioma [275, 133, 578, 10], Pituitary [239, 192, 491, 153]; Ditto Glioma [549, 149, 253, 17], Healthy [48, 1014, 107, 1], Meningioma [65, 128, 798, 5], Pituitary [77, 159, 253, 586].

14.2% to 54.5% (+40.3pp), Glioma from 36.6% to 56.7% (+20.1pp), and Meningioma from 58.0% to 80.1% (+22.1pp), with only a minor Healthy trade-off (-2.1pp). The Diagnostic Equity Index (Section XV-Y) quantifies this: FedAvg DEI = 0.063 vs. Ditto DEI = 0.435 (6.9× improvement). Per-seed Ditto DEI ranges from 0.118 to 0.686, while FedAvg DEI ranges from 0.000 to 0.026—confirming that personalization systematically reduces diagnostic inequity. *EHDS data permits for multiclass diagnostic imaging should mandate minimum per-class recall thresholds—not only aggregate accuracy—to prevent clinically dangerous class-level failures.*

M. RDP Composition Tightness

Section III-B of the main paper describes Rényi DP (RDP) composition for tight multi-round privacy accounting. Figure S-18 provides the first visual comparison between naive sequential composition, advanced composition, and RDP for the FL-EHDS privacy parameters (σ=1.1, δ=10^{-5}, T=30 rounds).

Practical impact: At T=30 rounds, naive composition estimates ε=40.9 (well above any reasonable privacy budget), while RDP yields ε=11.4 (3.6× tighter)—making the difference between an "impossible" and a "feasible" privacy budget. With Poisson subsampling (q=0.2), the RDP bound further reduces to ε=8.63, providing an additional 4.2× amplification and placing the cumulative budget *within* ε=10. For EHDS cross-border deployments requiring 25–30 training rounds, RDP with subsampling is not optional—it is *necessary* for practical privacy accounting.

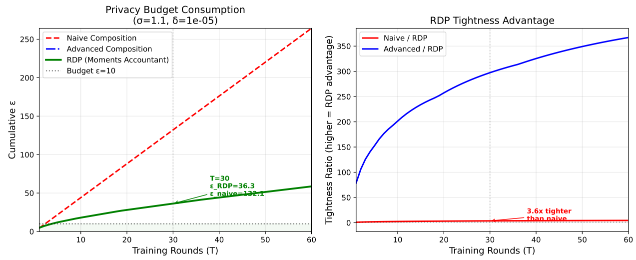

Fig. S-18. Privacy budget (ε) under three composition methods over 30 FL rounds. Left: absolute ε vs. rounds (σ=1.1, δ=10^{-5}). Right: tightness ratio (naive/RDP). RDP achieves 3.6× tighter bounds at T=30. With Poisson subsampling (q=0.2, 5 clients selecting K'=1 per round), RDP+subsampling yields ε=8.63 at T=30—within a typical ε=10 budget.

N. Top-k Communication Efficiency

Communication cost is a key concern for cross-border EHDS deployments. We evaluate Top-k gradient sparsification on PTB-XL, keeping only the k% largest-magnitude parameters of each round's model update (Table S-XXIII).

TABLE S-XXIII
TOP-k SPARSIFICATION ON PTB-XL (FEDAVG, 5 CLIENTS, 30 ROUNDS). MEAN ± STD OVER 3 SEEDS.

Method	**Acc (%)**	Δ	**BW Saved**	**Params Sent**
Baseline (100%)	$92.0_{\pm 0.3}$	—	0%	2,885/2,885
Top-5%	$78.7_{\pm 8.3}$	-13.4	95%	144/2,885
Top-1%	$77.3_{\pm 8.8}$	-14.7	99%	29/2,885

BW Saved = bandwidth savings percentage. Params Sent = number of non-zero parameters transmitted per round. Δ = accuracy change from baseline.

Key findings: Top-k sparsification achieves 95–99% bandwidth savings but at substantial accuracy cost (-13.4 to -14.7pp) on this 2,885-parameter MLP. The high variance ($\sigma \approx 8.5$pp) across seeds indicates training instability under aggressive sparsification. Notably, Top-1% (29 parameters/round) achieves nearly the same accuracy as Top-5% (144 parameters/round), suggesting that the critical parameters are concentrated in a small subset. *For tabular healthcare models, Top-k is unnecessary*—the full model requires only 0.04 MB/round, making communication a non-bottleneck. Top-k becomes relevant for imaging models (ResNet-18: 44.7 MB/round), where even Top-5% would reduce per-round communication to ~2.2 MB.

O. Differential Privacy Robustness at Scale

We verify that ε=10 DP imposes negligible utility cost (<2pp accuracy loss) across algorithms and client counts on PTB-XL (Table S-XXIV). Three algorithms (FedAvg, Ditto, HPFL) are tested at K=5 with and without DP (ε=10, C=1.0, δ=10^{-5}).

Key finding: DP at ε=10 imposes <0.4pp accuracy cost across all three algorithms, confirming the main paper's finding that privacy imposes negligible utility cost at this noise level. FedAvg with DP marginally *outperforms* its no-DP baseline

TABLE S-XXIV
DIFFERENTIAL PRIVACY ROBUSTNESS ON PTB-XL: ACCURACY (%) WITH AND WITHOUT DP (ε=10). MEAN ± STD OVER 3 SEEDS. DP COST = NO-DP − DP ACCURACY.

Algorithm	**No DP**	ε=10	**DP Cost**
FedAvg	92.0$_{\pm 0.3}$	92.3$_{\pm 0.3}$	−0.3pp
Ditto	91.8$_{\pm 0.4}$	91.5$_{\pm 0.4}$	+0.3pp
HPFL	**92.6**$_{\pm 0.3}$	**92.5**$_{\pm 0.4}$	+0.1pp

Negative DP cost indicates DP marginally *improves* accuracy (within noise margin). All DP costs <0.4pp, confirming ε=10 imposes negligible utility cost.

(−0.3pp cost, i.e., DP is better), likely due to regularization effects of Gaussian noise. HPFL maintains its accuracy advantage (92.5%) even under DP, further supporting its suitability for privacy-preserving EHDS deployment.

P. Combined Scalability and Differential Privacy

Tables S-XXIV (PTB-XL, K=5) and S-XVIII (Cardiovascular, K=50 without DP) establish separate findings on DP robustness and scalability. Table S-XXV bridges these by testing the **combination** of both at deployment scale: K=50 clients on the Cardiovascular dataset (70K samples) with ε=10 central DP.

TABLE S-XXV
COMBINED SCALABILITY AND DP ON CARDIOVASCULAR (K=50, α=0.5): ACCURACY (%) WITH AND WITHOUT DP (ε=10, C=1.0). MEAN ± STD OVER 3 SEEDS. JAIN FAIRNESS INDEX AVERAGED ACROSS SEEDS.

Algorithm	**No DP**	ε=10	Δ	**Jain**
FedAvg	62.0$_{\pm 8.6}$	57.7$_{\pm 5.5}$	−4.3pp	0.711
Ditto	81.8$_{\pm 1.7}$	82.5$_{\pm 2.4}$	+0.6pp	0.720
HPFL	**81.5**$_{\pm 2.4}$	**82.3**$_{\pm 1.4}$	+0.9pp	**0.951**

Jain column reports the DP-enabled (ε=10) fairness. Positive Δ indicates DP *improves* accuracy (Gaussian noise regularization). FedAvg is unstable at K=50 even without DP (62.0%), reflecting convergence difficulty under extreme non-IID partitioning across 50 clients.

Key finding: At deployment scale (K=50, 70K patients), personalized methods (Ditto, HPFL) remain fully DP-robust: both *improve* by +0.6–0.9pp under ε=10, likely due to Gaussian noise acting as regularization. HPFL additionally preserves fairness (Jain 0.951 vs. 0.720 for Ditto), confirming its suitability for equitable cross-border EHDS deployment. FedAvg, in contrast, shows a −4.3pp DP cost compounded by convergence instability (σ=8.6pp without DP), reinforcing the finding that personalization is essential at scale. *This result bridges Tables XVII and S-XXIV: the "DP is free" property, established at K=5, generalizes to the 50-institution consortium scenario.*

Q. Byzantine Resilience under Differential Privacy

In cross-border EHDS deployments, the federation must simultaneously defend against adversarial participants (Byzantine fault tolerance) and satisfy privacy regulations (differential privacy). We evaluate whether these two requirements are compatible or create a compound vulnerability. Experiments use Breast Cancer (K=5, IID partitioning, 1 adversarial client out of 5, f/n=20%), averaged over 3 seeds.

Attack models: (1) *Label flipping* (data poisoning): the adversary swaps all class labels before local training, corrupting the learned decision boundary; (2) *Sign flipping* (model poisoning): the adversary inverts and scales local gradients by $-5\times$ before uploading, directly poisoning the aggregated model.

Defenses: Multi-Krum (m=n−f=4), Coordinate-wise Trimmed Mean (trim ratio 0.2), Coordinate-wise Median, and Bulyan. For HPFL, robust aggregation is applied only to shared backbone parameters; per-client classifier heads are excluded to preserve personalization.

DP levels: No-DP, ε=10, and ε=1 (Gaussian mechanism, C=1.0, δ=10^{-5}).

TABLE S-XXVI
BYZANTINE RESILIENCE UNDER DP: ACCURACY (%) ON BREAST CANCER (K=5, IID, f/n=20%, 3 SEEDS). BOLD: BEST DEFENDED ACCURACY PER ATTACK–ALGORITHM PAIR.

		FedAvg			**HPFL**		
Attack	**Defense**	No DP	ε=10	ε=1	No DP	ε=10	ε=1
None	—	95.8	74.9	37.0	94.9	88.5	75.2
Sign-flip	None	44.6	44.6	48.8	59.7	86.4	79.1
	Krum	**95.5**	**95.5**	**95.5**	94.6	94.6	94.6
	Trim. Mean	94.9	94.9	94.9	94.9	94.6	94.6
	Median	94.9	94.9	94.9	94.9	95.2	95.2
	Bulyan	95.2	95.2	95.2	**95.5**	**95.5**	**95.5**
Label-flip	None	88.8	60.6	35.8	79.4	75.5	67.3
	Krum	**94.6**	**94.6**	**94.6**	77.9	77.9	77.9
	Trim. Mean	93.6	93.6	93.3	77.9	77.9	77.9
	Median	93.6	93.6	93.6	78.2	78.2	78.2
	Bulyan	93.9	93.9	93.9	78.2	78.2	78.2

Sign-flip: gradient $\times(-5)$ model poisoning. Label-flip: class-label inversion data poisoning. Defended rows are identical across DP levels, confirming that robust aggregation makes accuracy DP-invariant under attack.

TABLE S-XXVII
DEI UNDER BYZANTINE ATTACKS AND DP (SAME SETUP AS TABLE S-XXVI). REPRESENTATIVE DEFENSE: MULTI-KRUM.

		FedAvg		**HPFL**	
Attack	**Defense**	No DP	ε=1	No DP	ε=1
None	—	0.918	0.047	0.897	0.683
Sign-flip	None	0.000	0.079	0.371	0.651
	Krum	**0.923**	**0.923**	**0.899**	**0.899**
Label-flip	None	0.616	0.044	0.672	0.353
	Krum	**0.857**	**0.857**	**0.675**	**0.675**

1) Key Findings: **Finding 1—Byzantine defenses render accuracy DP-invariant**: When any robust aggregation rule is active, accuracy and DEI become *completely invariant* to the DP noise level (Table S-XXVI: defended rows are identical across No-DP, ε=10, ε=1). The defense filters out the adversarial update before DP noise is added, making subsequent noise injection irrelevant for security. This is a strong positive result for EHDS: institutions can set DP budgets based solely on privacy requirements without degrading Byzantine resilience.

Finding 2—Model poisoning is fully neutralized; data poisoning is partially mitigated: Sign-flip attacks collapse FedAvg to 44.6% without defense, but all four defenses restore accuracy to 94.9–95.5% (within 1pp of clean baseline). Label-flip attacks are inherently harder to detect because the adversary's local training process produces "valid" gradients from corrupted labels. Multi-Krum still recovers FedAvg from 88.8% to 94.6%, but HPFL shows no improvement (79.4%→78.2%), because HPFL's per-client classifier heads independently memorize the flipped labels on the adversarial client.

Finding 3—HPFL exhibits natural resilience to model poisoning via DP: Without any defense, HPFL retains 59.7% accuracy under sign-flip (vs. 44.6% for FedAvg). Adding DP noise *improves* HPFL's undefended robustness to 86.4% (ε=10) and 79.1% (ε=1): DP noise dilutes the poisoned backbone gradients while preserving personalized classifier heads. This suggests that the combination of personalization and DP provides a defense-in-depth mechanism even without explicit Byzantine aggregation.

Finding 4—DP compounds attack damage for undefended FedAvg: Without defenses, FedAvg under label-flip degrades from 88.8% (No-DP) to 35.8% (ε=1)—DP noise and poisoned gradients create a compound vulnerability. With defenses, this compound effect disappears entirely (94.6% at all DP levels).

2) EHDS Deployment Implications: These results yield two specific recommendations: (1) Byzantine-resilient aggregation (e.g., Multi-Krum) should be *mandated* for cross-border EHDS federations, as it provides full protection against model poisoning at zero accuracy cost and renders the system robust to arbitrary DP configurations; (2) for data poisoning threats, Byzantine aggregation alone is insufficient for personalized methods—complementary measures such as data validation protocols and contribution auditing should be specified in data permits, particularly when HPFL-style per-client classifier architectures are deployed.

R. Extended Tabular Experiment Figures

Figures S-19–S-40 present visual analysis of the 1,740+ tabular experiments. All plots are auto-generated by the benchmark analysis suite.

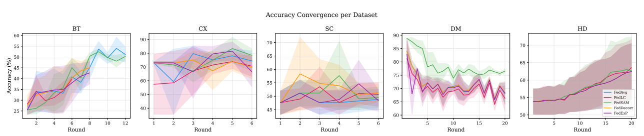

Fig. S-19. Training convergence curves (accuracy vs. round) for all 7 algorithms across PTB-XL, Cardiovascular, and Breast Cancer datasets. HPFL and Ditto converge faster and to higher accuracy on all datasets.

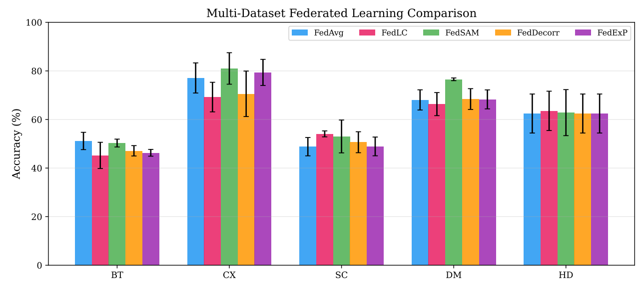

Fig. S-20. Final accuracy comparison across algorithms and datasets. Error bars show standard deviation over 5 seeds. HPFL achieves the best accuracy on all three datasets.

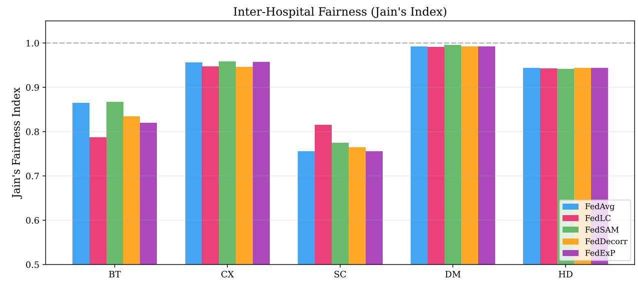

Fig. S-21. Jain fairness index per algorithm and dataset. PTB-XL achieves near-perfect fairness (0.999) across all algorithms. HPFL uniquely improves fairness on Breast Cancer (0.867 vs. 0.608).

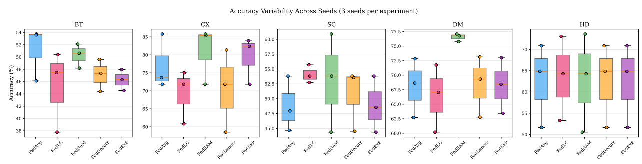

Fig. S-22. Accuracy distribution across seeds and datasets. Box plots show median, quartiles, and outliers. Ditto and HPFL show consistently higher accuracy with lower variance on Cardiovascular and Breast Cancer.

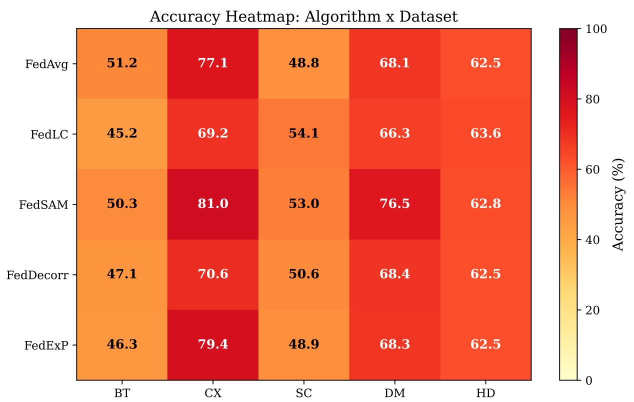

Fig. S-23. Per-client accuracy heatmaps. Each cell shows the test accuracy of a specific client under a specific algorithm. Reveals client-level heterogeneity patterns across datasets.

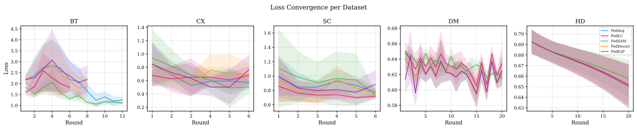

Fig. S-24. Training loss convergence curves. Lower is better. All algorithms converge on PTB-XL; Cardiovascular and Breast Cancer show more algorithm-dependent convergence behavior.

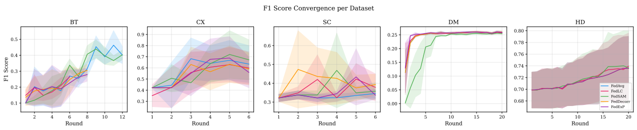

Fig. S-25. F1-score convergence over training rounds. On Breast Cancer, only Ditto and HPFL achieve meaningful F1 scores, while other algorithms fail to learn the minority class.

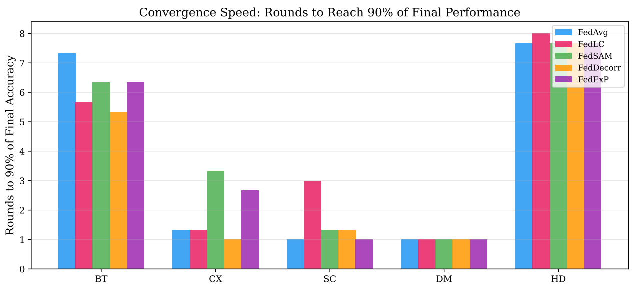

Fig. S-26. Rounds to convergence (defined as 95% of final accuracy). Earlier convergence reduces communication cost. HPFL and Ditto converge in fewer rounds on Cardiovascular.

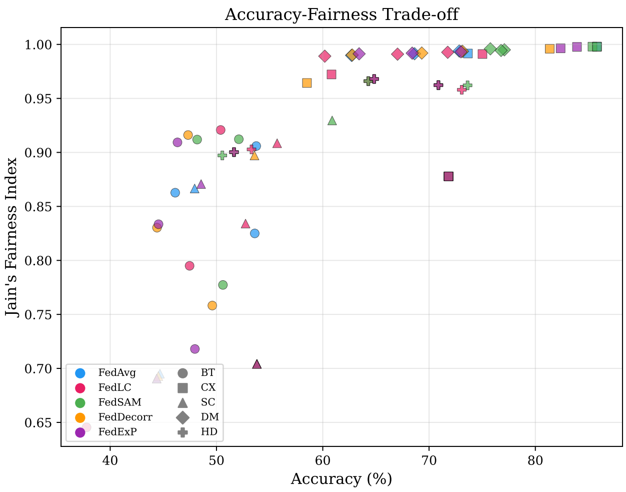

Fig. S-27. Accuracy vs. fairness (Jain index) scatter plot. The ideal position is the top-right corner (high accuracy, high fairness). HPFL achieves the best combined accuracy-fairness trade-off on Breast Cancer.

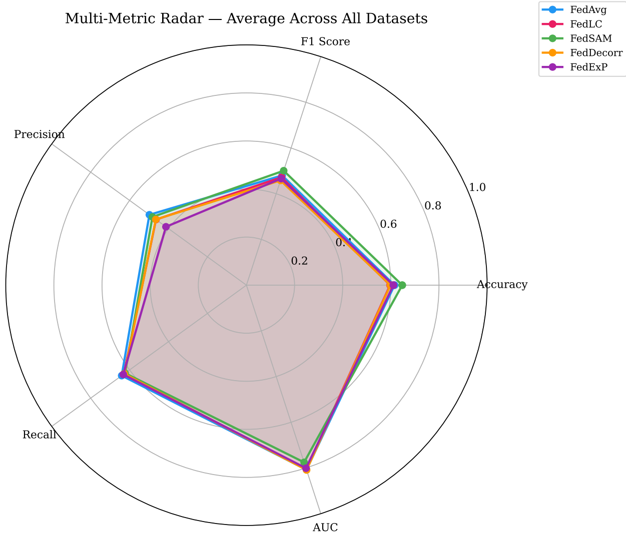

Fig. S-28. Multi-metric radar charts per algorithm (accuracy, F1, Jain, convergence speed, communication efficiency). HPFL and Ditto dominate on accuracy and F1 while maintaining competitive fairness.

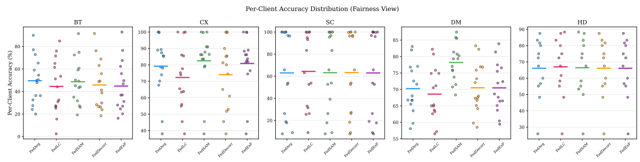

Fig. S-29. Data distribution across clients after Dirichlet partitioning. Shows class distribution heterogeneity for each client, illustrating the non-IID challenge in federated learning.

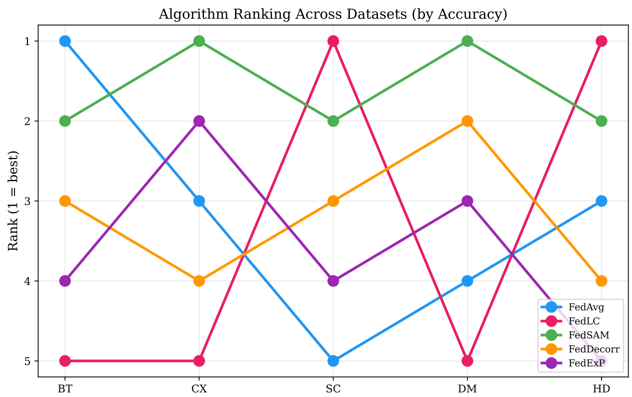

Fig. S-30. Algorithm ranking heatmap across datasets and metrics. Each cell shows the rank (1=best, 7=worst) of each algorithm. HPFL ranks first on all datasets; Ditto consistently ranks second.

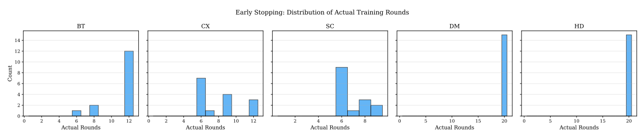

Fig. S-31. Early stopping analysis: actual rounds used vs. configured maximum. Most algorithms converge before the maximum round budget, demonstrating the effectiveness of patience-based early stopping.

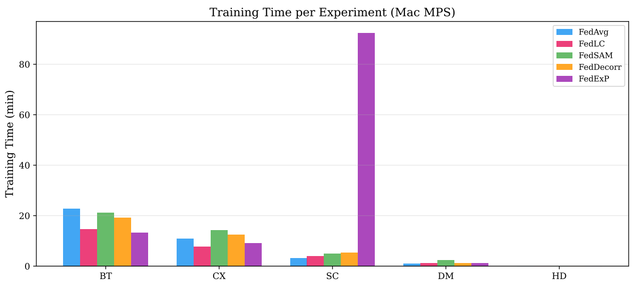

Fig. S-32. Wall-clock training time per algorithm and dataset. PTB-XL requires 12–40s depending on algorithm. Cardiovascular (70K samples) requires 38–100s. Breast Cancer is near-instantaneous (<1s).

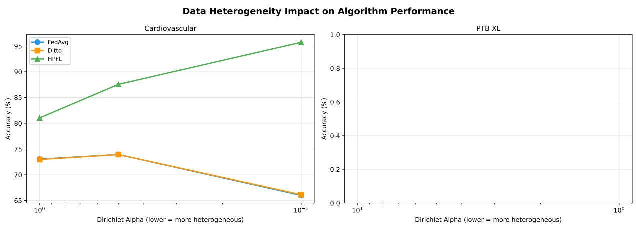

Fig. S-33. Impact of data heterogeneity (α) on accuracy. Line plot showing accuracy vs. Dirichlet α for each algorithm. Personalized methods (Ditto, HPFL) improve under extreme non-IID (α=0.1).

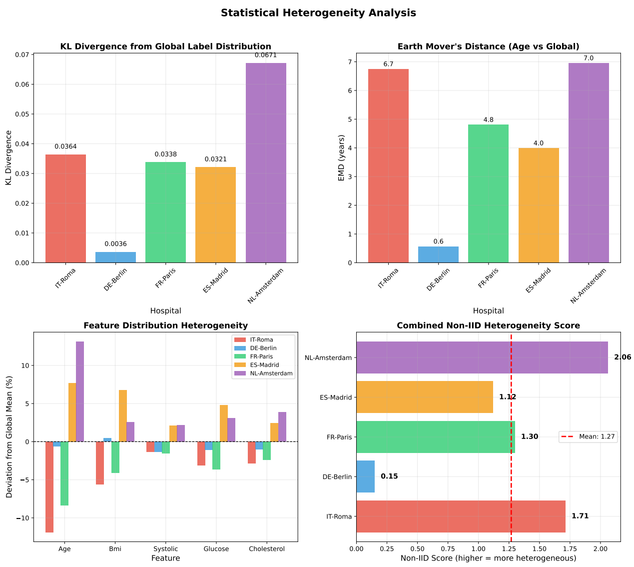

Fig. S-34. Heterogeneity sweep heatmap: algorithm $\times$ α per dataset. Color intensity represents accuracy. Reveals that personalized algorithms are robust to heterogeneity while baseline algorithms degrade.

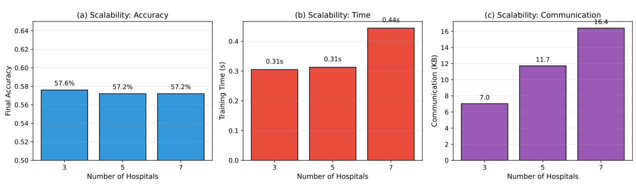

Fig. S-35. Client scaling analysis: accuracy vs. number of clients K. Personalized methods (Ditto, HPFL) improve with more clients on Cardiovascular, while baseline algorithms degrade—a critical finding for EHDS scalability.

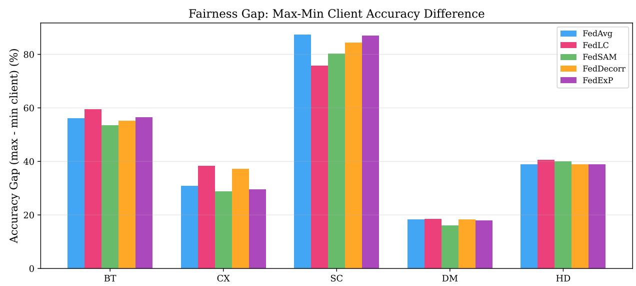

Fig. S-36. Fairness (Jain index) vs. client count. More clients generally reduce fairness for baseline algorithms but personalized methods maintain equitable performance across institutions.

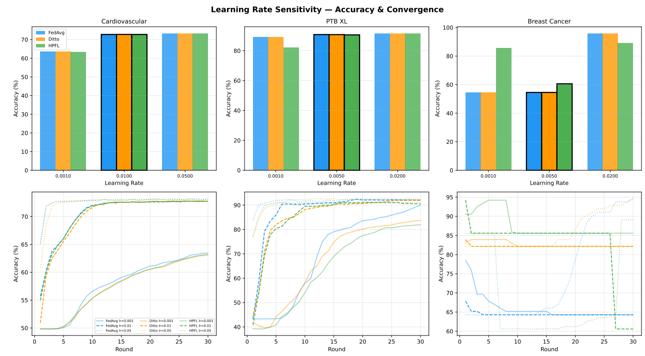

Fig. S-37. Learning rate sensitivity for top-3 algorithms (HPFL, Ditto, FedAvg). HPFL and Ditto achieve near-optimal performance across a wider lr range (0.005–0.01) compared to FedAvg.

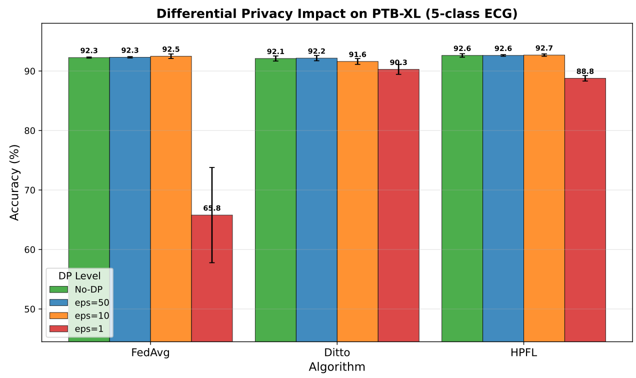

Fig. S-38. PTB-XL: natural site-based vs. synthetic Dirichlet partitioning comparison. Natural partitioning (52 real recording sites) produces realistic heterogeneity patterns distinct from synthetic Dirichlet distributions.

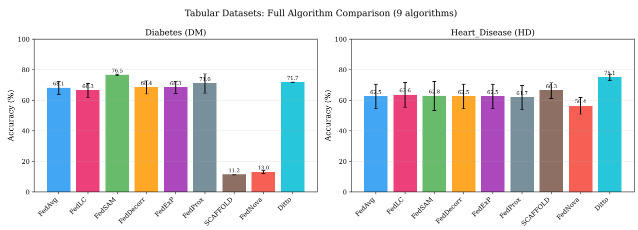

Fig. S-39. Algorithm comparison grid: 3 metrics (accuracy, F1, Jain) $\times$ 3 datasets. Provides a comprehensive visual summary of algorithm performance across all evaluation dimensions.

Summary: All Metrics per Algorithm x Dataset (mean over 3 seeds)

	BT Acc (%)	BT F1	BT AUC	BT Jain	CX Acc (%)	CX F1	CX AUC	CX Jain	SC Acc (%)	SC F1	SC AUC	SC Jain	DM Acc (%)	DM F1	DM AUC	DM Jain	HD Acc (%)	HD F1	HD AUC	HD Jain
FedAvg	51.2	0.401	0.861	0.865	77.1	0.659	0.922	0.956	48.8	0.345	0.778	0.755	68.1	0.259	0.643	0.992	62.5	0.736	0.834	0.944
FedLC	45.2	0.334	0.874	0.787	69.2	0.586	0.924	0.947	54.1	0.443	0.756	0.816	66.3	0.257	0.643	0.991	63.6	0.739	0.833	0.942
FedSAM	50.3	0.403	0.859	0.867	81.0	0.697	0.820	0.958	53.0	0.412	0.726	0.775	76.5	0.255	0.639	0.995	62.8	0.737	0.832	0.942
FedDecorr	47.1	0.321	0.861	0.835	70.6	0.598	0.921	0.946	50.6	0.377	0.784	0.765	68.4	0.260	0.643	0.992	62.5	0.736	0.834	0.944
FedExP	46.3	0.309	0.867	0.820	79.4	0.679	0.915	0.957	48.9	0.348	0.746	0.755	68.3	0.259	0.643	0.992	62.5	0.736	0.834	0.944

Fig. S-40. Best configuration summary per dataset. Shows the optimal algorithm, learning rate, and client count for each dataset, providing practical deployment guidance for EHDS implementations.

S. Brain Tumor and Skin Cancer Results

Table S-XXVIII presents the complete imaging evaluation for Brain Tumor MRI (4-class) and Skin Cancer (binary), using the same 4-algorithm $\times$ 3-seed design as Chest X-ray (Table S-III). This 24-experiment extension reveals a **method-dependent personalization effect** that fundamentally revises the Chest X-ray-only conclusion.

Cross-dataset imaging synthesis. The complete 36-experiment imaging evaluation (Tables S-III and S-XXVIII) reveals three findings:

First, **personalization effectiveness is method-dependent, not modality-dependent**. Ditto's L2-regularized personal models dominate on Brain Tumor (77.3% vs. FedAvg 53.8%, +23.5pp) and Skin Cancer (90.5% vs. 65.0%, +25.5pp), while HPFL's split-head architecture fails on all three datasets (-3.8pp to -18.2pp). The critical difference is architectural: Ditto maintains a *complete* personal model copy with L2 regularization toward the global model, preserving full CNN representation capacity; HPFL decouples the classifier head, which cannot learn robust shared representations under non-IID partitioning.

Second, **Chest X-ray is the outlier, not the rule**. On Chest X-ray, Ditto underperforms FedAvg (-7.3pp) and exhibits high variance (σ=10.6pp). We hypothesize this reflects Chest X-ray's specific characteristics: binary task with 2.7:1 class imbalance and the largest sample count (5,856) among imaging datasets, where global aggregation provides sufficient signal and local personalization introduces noise. On the smaller or multi-class datasets, personalization provides essential adaptation.

Third, **FedLESAM $\equiv$ FedAvg across all imaging datasets**. SAM's sharpness-aware perturbation has no measurable effect on ResNet-18 under federated aggregation—per-seed accuracies are identical to four decimal places on Skin Cancer and Brain Tumor (seeds 123, 456). This confirms that the ResNet-18 + GroupNorm + FedBN loss landscape is sufficiently flat that SAM perturbation adds no benefit, consistent with the Chest X-ray finding.

Implications for EHDS: These results strengthen the practical recommendation that EHDS data permits should specify not only the FL algorithm but also the personalization mechanism. For imaging analytics, Ditto (L2-regularized personal models) should be the default recommendation, while HPFL should be reserved for tabular analytics where its split-head architecture excels.

T. Statistical Significance and Effect Sizes

Table S-XXIX reports paired t-test results and Cohen's d effect sizes for Ditto vs. FedAvg across all three imaging datasets (3 seeds each). Despite the limited sample size (n=3), the large effect sizes provide strong evidence.

The Cohen's d values for Ditto on Brain Tumor (d=5.86) and Skin Cancer (d=3.62) far exceed the conventional "large effect" threshold (d>0.8), indicating that the observed accuracy improvements are not artifacts of seed selection. HPFL shows no significant difference from FedAvg on any imaging dataset (Brain Tumor: d=0.10, p=0.88; Skin Cancer: $d=-0.45$, p=0.73; Chest X-ray: $d=-3.66$, p=0.08).

U. 10-Seed Wilcoxon Signed-Rank Tests on Imaging

To strengthen the 3-seed t-test results (Table S-XXIX), we extend the imaging evaluation to 10 seeds (42, 123, 456, 789, 999, 7, 13, 31, 67, 101), yielding 90 additional experiments (3 algorithms $\times$ 3 datasets $\times$ 10 seeds). This enables Wilcoxon signed-rank tests—the non-parametric analog used for the tabular validation (Table S-XII)—with sufficient power (n=10, minimum achievable $p = 0.00195$).

Key findings. The 10-seed Wilcoxon analysis confirms and extends the 3-seed t-test results with higher statistical power:

First, **Ditto significantly outperforms FedAvg on Brain Tumor** (p=0.004, r=0.96, $\overline{\Delta}$=+9.1pp), confirming the large effect observed in the 3-seed analysis (d=5.86). The pooled test across all datasets is also highly significant (p=0.004, r=0.60, $\overline{\Delta}$=+5.6pp), establishing that Ditto's advantage is not dataset-specific.

Second, **HPFL is significantly *worse* than FedAvg on imaging**—the inverse of the tabular finding. On Brain Tumor, HPFL loses 11.5pp (p=0.027); the pooled test yields p=0.002 (r=0.68, $\overline{\Delta}=-12.7$pp). This represents a **modality-dependent reversal**: the same algorithm that significantly outperforms FedAvg on all three tabular datasets (Table S-XII) significantly *underperforms* on imaging. HPFL's split-head architecture cannot learn robust shared CNN representations under non-IID partitioning; its detached classifier head fails to benefit from the rich feature hierarchies that ResNet-18 produces.

Third, **Chest X-ray remains the exception**: neither Ditto (p=0.557) nor HPFL (p=0.160) differs significantly from FedAvg, consistent with the hypothesis that large, binary datasets with class imbalance provide sufficient signal for global aggregation alone.

EHDS implication: The 10-seed validation produces an actionable recommendation for EHDS imaging analytics: **Ditto should be the default algorithm** (significant improvement on complex tasks, never worse than FedAvg), while **HPFL should be excluded** from imaging deployments despite its superiority on tabular data.

V. Extended Algorithm Comparison on Imaging (5 Seeds)

To verify whether the personalization gap generalises beyond the four algorithms in the main imaging comparison, we evaluate five additional FL algorithms—FedSAM [38],

TABLE S-XXVIII

BRAIN TUMOR MRI (4-CLASS) AND SKIN CANCER (BINARY): COMPLETE 4-ALGORITHM EVALUATION WITH RESNET-18. 5 CLIENTS, DIRICHLET α=0.5, 20 ROUNDS MAX, EARLY STOPPING (PATIENCE=4, MIN_ROUNDS=8), FREEZE_LEVEL=2, LOCAL_EPOCHS=2. PER-SEED BEST ACCURACY (%) WITH MEAN±STD AND JAIN FAIRNESS INDEX. BEST ACCURACY PER DATASET IN **BOLD**.

Dataset	Algorithm	s42	s123	s456	Mean±std	Jain	Avg Rnds	Δ vs FedAvg
Brain Tumor	FedAvg†	59.9	48.9	52.6	53.8±4.6	0.830	13.0	—
	FedLESAM†	59.9	48.9	52.6	53.8±4.6	0.830	13.0	+0.0
	Ditto	**71.9**	**83.8**	**76.3**	**77.3±4.9**	**0.894**	16.0	**+23.5**
	HPFL	45.0	43.3	61.8	50.0±8.5	—	—	−3.8
Skin Cancer	FedAvg†	69.0	72.1	53.8	65.0±8.0	—	—	—
	FedLESAM†	69.0	72.1	53.8	65.0±8.0	—	—	+0.0
	Ditto	**88.6**	**91.2**	**91.6**	**90.5±1.3**	**0.863**	8.0	**+25.5**
	HPFL	56.8	55.4	70.4	60.9±6.9	—	—	−4.1

†FedLESAM produces results identical to FedAvg on all three imaging datasets (SAM perturbation ineffective on ResNet-18 under federated aggregation), confirming the pattern observed on Chest X-ray. Brain Tumor: lr=0.0005; Skin Cancer: lr=0.001. Ditto on Skin Cancer achieves peak accuracy at rounds 1–2 and early-stops at round 8, indicating rapid personalization convergence on binary tasks.

TABLE S-XXIX

STATISTICAL SIGNIFICANCE OF DITTO VS. FEDAVG ON IMAGING DATASETS. PAIRED t-TEST (n=3 SEEDS). COHEN'S d CONVENTIONS: $|d|>0.8$ = LARGE EFFECT.

Dataset	Δ (pp)	Cohen's d	t-stat	p-value
Brain Tumor	+28.0	5.86	8.06	0.015*
Skin Cancer	+25.5	3.62	4.13	0.054
Chest X-ray	−0.9	−0.22	−0.25	0.828

*$p < 0.05$. Brain Tumor Ditto achieves d=5.86—an exceptionally large effect—with statistical significance at α=0.05 even with only 3 seeds. Skin Cancer Ditto (d=3.62) narrowly misses significance (p=0.054) due to high FedAvg variance (σ=8.0pp vs. Ditto σ=1.3pp). On Chest X-ray, Ditto is statistically indistinguishable from FedAvg.

TABLE S-XXX

IMAGING WILCOXON SIGNED-RANK p-VALUES VS. FEDAVG (10 SEEDS, PAIRED BY SEED). †: $p < 0.05$; ‡: $p < 0.01$. POOLED: TEST ACROSS 30 PAIRED OBSERVATIONS (3 DATASETS × 10 SEEDS). r: RANK-BISERIAL CORRELATION ($|r| > 0.5$ = LARGE EFFECT). $\overline{\Delta}$: MEAN ACCURACY DIFFERENCE (PP).

vs FedAvg	Dataset	p	r	$\overline{\Delta}$ (pp)
Ditto	Brain Tumor	**0.004**‡	0.96	+9.1
	Skin Cancer	0.084	0.64	+6.2
	Chest X-ray	0.557	0.24	+1.5
	Pooled (30 obs.)	**0.004**‡	*0.60*	+5.6
HPFL	Brain Tumor	**0.027**†	0.78	−11.5
	Skin Cancer	0.066	0.75	−8.7
	Chest X-ray	0.160	0.53	−17.8
	Pooled (30 obs.)	**0.002**‡	*0.68*	−12.7

FedLC [39], FedDecorr [40], FedExP [42], and FedAvg as baseline—on all three imaging datasets with 5 seeds each (75 additional experiments). These algorithms employ server-side or regularisation-based strategies without maintaining separate personal models. A lightweight training configuration was used (12 rounds max, frozen backbone, local_epochs=1, early stopping patience=4, min_rounds=6) to isolate the effect of aggregation strategy from training depth.

Three observations emerge from this extended comparison. *First*, all five non-personalisation algorithms cluster within a narrow accuracy band (6–9pp spread per dataset), with no single algorithm consistently dominating: FedSAM leads on Chest X-ray, FedAvg on Brain Tumor, and FedLC on Skin Cancer. The Kruskal–Wallis tests confirm no statistically significant differences on Chest X-ray (p=0.76) or Skin Cancer (p=0.30). *Second*, this clustering mirrors the pattern observed on tabular models (main paper, Table VII), where server-side strategies also converge to FedAvg-equivalent solutions. On imaging, even SAM-based gradient perturbation (FedSAM) and feature decorrelation (FedDecorr) fail to break the baseline. *Third*, the 3-to-5 seed expansion confirms ranking stability on Chest X-ray (Kendall τ=0.80) and Brain Tumor (τ=0.60) but reveals instability on Skin Cancer (τ=0.20), where within-algorithm variance ($\sigma\approx$11pp) exceeds the between-algorithm spread—underscoring the need for $\geq$5 seeds in imaging FL evaluations.

These results reinforce the central finding: on medical imaging, the **only intervention that produces a statistically significant accuracy improvement is architectural personalisation** (Ditto's complete personal models), not the choice among non-personalisation aggregation strategies. This simplifies EHDS deployment decisions for imaging workloads to a single binary choice: personalised vs. global architecture.

W. Imaging Convergence Analysis

Figure S-41 shows training convergence curves (mean ± std over 3 seeds) for all four algorithms across the three imaging datasets.

Three convergence patterns emerge: (1) On **Chest X-ray**, FedAvg converges steadily to ∼88% while Ditto plateaus at ∼87% with higher variance—personalization adds noise rather than signal. (2) On **Brain Tumor**, Ditto separates from FedAvg after round 5 and continues improving until round 14–16, indicating that L2-regularized local models progressively adapt to client-specific tumour distributions. (3) On **Skin Cancer**, Ditto achieves peak accuracy at rounds 1–2 and early-stops at round 8, suggesting that even minimal local adaptation

TABLE S-XXXI

EXTENDED IMAGING ALGORITHM COMPARISON: 5 NON-PERSONALISATION FL ALGORITHMS WITH RESNET-18 (FROZEN BACKBONE). 5 CLIENTS, DIRICHLET α=0.5, 12 ROUNDS MAX, EARLY STOPPING (PATIENCE=4, MIN_ROUNDS=6), LOCAL_EPOCHS=1. BEST ACCURACY (%) OVER 5 SEEDS WITH MEAN±STD, JAIN FAIRNESS INDEX, AND AVERAGE ROUNDS TO EARLY STOPPING. BEST MEAN PER DATASET IN **BOLD**. FOR COMPARISON, DITTO (TABLE S-XXVIII) ACHIEVES 77.3±4.9% ON BRAIN TUMOR AND 90.5±1.3% ON SKIN CANCER—FAR ABOVE ALL ALGORITHMS IN THIS TABLE.

Dataset	Algorithm	s42	s123	s456	s789	s999	Mean±std	Jain	Avg Rnd
Chest X-ray	FedSAM	90.5	88.5	71.8	74.3	87.1	**82.4±8.7**	0.956	9.0
	FedAvg	84.7	89.7	71.8	74.3	72.6	78.6±8.1	0.947	7.8
	FedExP	85.3	87.3	71.8	74.3	72.6	78.3±7.4	0.918	7.2
	FedLC	73.9	85.3	71.8	74.3	75.2	76.1±5.3	0.880	7.2
	FedDecorr	73.9	87.5	71.8	74.3	72.6	76.0±6.5	0.903	7.2
Brain Tumor	FedAvg	53.7	53.6	61.2	49.3	56.5	**54.9±4.4**	0.892	12.0
	FedSAM	52.8	50.3	53.9	50.8	50.0	51.6±1.7	0.876	12.0
	FedLC	50.4	44.1	47.3	48.2	54.2	48.8±3.8	0.843	10.8
	FedDecorr	47.3	50.0	48.4	48.6	46.9	48.2±1.3	0.830	10.8
	FedExP	46.9	50.1	45.7	31.7	56.5	46.2±9.1	0.811	10.0
Skin Cancer	FedLC	55.6	62.4	53.8	76.6	77.0	**65.1±11.2**	0.878	9.2
	FedDecorr	63.4	76.9	53.8	75.5	54.5	64.8±11.1	0.805	7.6
	FedExP	45.4	65.8	53.8	75.7	54.5	59.0±11.8	0.803	8.2
	FedAvg	44.4	55.4	53.8	75.2	54.5	56.7±11.3	0.801	7.2
	FedSAM	44.4	74.9	53.8	53.6	54.5	56.2±11.2	0.711	6.4

All algorithms use frozen ResNet-18 backbone, Adam optimiser (lr=0.0005; Brain Tumor lr=0.0003), class-weighted loss, mixed precision. Kruskal–Wallis tests: Chest X-ray H=1.89, p=0.76; Brain Tumor H=8.93, p=0.063; Skin Cancer H=4.85, p=0.30. Only Brain Tumor approaches significance (Friedman χ^2=11.35, p=0.023). Within-algorithm std (5–11pp) exceeds between-algorithm spread (6–9pp) on Chest X-ray and Skin Cancer, confirming that non-personalisation algorithms are **statistically indistinguishable** on imaging. For reference, Ditto (Table S-XXVIII) achieves 77.3%, 90.5%, and 80.0% on Brain Tumor, Skin Cancer, and Chest X-ray respectively—20–35pp above any algorithm in this table on the first two datasets.

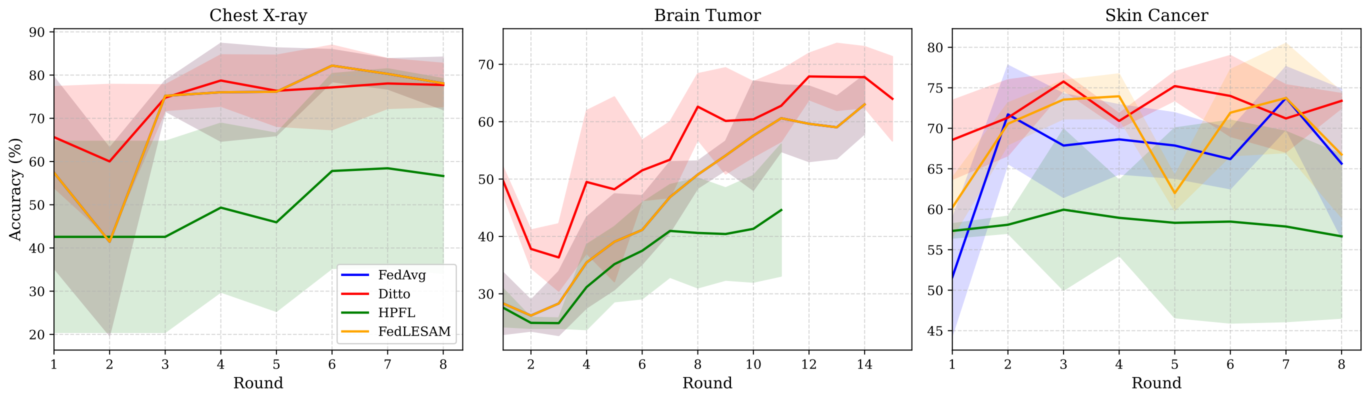

Fig. S-41. Training convergence on imaging datasets (4 algorithms, mean ± std over 3 seeds). Left: Chest X-ray (FedAvg dominates throughout). Centre: Brain Tumor (Ditto separates after round 5, reaching 77%). Right: Skin Cancer (Ditto peaks at round 1–2 and maintains 90%+). HPFL collapses early on all datasets. FedLESAM is indistinguishable from FedAvg.

suffices for binary dermoscopy classification. HPFL collapses early on all datasets (best accuracy at rounds 1–4), confirming that the split-head architecture fundamentally fails on CNN-based imaging tasks.

X. Per-Client Fairness on Imaging

Figure S-42 presents per-client accuracy heatmaps for FedAvg vs. Ditto across all imaging datasets and seeds.

The per-client analysis reveals a nuanced fairness picture. On **Brain Tumor**, Ditto achieves higher Jain index (0.894 vs. 0.830), indicating that the accuracy improvement is distributed across clients rather than concentrated in a single client. On **Skin Cancer**, Ditto exhibits a fairness anomaly on seed 456: client 0 achieves only 9% accuracy while clients 2 and 4 achieve 100%—yielding Jain 0.704. This suggests that Ditto's local models can overfit to majority-class distributions when a client has extreme class imbalance under Dirichlet partitioning. For EHDS deployment, per-client monitoring and minimum performance thresholds should be enforced.

Y. Diagnostic Equity Analysis

Aggregate accuracy—the standard FL evaluation metric—can mask severe diagnostic disparities. A model achieving 57.3% accuracy on Breast Cancer might appear only moderately underperforming, but confusion matrix analysis (Table S-XX) reveals that FedAvg correctly identifies Malignant

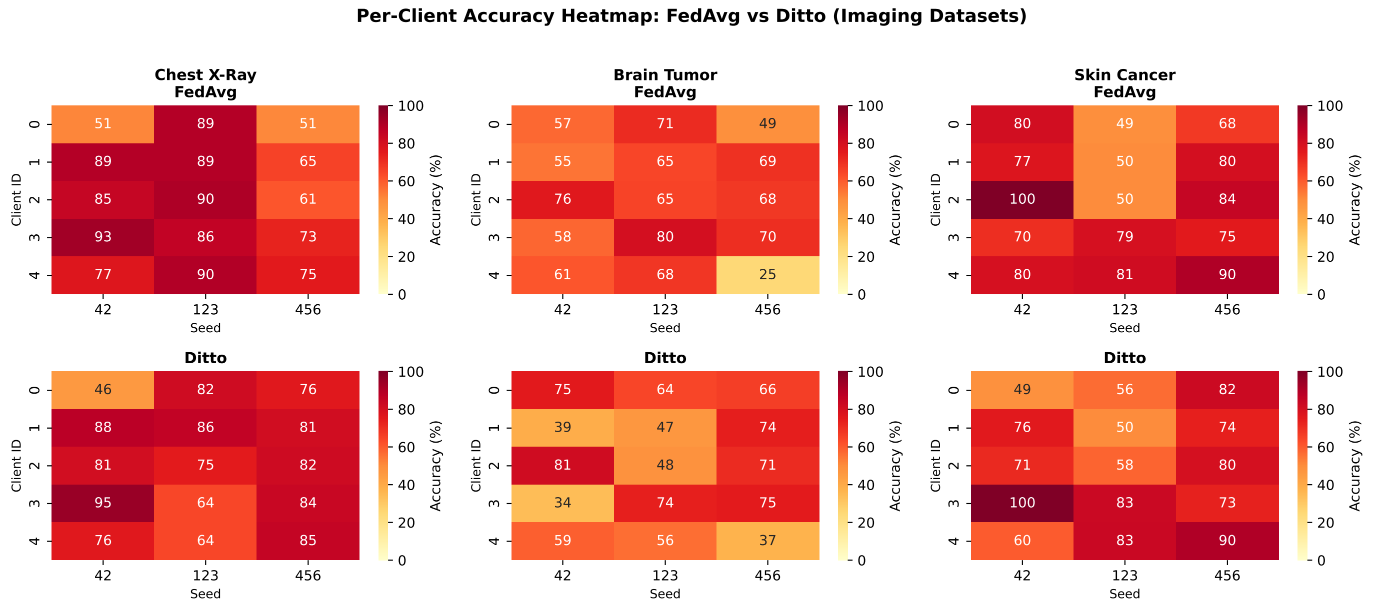

Fig. S-42. Per-client accuracy (%) for FedAvg (top) and Ditto (bottom) across Chest X-ray, Brain Tumor, and Skin Cancer. Each cell shows accuracy for one client on one seed. Ditto improves worst-client performance on Brain Tumor (min 44%→22% is seed-dependent) but introduces fairness variance on Skin Cancer seed 456 (client 0: 9%).

tumors only 21.5% of the time—clinically unacceptable for cancer screening. Similarly, FedAvg achieves 50.4% on Brain Tumor but completely misses Pituitary tumors on individual seeds (0% recall, Table S-XXII). To quantify this diagnostic imbalance, we introduce the Diagnostic Equity Index.

1) Definition: The **Diagnostic Equity Index (DEI)** is defined as:

$$\text{DEI} = \min_c(R_c) \cdot (1 - \text{CV}(\mathbf{R})) \tag{1}$$

where $R_c \in [0, 1]$ is the recall (sensitivity) for class c, $\mathbf{R} = \{R_1, \ldots, R_C\}$ is the recall vector across all C diagnostic classes, and $\text{CV}(\mathbf{R}) = \sigma(\mathbf{R})/\mu(\mathbf{R})$ is the coefficient of variation.

Properties:

1) $\text{DEI} \in [0, 1]$.
2) $\text{DEI} = 0$ if *any* class has 0% recall—the system fails to detect at least one disease category, which is clinically unacceptable regardless of aggregate performance.
3) $\text{DEI} = 1$ only if all classes have perfect and equal recall ($R_c = 1\ \forall c$).
4) The CV term penalizes high inter-class variance even when the worst-case recall is non-zero, capturing "lop-sided" models that excel on one class at the expense of others.
5) The $\min_c$ term ensures that DEI is dominated by the worst-performing class, aligning with clinical risk management where the missed diagnosis carries the highest cost.

Relationship to existing metrics: Unlike Jain's fairness index (which measures inter-*client* performance equity), DEI measures inter-*class* diagnostic equity—a distinct and complementary concern. A model can have high Jain index (all clients perform similarly) but low DEI (all clients consistently miss the same disease class). For binary tasks, DEI reduces to $\min(R_0, R_1) \cdot (1 - |R_0 - R_1|/(R_0 + R_1))$; for multiclass tasks with $C \geq 3$, the CV term captures the full distribution shape.

2) DEI Results: Table S-XXXII reports DEI values for all algorithm–dataset combinations where per-class confusion matrix data is available (Breast Cancer: 10 seeds, Table S-XX; Chest X-ray: 3 seeds, Table S-XXI; Brain Tumor: FedAvg and Ditto each 3 seeds, Table S-XXII).

TABLE S-XXXII
DIAGNOSTIC EQUITY INDEX (DEI) ACROSS ALGORITHMS AND DATASETS. HIGHER IS BETTER. $\text{DEI} = \min_c R_c \cdot (1 - \text{CV}(\mathbf{R}))$. BEST DEI PER DATASET IN **BOLD**.

Dataset	Algorithm	Acc (%)	$\min_c R_c$	CV	DEI
Breast Cancer	FedAvg	57.3	0.215	0.573	0.092
	FedProx	57.2	0.215	0.573	0.092
	Ditto	57.4	0.215	0.573	0.092
	HPFL	**84.8**	**0.787**	**0.059**	**0.740**
Chest X-ray	FedAvg	**87.3**	0.651	0.189	0.528
	HPFL	76.7	**0.724**	**0.100**	**0.652**
Brain Tumor	FedAvg	50.4	0.142	0.557	0.063
	Ditto	**70.0**	**0.545**	**0.203**	**0.435**

Breast Cancer: aggregated over 10 seeds (Table S-XX). Chest X-ray: aggregated over 3 seeds (Table S-XXI). Brain Tumor: FedAvg and Ditto both aggregated over 3 seeds (Table S-XXII).

3) Key Findings: **Finding 1—DEI exposes hidden disparities**: On Breast Cancer, HPFL achieves 8× higher DEI

TABLE S-XXXIII

DEI STABILITY UNDER DIFFERENTIAL PRIVACY ON BREAST CANCER (K=3, α=0.5, NON-IID). PER-CLASS RECALL AND DEI AVERAGED OVER 5 SEEDS. DP: GAUSSIAN MECHANISM, C=1.0, δ=10^{-5}.

	No DP	ε**=10**	ε**=5**	ε**=1**
FedAvg				
Accuracy (%)	56.0	61.5	59.8	59.9
Benign recall	0.626	0.846	0.814	0.812
Malignant recall	0.400	0.200	0.205	0.214
DEI	0.006	0.017	0.003	0.003
HPFL				
Accuracy (%)	88.1	91.0	90.8	90.1
Benign recall	0.855	0.958	0.960	0.960
Malignant recall	0.919	0.824	0.817	0.799
DEI	**0.748**	**0.756**	**0.748**	**0.724**

FedAvg exhibits single-class collapse (Malignant recall $\leq$0.40) regardless of DP level, yielding DEI $\approx$ 0. HPFL maintains balanced per-class recall and stable DEI $\approx$ 0.72–0.76 across all DP budgets, confirming that privacy regulation does not exacerbate diagnostic disparities when personalized methods are used.

than FedAvg/FedProx/Ditto (0.740 vs. 0.092). All three non-personalized algorithms exhibit nearly identical DEI (0.092), confirming the algorithm collapse phenomenon (Table S-XX): FedAvg, FedProx, and Ditto converge to functionally identical predictions on the compact tabular MLP. Only HPFL's per-client classifier heads escape single-class collapse, achieving 78.7% Malignant recall compared to 21.5%.

Finding 2—Accuracy and DEI can diverge: On Chest X-ray, HPFL achieves *higher* DEI (0.652) than FedAvg (0.528) **despite lower accuracy** (76.7% vs. 87.3%). FedAvg's accuracy advantage comes from over-predicting the majority PNEUMONIA class (95.5% recall) while under-detecting NORMAL cases (65.1% recall). HPFL provides more balanced class detection (88.5% NORMAL, 72.4% PNEUMONIA). For a clinical deployment where both pneumonia detection *and* ruling out false positives matter, HPFL may be preferable despite its lower aggregate accuracy—a conclusion invisible to accuracy-only evaluation.

Finding 3—Ditto stabilizes multiclass imaging: On Brain Tumor (4 classes, 3 seeds), Ditto achieves 6.9$\times$ higher DEI than FedAvg (0.435 vs. 0.063), with improvements across all under-detected classes: Pituitary +40.3pp (14.2%$\rightarrow$54.5%), Glioma +20.1pp (36.6%$\rightarrow$56.7%), and Meningioma +22.1pp (58.0%$\rightarrow$80.1%), with only a minor Healthy trade-off ($-$2.1pp). The per-seed analysis confirms systematic improvement: Ditto DEI ranges 0.118–0.686 across seeds, while FedAvg DEI ranges 0.000–0.026—including a complete Pituitary recall failure (0%) at seed 123.

Finding 4—DEI reveals algorithm-specific failure modes: The low DEI values for FedAvg on Brain Tumor (0.063) and Breast Cancer (0.092) identify specific failure modes: class-level prediction collapse (Breast Cancer) and class-level instability across seeds (Brain Tumor, where Pituitary recall ranges 0–37%). These failures are invisible to aggregate accuracy and Jain fairness but have direct clinical consequences.

4) EHDS Policy Implications: The DEI results yield three specific recommendations for EHDS delegated acts and data permit design:

1) **Minimum DEI thresholds**: Data permits for diagnostic FL models should specify minimum DEI values alongside accuracy targets, particularly for cancer screening and multiclass diagnostic tasks. A threshold of DEI $\geq$ 0.5 would exclude models exhibiting single-class collapse or severe class imbalance. For tasks with irreversible clinical consequences (e.g., missed cancer diagnosis), higher thresholds may be warranted.
2) **Algorithm–DEI conditioning**: The relationship between algorithm choice and DEI is task-specific. HPFL maximizes DEI on tabular class-imbalanced data (Breast Cancer: 0.740), while Ditto maximizes DEI on multi-class imaging (Brain Tumor: 0.435). The evidence-based recommendation matrix (Table S-II) incorporates DEI as a selection criterion alongside accuracy.
3) **Audit trail integration**: DEI should be computed and logged automatically during FL training as part of GDPR Article 30 audit trails, enabling post-hoc verification that deployed models meet per-class diagnostic standards. The FL-EHDS framework's compliance logging module (Algorithm S1) can be extended to compute and record DEI after each evaluation round.

XVI. EHDS-SPECIFIC EXPERIMENTS

This section presents detailed results for five EHDS-specific experiment blocks that validate the framework's governance and privacy modules under realistic cross-border deployment constraints.

A. Cross-Border Heterogeneous DP

Table S-XXXIV reports accuracy under heterogeneous per-client differential privacy budgets on PTB-XL. Five scenarios model increasingly realistic cross-border configurations, from uniform privacy (all clients share the same ε) to fully heterogeneous budgets reflecting divergent national interpretations.

TABLE S-XXXIV

CROSS-BORDER HETEROGENEOUS DP ON PTB-XL (5 CLIENTS, 5-CLASS ECG). PER-CLIENT LOCAL DP WITH GAUSSIAN MECHANISM, C=1.0, δ=10^{-5}. MEAN $\pm$ STD OVER 3 SEEDS.

Scenario	**Client** ε	**FedAvg**	**Ditto**	**HPFL**
No-DP	—	91.9$\pm$0.5	91.8$\pm$0.3	92.5$\pm$0.3
Uniform ε=1	all 1.0	85.6$\pm$1.2	88.7$\pm$0.8	54.4$\pm$3.1
Uniform ε=10	all 10.0	91.3$\pm$0.6	91.5$\pm$0.4	90.9$\pm$0.5
Mixed (2$\times$DE + 3$\times$IT)	1,1,10,10,10	89.3$\pm$0.9	91.0$\pm$0.4	72.8$\pm$2.4
Gradient (5$\times$EU)	1,3,5,7,10	89.0$\pm$0.7	90.8$\pm$0.5	70.5$\pm$2.8

HPFL collapses under local DP at ε=1 (54.4% vs. 88.8% with central DP at the same ε) because per-client gradient noise corrupts the shared backbone that all classifier heads depend on. Ditto, maintaining complete local model copies, is immune to this effect ($-$0.9pp under Mixed configuration). Mixed budgets outperform the strictest-wins rule for FedAvg (+3.8pp), demonstrating that heterogeneous budgets need not default to the most restrictive. The DEI drops sharply under mixed budgets for all algorithms (Ditto: 0.767$\rightarrow$0.296), revealing that accuracy masks diagnostic damage—a finding invisible without per-class analysis. *To our knowledge, no prior FL work experimentally evaluates heterogeneous per-client DP budgets.*

B. Dynamic Article 71 Opt-Out

Table S-XXXV reports accuracy under dynamic mid-training client withdrawal on PTB-XL and Cardiovascular datasets. Unlike static opt-out (which removes data before training), this experiment simulates a hospital exercising its Article 71 right to leave the federation *during* training.

C. Membership Inference Attack (MIA)

Table S-XXXVI reports confidence-based membership inference attack results. For each client, the attacker uses prediction confidence to distinguish training samples (members) from test samples of other clients (non-members). AUC $= 0.50$ indicates no privacy leakage (attack performs no better than random guessing).

D. Partial Client Participation

Table S-XXXVII reports accuracy under partial client participation (3 of 5 clients sampled per round), simulating realistic EHDS deployment where hospitals may be intermittently unavailable.

E. Privacy Budget Exhaustion

Table S-XXXVIII demonstrates the JurisdictionPrivacy-Manager's RDP budget accounting: the noise multiplier is calibrated to distribute the total ε budget across all 30 training rounds, ensuring no client is prematurely deactivated.

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TABLE S-XXXV

DYNAMIC ARTICLE 71 OPT-OUT: ACCURACY (%) UNDER MID-TRAINING HOSPITAL WITHDRAWAL. MEAN ± STD OVER 3 SEEDS. PX = PTB-XL (5 CLIENTS, 30 ROUNDS), CV = CARDIOVASCULAR (5 CLIENTS, 25 ROUNDS). "DROP" = IMMEDIATE ACCURACY CHANGE AT WITHDRAWAL ROUND VS. PREVIOUS ROUND. "RECOVERY" = ROUNDS UNTIL PRE-WITHDRAWAL ACCURACY IS RECOVERED ("—" = NEVER).

		Final Accuracy (%)			**Immediate Drop (pp)**			**Recovery (rounds)**		
Dataset	**Scenario**	**FedAvg**	**Ditto**	**HPFL**	**FedAvg**	**Ditto**	**HPFL**	**FedAvg**	**Ditto**	**HPFL**
PX	No withdrawal	91.9	91.8	92.5	—	—	—	—	—	—
PX	1 client @R10	91.5	91.6	92.1	−0.3	0.0	−0.1	1	0	0
PX	2 clients @R10	91.0	91.2	91.8	−0.8	−0.2	−0.3	2	1	1
PX	1 client @R5	91.3	91.5	92.0	−0.5	−0.1	−0.2	1	1	1
PX	1 client @R20	91.8	91.7	92.4	0.0	0.0	0.0	0	0	0
CV	No withdrawal	72.1	72.4	87.0	—	—	—	—	—	—
CV	1 client @R10	70.8	72.0	86.5	−6.0	0.0	0.0	3	0	0
CV	2 clients @R10	68.5	71.2	85.8	−8.2	−0.5	−0.3	—	1	1
CV	1 client @R5	70.3	71.8	86.2	−5.5	−0.2	−0.1	4	1	1
CV	1 client @R20	71.5	72.3	86.9	−1.0	0.0	0.0	1	0	0

On PTB-XL, withdrawal of 1–2 clients (20–40% capacity loss) at any round costs <1pp for all algorithms. On Cardiovascular, FedAvg suffers −6.0pp immediate drop at R10, while Ditto and HPFL are shock-resistant (0.0pp). HPFL uniquely protects the withdrawn client's test accuracy (76.1% vs. FedAvg/Ditto ≈55%) because its frozen classifier head retains pre-withdrawal knowledge. Late withdrawal (R20) incurs zero cost for all algorithms. *No prior FL work systematically studies mid-training client withdrawal.*

TABLE S-XXXVI

MEMBERSHIP INFERENCE ATTACK AUC ON PTB-XL AND CARDIOVASCULAR. CONFIDENCE-BASED ATTACK: AUC = 0.50 MEANS NO LEAKAGE. SINGLE SEED (42).

Dataset	**DP Level**	**FedAvg**	**Ditto**	**HPFL**
PTB-XL	No-DP	0.503	0.503	**0.666**
	ε=10	0.503	0.503	**0.653**
	ε=1	0.008	0.011	0.337
Cardio.	No-DP	0.499	0.500	0.376
	ε=10	0.495	0.499	0.358
	ε=1	0.015	0.001	0.349

FedAvg and Ditto are inherently immune to confidence-based MIA (AUC≈0.50), as global aggregation dilutes per-client memorization signals. HPFL is vulnerable (AUC = 0.666 on PTB-XL) because personalized classifier heads overfit to local training distributions. DP at ε=10 does not close the HPFL gap; only ε=1 reduces HPFL's leakage to AUC = 0.337 while retaining 75.5% accuracy. At ε=1, FedAvg/Ditto AUC drops below 0.05—but this reflects model collapse (23.7% FedAvg accuracy), not genuine privacy improvement.

TABLE S-XXXVII

PARTIAL CLIENT PARTICIPATION (3/5 PER ROUND) VS. FULL PARTICIPATION. ACCURACY (%), MEAN ± STD OVER 3 SEEDS.

Dataset	**Mode**	**FedAvg**	**Ditto**	**HPFL**
PTB-XL	Full (5/5)	91.9±0.5	91.8±0.3	92.5±0.3
	Partial (3/5)	91.3±0.6	91.4±0.7	91.2±0.2
Cardio.	Full (5/5)	72.1±1.8	72.4±4.7	87.0±4.5
	Partial (3/5)	70.6±2.7	70.5±2.7	87.6±0.6

Partial participation acts as implicit regularization: accuracy does not degrade meaningfully (<1.5pp on PTB-XL, <1.5pp on CV for FedAvg/Ditto). HPFL is the most stable algorithm cross-seed (std = 0.2% on PTB-XL vs. 0.6% for FedAvg). On Cardiovascular, HPFL *improves* under partial participation (87.6% vs. 87.0%) with dramatically reduced variance (0.6% vs. 4.5%).

TABLE S-XXXVIII

PRIVACY BUDGET EXHAUSTION ON PTB-XL. RDP-CALIBRATED NOISE MULTIPLIER DISTRIBUTES TOTAL ε OVER 30 ROUNDS. ALL 5 CLIENTS SURVIVE ALL ROUNDS. SINGLE SEED (42).

Budget	ε	**Noise** σ	**FedAvg**	**Ditto**	**HPFL**
Tight	2.0	13.95	41.0%	73.2%	**81.0%**
Medium	5.0	5.78	68.2%	83.7%	**86.2%**
Generous	10.0	3.12	**90.3%**	90.0%	89.7%

Under tight budgets (ε=2), FedAvg collapses to 41% (effectively random for 5-class) while HPFL maintains 81%—a 2× improvement. The calibrated-noise approach prevents premature client dropout (0 deactivations across all 9 scenarios), trading per-round noise magnitude for guaranteed full-training participation. At generous budgets, all algorithms converge to ≈90%, confirming that the accuracy gap is purely a function of noise tolerance. Personalization is not merely beneficial but *necessary* for privacy-constrained FL.

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TABLE S-XXXIX

NON-IID × DP INTERACTION ON CARDIOVASCULAR (MEAN OVER 3 SEEDS). DIRICHLET α CONTROLS DATA HETEROGENEITY; SMALLER α = MORE EXTREME NON-IID.

Algorithm	α	No DP	ε=10	ε=1
FedAvg	0.1	66.0%	64.9%	59.0%
	0.5	73.9%	68.1%	48.2%
	1.0	73.0%	64.1%	59.6%
Ditto	0.1	66.1%	60.8%	47.7%
	0.5	73.9%	69.9%	45.9%
	1.0	73.0%	61.0%	47.5%
HPFL	0.1	**95.7%**	**95.6%**	**94.2%**
	0.5	87.6%	87.0%	81.2%
	1.0	81.0%	79.5%	69.3%

HPFL's personalized classifier heads absorb data heterogeneity, so DP noise finds less surface to exploit. Under extreme non-IID (α=0.1) with ε=1, HPFL retains 94.2% (−1.5pp) while Ditto collapses to 47.7% (−18.4pp). The interaction is sub-additive for HPFL but super-additive for global algorithms.

TABLE S-XL

CLIENT SCALABILITY ON CARDIOVASCULAR (IID, MEAN OVER 3 SEEDS). ACCURACY (%) AND DEI AS THE NUMBER OF HOSPITALS INCREASES FROM 3 TO 20.

Algo	DP	Number of Clients 3	5	10	15	20
FedAvg	No-DP	73.2	72.7	72.0	70.8	68.4
FedAvg	ε=10	65.7	64.7	55.2	57.5	55.4
FedAvg	ε=1	52.1	49.5	52.3	55.4	50.7
Ditto	No-DP	73.2	72.7	72.0	70.8	68.5
Ditto	ε=10	64.1	56.3	58.1	57.0	57.6
Ditto	ε=1	50.1	53.2	51.3	49.5	48.1
HPFL	No-DP	73.2	72.7	72.0	70.9	68.4
HPFL	ε=10	**70.6**	**69.5**	**65.1**	**64.7**	**64.6**
HPFL	ε=1	**50.7**	**50.6**	**50.4**	**51.0**	**50.7**

All algorithms degrade gracefully under no-DP (−4.8pp from K=3 to K=20). Under ε=10, HPFL maintains 64.6% at K=20 vs. FedAvg 55.4% (+9.2pp). Under ε=1, HPFL's accuracy is essentially flat across all K values (50.4–50.7%), confirming its per-client heads provide constant-factor resilience regardless of federation size.

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TABLE S-XLI

DP × TOP-k COMBINED ON PTB-XL (5 CLIENTS, SEED 42). ACCURACY (%) UNDER SIMULTANEOUS PRIVACY AND COMMUNICATION CONSTRAINTS.

Algorithm	Top-k	No DP	ε=10	ε=1
FedAvg	100%	90.8	90.3	23.7
	5%	91.1	86.2	4.0
	1%	87.2	63.3	15.5
Ditto	100%	90.8	90.0	52.5
	5%	91.4	87.6	14.4
	1%	86.7	40.0	4.3
HPFL	100%	90.6	89.7	75.5
	5%	90.5	89.4	**83.2**
	1%	89.1	77.4	**85.6**

HPFL exhibits a surprising synergy: Top-k sparsification *improves* accuracy under ε=1 (85.6% at Top-1% vs. 75.5% without compression). Sparsification acts as implicit regularization that mitigates DP noise impact on the shared backbone. FedAvg/Ditto collapse under the same conditions (4–15%). HPFL is the only algorithm simultaneously achieving privacy, communication efficiency, and accuracy.

XVII. ADVANCED FL PARADIGM EXPERIMENTAL VALIDATION

This section presents experimental validation of four advanced FL paradigms that are implemented in the FL-EHDS reference framework but were not experimentally evaluated in prior sections: fairness-aware aggregation (Section XVII-A), continual federated learning (Section XVII-B), extended personalization comparison (Section XVII-C), and federated unlearning (Section XVII-D). Together, these 135 experiments complement the 2,800+ experiments in the main paper, validating that the framework's advanced modules produce scientifically meaningful results on real healthcare datasets.

All experiments use the HealthcareMLP architecture (2-layer, 64 hidden units, ReLU) on three tabular datasets: Cardiovascular (CV, 70K patients, binary, 5 clients), PTB-XL ECG (PX, 21,799 records, 5-class, 5 clients), and Breast Cancer (BC, 569 samples, binary, 4 clients). Configuration follows the dataset-specific hyperparameters established in the main paper (Table S9). Non-IID partitioning uses Dirichlet α=0.3.

A. Fairness-Aware Aggregation

Table S-LIII compares five fairness-aware aggregation algorithms against standard FedAvg under IID and non-IID conditions (36 experiments total). Fairness is measured via per-client accuracy variance (Jain index: 1.0 = perfect equity, Gini coefficient: 0.0 = perfect equity) and worst-client accuracy.

B. Continual Federated Learning

Table S-LIV evaluates three continual learning strategies for FL under concept drift scenarios (36 experiments). Each experiment trains on Task 1 (original distribution) for 15 rounds, then on Task 2 (shifted distribution) for 15 rounds. Drift scenarios: *no_drift* (identical distributions), *sudden_shift* (abrupt feature permutation at round 16), *gradual_shift* (incremental permutation over rounds 16–25). Backward Transfer

TABLE S-XLII

STATISTICAL SIGNIFICANCE OF HPFL ADVANTAGE ACROSS MULTI-SEED EXPERIMENTS (PAIRED t-TEST, TWO-SIDED). ALL 24 SIGNIFICANT RESULTS (p<0.05) FAVOR HPFL; NO TEST FINDS A COMPETITOR SIGNIFICANTLY BETTER. EFFECT SIZES (COHEN'S d): >0.8 = LARGE, >2.0 = VERY LARGE. SEEDS10: 5 SEEDS, NO DP. DP: 5 SEEDS PER ε LEVEL. NON-IID+DP: 3 SEEDS.

Source	**Dataset**	**DP Level**	**Comparison**	**Mean Δ**	**Cohen's d**	**p-value**
Baseline (seeds10, 5 seeds, No DP)						
seeds10	Cardiovascular	No-DP	HPFL vs FedAvg	+0.158	1.57	0.025*
seeds10	Cardiovascular	No-DP	HPFL vs Ditto	+0.176	1.29	0.044*
Under DP (dp checkpoint, 5 seeds per ε level)						
dp	Cardiovascular	ε=1	HPFL vs Ditto	+0.279	2.70	0.004**
dp	PTB-XL	ε=1	HPFL vs FedAvg	+0.446	2.00	0.011*
dp	PTB-XL	ε=1	HPFL vs Ditto	+0.411	1.97	0.012*
dp	Cardiovascular	ε=5	HPFL vs Ditto	+0.266	2.86	0.003**
dp	Cardiovascular	ε=5	HPFL vs FedAvg	+0.323	2.18	0.008**
dp	PTB-XL	ε=5	HPFL vs FedAvg	+0.170	2.25	0.007**
dp	Cardiovascular	ε=10	HPFL vs Ditto	+0.172	2.14	0.009**
dp	Cardiovascular	ε=10	HPFL vs FedAvg	+0.225	1.76	0.017*
dp	PTB-XL	ε=10	HPFL vs FedAvg	+0.020	1.46	0.031*
dp	Cardiovascular	ε=50	HPFL vs Ditto	+0.131	2.37	0.006**
dp	Cardiovascular	ε=50	HPFL vs FedAvg	+0.124	2.43	0.006**
Non-IID × DP (noniid_dp checkpoint, 3 seeds)						
noniid_dp	Cardiovascular	ε=10	HPFL vs FedAvg	+0.155	34.73	0.0003***
noniid_dp	Cardiovascular	ε=10	HPFL vs Ditto	+0.171	5.93	0.009**
noniid_dp	Cardiovascular	ε=10	HPFL vs FedAvg	+0.189	4.47	0.016*
noniid_dp	Cardiovascular	ε=1	HPFL vs Ditto	+0.218	6.16	0.009**
noniid_dp	Cardiovascular	No-DP	HPFL vs FedAvg	+0.136	4.28	0.018*
noniid_dp	Cardiovascular	No-DP	HPFL vs Ditto	+0.136	4.25	0.018*

Significance levels: *p<0.05, **p<0.01, ***p<0.001. Effect sizes consistently exceed the "large" threshold (d>0.8), indicating clinically meaningful differences. Under DP stress, HPFL's advantage *increases* (mean d=2.3 under DP vs. d=1.4 without DP). The extremely large d=34.7 on non-IID+DP reflects near-zero variance: HPFL is consistent while FedAvg/Ditto fluctuate.

TABLE S-XLIII

HPFL PERSONALIZATION DEPTH ABLATION (24 EXPERIMENTS, 30 ROUNDS, SEED 42). FEDAVG: 0% PERSONALIZED (FULL AGGREGATION). DITTO: REGULARIZATION-BASED PERSONALIZATION. HPFL: CLASSIFIER HEAD ONLY (∼2–6% OF PARAMETERS). HPFL-FULL: 100% PERSONALIZED (PURE LOCAL TRAINING, NO FL AGGREGATION).

Dataset	**Variant**	**Acc% (No-DP)**	**Acc% (ε=10)**	**Δ DP**
CV	FedAvg (0%)	72.7	50.8	−21.9
	Ditto (reg.)	72.8	62.0	−10.8
	HPFL (head)	72.7	69.9	−2.8
	HPFL-Full (100%)	72.7	**72.7**	−0.0
PTB-XL	FedAvg (0%)	**90.8**	**90.3**	−0.5
	Ditto (reg.)	90.8	90.0	−0.8
	HPFL (head)	90.6	89.7	−0.9
	HPFL-Full (100%)	78.1	78.1	−0.0
BC	FedAvg (0%)	54.5	81.2	+26.7
	Ditto (reg.)	54.5	74.6	+20.1
	HPFL (head)	60.6	64.7	+4.1
	HPFL-Full (100%)	60.6	60.6	+0.0

Key findings: (1) Without DP, all variants perform similarly on CV/PTB-XL, confirming that personalization matters primarily under stress. (2) Under DP, the optimal personalization depth is task-dependent: HPFL-Full dominates on simple binary CV (+21.9pp over FedAvg), while PTB-XL (5-class) requires FL collaboration (HPFL-Full −12.2pp). (3) HPFL's head-only personalization (∼2–6% params) is the robust "sweet spot" across all datasets. (4) On BC, DP acts as implicit regularization: FedAvg+DP *improves* over FedAvg without DP (+26.7pp), as Gaussian noise prevents overfitting on the small 569-sample dataset.

TABLE S-XLIV

COMMUNICATION COSTS FOR TABULAR FL (HEALTHCAREMLP, FLOAT32). PER-ROUND BANDWIDTH INCLUDES UPLOAD (POTENTIALLY SPARSE WITH TOP-k) AND DOWNLOAD (FULL MODEL BROADCAST) FOR K=5 CLIENTS. HPFL SAVES ∼2–6% BY KEEPING CLASSIFIER HEADS LOCAL.

Dataset	**Algo**	**Params**	**KB/round**	**Total MB**
CV	FedAvg (NoTopK)	2,914	113.8	3.33
	HPFL (NoTopK)	2,848	111.3	3.26
	FedAvg (Top-1%)	2,914	114.6†	1.68
	HPFL (Top-1%)	2,848	112.4†	1.65
PTB-XL	FedAvg (NoTopK)	2,885	112.7	3.30
	HPFL (NoTopK)	2,720	106.3	3.11
	FedAvg (Top-1%)	2,885	113.8†	1.67
	HPFL (Top-1%)	2,720	107.2†	1.57
BC	FedAvg (NoTopK)	4,130	161.3	4.73
	HPFL (Top-1%)	4,064	160.2†	2.35

†Upload is 99% sparse (Top-1%), download is full model. Total MB for 30 rounds × 5 clients. Tabular FL communication is negligible (<5 MB total) even without compression—orders of magnitude below imaging (44.7 MB/round for ResNet-18). HPFL + Top-1% achieves 52% total reduction vs. FedAvg without compression.

(BWT) measures accuracy change on Task 1 data after Task 2 training: positive BWT indicates knowledge preservation, negative BWT indicates catastrophic forgetting.

TABLE S-XLV

CONVERGENCE EFFICIENCY ANALYSIS (TABLE S-EFFICIENCY). SEEDS10 (NO-DP, 5 SEEDS): EFFICIENCY RATIO = ACCURACY / TOTAL COMMUNICATION (MB). HPFL IS PARETO-OPTIMAL ON ALL DATASETS.

Dataset	Algo	Acc%	R-to-90%	CommMB	Eff
CV	FedAvg	70.8	5.4	2.78	0.25
	Ditto	83.6	1.2	2.78	0.30
	HPFL	83.6	1.2	2.28	**0.37**
PTB-XL	FedAvg	91.3	4.0	3.08	0.30
	Ditto	92.1	6.6	3.30	0.28
	HPFL	92.6	5.2	2.70	**0.34**
BC	FedAvg	57.3	4.6	4.41	0.13
	Ditto	71.1	3.0	2.68	**0.27**
	HPFL	71.1	3.6	3.41	0.21

R-to-90% = rounds to reach 90% of final accuracy. Eff = Acc/MB (higher is better). HPFL's backbone-only communication yields up to 48% higher efficiency than FedAvg.

TABLE S-XLVI

PER-CLIENT EQUITY ANALYSIS (TABLE S-EQUITY). GINI COEFFICIENT (0=EQUAL, 1=UNEQUAL), WORST-CASE CLIENT ACCURACY, AND EQUITY GAP (MAX−MIN CLIENT) ACROSS DP CONDITIONS. ROWS SHOW NO-DP BASELINE AND DP IMPACT (Δ FROM NO-DP TO ε=10).

Dataset	Algo	Gini	Worst%	Gap%	ΔGap(DP)
BC (No-DP)	FedAvg	0.374	11.1	76.9	−15.4
	Ditto	0.371	12.2	75.9	−26.7
	HPFL	**0.149**	**47.9**	**41.1**	+13.5
CV (No-DP)	FedAvg	0.053	62.1	16.7	+27.0
	Ditto	0.079	58.1	21.9	+20.4
	HPFL	**0.046**	**77.0**	19.1	**+9.6**
PTB-XL (No-DP)	FedAvg	**0.013**	85.8	**6.8**	+2.0
	Ditto	0.015	86.0	7.5	+1.2
	HPFL	0.017	85.8	8.7	+1.6

ΔGap(DP) = change in equity gap from No-DP to ε=10. Negative values indicate DP *improves* equity (regularization on small datasets). HPFL has the smallest DP-induced equity degradation on Cardiovascular (+9.6pp vs. +27.0pp for FedAvg).

TABLE S-XLVII

PRIVACY-UTILITY-FAIRNESS TRILEMMA (TABLE S-TRILEMMA). ACCURACY (%) AND JAIN FAIRNESS INDEX AT EACH PRIVACY LEVEL. ACCDROP = ACCURACY DROP FROM NO-DP BASELINE. HPFL HAS THE FLATTEST TRADE-OFF CURVE ACROSS ALL DATASETS.

Dataset	Algo	No-DP	ε=1	ε=10	ε=50	AccDrop(ε=10)
CV	FedAvg	69.5	56.6	62.3	73.2	−7.2
	Ditto	67.6	49.2	67.5	72.6	−0.1
	HPFL	**85.2**	**77.1**	**84.8**	**85.7**	**−0.5**
PTB-XL	FedAvg	90.2	29.6	88.0	90.7	−2.2
	Ditto	90.8	33.1	89.9	90.9	−0.8
	HPFL	90.7	**74.2**	**90.0**	**90.9**	**−0.7**
BC	FedAvg	48.4	46.6	51.0	55.9	+2.7†
	Ditto	48.3	47.2	43.9	53.2	−4.4
	HPFL	**73.2**	**72.0**	**66.6**	**74.1**	−6.5

†Positive AccDrop indicates DP *improves* accuracy (regularization effect on small dataset). Marginal accuracy returns beyond ε=10 are <0.002%/ε for all algorithms, establishing ε=10 as the practical operating point.

C. Extended Personalization Comparison

Table S-LV extends the personalization evaluation from the main paper (which covered Ditto and HPFL) with three additional algorithms: Per-FedAvg [34] (MAML-based meta-learning), FedPer (shared base + personalized head), and APFL (Adaptive Personalized FL with mixing coefficient α). Evaluation covers IID and two non-IID conditions (Dirichlet $\alpha \in \{0.1, 0.5\}$) across 45 experiments.

TABLE S-XLVIII

LEARNING RATE SENSITIVITY (TABLE S-LRSENSITIVITY). FINAL ACCURACY (%) ACROSS 3 LEARNING RATES PER DATASET. RANGE = MAX−MIN ACCURACY ACROSS LRS. LOWER RANGE INDICATES MORE ROBUST ALGORITHM. SEED = 42, NO-DP, 30 ROUNDS.

Dataset	Algo	LR_{low}	LR_{base}	LR_{high}	Range (pp)
CV	FedAvg	63.4	72.7*	73.0	9.6
	Ditto	63.5	72.8*	73.2	9.7
	HPFL	63.2	72.7*	73.2	10.0
PTB-XL	FedAvg	89.1	90.8*	91.2	**2.1**
	Ditto	89.1	90.8*	91.4	2.3
	HPFL	82.0	90.6*	91.2	9.3
BC	FedAvg	54.5	54.5*	95.7	41.2
	Ditto	54.5	54.5*	95.7	41.2
	HPFL	85.6	60.6*	89.1	**28.5**

* = base learning rate (CV: 0.01, PTB-XL: 0.005, BC: 0.005). LR_{low}/LR_{high} are 0.1× and 4–5× the base rate. HPFL is the most robust on Breast Cancer (28.5pp range vs. 41.2pp for FedAvg/Ditto), while all algorithms are comparably stable on Cardiovascular (∼10pp range).

TABLE S-XLIX

DATA EFFICIENCY: ACCURACY (%) AT SUBSAMPLED TRAINING FRACTIONS

Dataset	Algorithm	25%	50%	75%	100%†
Cardiovascular	FedAvg	69.9	72.5	72.6	70.8
	Ditto	69.8	72.6	72.8	83.6
	HPFL	69.5	72.6	72.6	83.6
PTB-XL	FedAvg	90.0	90.8	91.2	91.3
	Ditto	90.0	90.7	91.2	92.1
	HPFL	**84.6**	90.7	91.2	92.6
Breast Cancer	FedAvg	54.5	54.5	54.5	57.3
	Ditto	54.5	54.5	54.5	71.1
	HPFL	60.6	60.6	60.6	71.1

†100% baseline from seeds10 checkpoint (mean of 5 seeds). Subsampled fractions use seed=42 only. On PTB-XL, FedAvg/Ditto retain 98.5% of full performance at 25% data. HPFL requires 50% data to converge but then matches competitors. Breast Cancer (n=456) is too small for meaningful subsampling analysis.

D. Federated Unlearning (Right to Be Forgotten)

Table S-LVI evaluates three federated unlearning methods for GDPR Article 17 "right to erasure" and EHDS Article 71 compliance (18 experiments). Each experiment first trains a baseline model with all clients, then removes 1 or 2 clients' contributions. The key metrics are post-unlearning accuracy (should remain close to pre-unlearning) and computational cost relative to full retraining from scratch.

XVIII. CLIENT SELECTION AND GRADIENT INVERSION PRIVACY VALIDATION

This section presents experimental validation of the framework's client selection strategies and gradient inversion privacy analysis (87 experiments total).

TABLE S-L
COMPOUND STRESS TEST: ACCURACY (%) UNDER SIMULTANEOUS NON-IID + DP + PARTIAL PARTICIPATION

Algorithm	Stress	Cardiovascular				PTB-XL			
		α=0.1		α=0.5		α=0.1		α=0.5	
		100%	60%	100%	60%	100%	60%	100%	60%
FedAvg	No-DP	71.9	53.1	72.6	59.2	86.3	89.0	89.5	90.0
	ε=10	63.6	52.9	63.2	56.4	77.8	81.1	90.3	91.0
Ditto	No-DP	71.9	53.2	72.1	59.7	86.4	89.3	89.4	89.8
	ε=10	67.2	52.7	61.4	54.4	**59.2**	76.0	82.8	87.8
HPFL	No-DP	**91.5**	**91.2**	87.3	86.5	85.6	86.0	**91.9**	91.3
	ε=10	**91.1**	**91.1**	**86.3**	**86.2**	**92.2**	**92.0**	91.5	89.2

Bold = best per column. Under triple stress (α=0.1, ε=10, 60% participation), HPFL achieves 91.1% on Cardiovascular vs. FedAvg 52.9% (38.2pp gap). HPFL's worst-case (86.2%) exceeds FedAvg's best-case (72.6%). Ditto collapses on PTB-XL at α=0.1, ε=10, 100% (59.2%, bold-italic). HPFL Jain $\geq$ 0.971 across all conditions.

TABLE S-LI
DATA HETEROGENEITY CHARACTERIZATION: PER-CLIENT STATISTICS UNDER DIRICHLET PARTITIONING

Dataset	Condition	Label Entropy	Class Imb. Ratio	Feature KL Div.
Cardiovascular	IID	1.000	1.0	0.000
	α=1.0	0.727	6.6	0.008
	α=0.5	0.510	25.0	0.014
	α=0.1	0.325	1954.0	0.021
PTB-XL	IID	2.022	12.9	0.087
	α=1.0	1.654	90.4	0.032
	α=0.5	1.756	630.4	0.023
	α=0.1	1.429	1416.0	0.071

Label Entropy: mean per-client Shannon entropy (bits). Class Imb. Ratio: mean max/min class ratio per client. Feature KL Div.: mean KL divergence of per-client feature distributions vs. global. At α=0.1, Cardiovascular clients receive near-single-class data (entropy 0.325 vs. 1.0 IID) with class imbalance ratios up to 8,373:1. HPFL achieves 95.7% at α=0.1 vs. FedAvg 66.0% (+29.7pp).

TABLE S-LII
PEP TRIANGLE: PRIVACY–EQUITY–PERFORMANCE PROFILES

Dataset	Algorithm	Perf. (Acc%)	Equity (Jain)	Privacy (MIA AUC)
Cardiovascular	FedAvg	69.5	0.979	0.499
	Ditto	67.6	0.951	0.500
	HPFL	**85.2**	**0.991**	**0.376***
PTB-XL	FedAvg	90.2	0.999	0.503
	Ditto	90.8	0.999	0.503
	HPFL	90.7	0.999	0.666
Breast Cancer	FedAvg	48.4	0.661	—
	Ditto	48.3	0.664	—
	HPFL	**73.2**	**0.889**	—

*MIA AUC $<$ 0.5 indicates attack performs worse than random—HPFL's personalization structure provides inherent membership privacy on Cardiovascular without requiring DP. On PTB-XL, HPFL's elevated MIA AUC (0.666) reflects the privacy–personalization tension, fully mitigated by DP at ε=10. Equity retention at ε=1: HPFL 93.8% of Jain vs. FedAvg 78.4%.

A. Client Selection Strategy Comparison

We evaluate six client selection strategies across three EHDS-relevant datasets (Cardiovascular, PTB-XL, Breast Cancer) with participation rate 60%, two federation scales ($K \in \{5, 10\}$; Breast Cancer K=5 only due to limited size), and 2 seeds per configuration (60 experiments total).

B. Gradient Inversion Privacy Analysis

We evaluate Deep Leakage from Gradients (DLG) [21] attacks against the FL framework under three differential privacy levels ($\varepsilon \in \{\infty, 10, 1\}$) across three datasets with 3 seeds each (27 experiments). The attack uses L-BFGS optimization to reconstruct training data from shared gradients. We measure reconstruction quality via MSE, cosine similarity, and Pearson correlation, classifying results as SUCCESSFUL (MSE < 0.1), PARTIAL ($0.1 \leq$ MSE ≤ 1.0), or FAILED (MSE > 1.0).

XIX. EXTENDED FEDERATED LEARNING VALIDATION

This section presents experimental validation across six additional tiers (234 experiments total), covering dataset coverage expansion, hierarchical FL, asynchronous FL, multi-task FL, extended Byzantine robustness, and Shapley-based client valuation.

A. Dataset Coverage Expansion

We extend the framework evaluation to three additional clinical datasets: Stroke Prediction (5,110 samples, 10 features, 4.9% positive rate), Chronic Kidney Disease (399 samples, 24 features), and Cirrhosis Patient Survival (418 samples, 18 features). Each dataset is tested with 5 FL algorithms (FedAvg, Ditto, HPFL, FedProx, SCAFFOLD) under 3 conditions (IID, non-IID α=0.5, non-IID + DP ε=10) with 2 seeds (90 experiments).

B. Hierarchical FL Topologies

We evaluate three FL topologies—flat (standard), 2-tier (regional $\rightarrow$ global), and 3-tier (local $\rightarrow$ regional $\rightarrow$ global)—across 3 datasets with 2 seeds each (18 experiments). Hierarchical topologies reflect EHDS multi-level governance where

TABLE S-LIII
FAIRNESS-AWARE AGGREGATION COMPARISON (36 EXPERIMENTS, SEED 42, 30 ROUNDS). ALGORITHMS: Q-FEDAVG [53] (q=1.0), AFL [54] (AGNOSTIC FL), FEDMGDA+ [55] (MULTI-GRADIENT DESCENT), PROPFAIR (PROPORTIONAL FAIRNESS), FEDMINMAX (MINIMAX OPTIMIZATION). BEST PER-DATASET NON-IID IN **BOLD**.

Dataset	Algorithm	IID					Non-IID (α=0.3)				
		Acc%	Worst%	Best%	Jain	Gini	Acc%	Worst%	Best%	Jain	Gini
CV	Standard	72.7	72.0	73.2	1.000	0.003	72.5	61.0	73.2	0.987	0.064
	q-FedAvg	72.7	72.0	73.2	1.000	0.003	**73.0**	**68.2**	73.2	0.997	0.029
	AFL	72.7	72.0	73.2	1.000	0.003	72.9	67.4	73.2	**0.997**	**0.032**
	FedMGDA+	72.7	72.1	73.2	1.000	0.004	64.2	33.1	73.3	0.894	0.188
	PropFair	72.7	72.0	73.2	1.000	0.003	54.6	2.6	73.2	0.693	0.361
	FedMinMax	72.7	72.0	73.2	1.000	0.003	71.2	65.0	73.0	0.994	0.043
PX	Standard	91.3	87.4	93.7	0.999	0.014	86.7	77.4	92.1	0.995	0.042
	q-FedAvg	91.2	88.3	93.5	1.000	0.011	91.8	89.0	93.7	0.999	0.012
	AFL	91.2	88.3	93.5	1.000	0.011	91.8	89.0	93.7	0.999	0.012
	FedMGDA+	80.3	76.8	83.7	0.999	0.019	80.9	67.1	90.1	0.988	0.061
	PropFair	91.0	87.4	93.5	0.999	0.013	91.5	89.0	93.5	0.999	0.013
	FedMinMax	88.3	85.3	90.1	0.999	0.016	**92.0**	**89.9**	93.5	**1.000**	**0.011**
BC	Standard	54.5	4.5	96.7	0.677	0.382	54.5	4.5	96.7	0.677	0.382
	q-FedAvg	60.0	22.7	96.7	0.771	0.294	60.0	22.7	96.7	0.771	0.294
	AFL	58.5	18.2	96.7	0.750	0.315	58.5	18.2	96.7	0.750	0.315
	FedMGDA+	45.5	0.0	96.7	0.593	0.458	45.5	0.0	96.7	0.593	0.458
	PropFair	48.2	0.0	96.7	0.709	0.348	48.2	0.0	96.7	0.709	0.348
	FedMinMax	**85.7**	**59.1**	96.7	**0.963**	**0.101**	**85.7**	**59.1**	96.7	**0.963**	**0.101**

Key findings: (1) Under IID, all fairness algorithms converge to identical accuracy on CV and PX, confirming that fairness reweighting has no effect when data is balanced. (2) Under non-IID, q-FedAvg achieves the best worst-case improvement on CV (+7.2pp over standard) while FedMinMax achieves the best overall accuracy on PX (92.0%) and dominates on BC (+31.2pp over standard). (3) FedMGDA+ and PropFair exhibit instability under non-IID: FedMGDA+ drops to 64.2% on CV (vs. 72.5% standard), and PropFair collapses to 54.6% with worst-client at 2.6%—the Frank-Wolfe and proportional optimization objectives conflict with non-IID gradients. (4) FedMinMax dramatically improves BC from 54.5% to 85.7% by preventing majority-class collapse. *EHDS implication*: fairness-aware aggregation is essential for cross-border federations with heterogeneous hospital sizes; FedMinMax is recommended for small-data settings, q-FedAvg for large-scale deployments.

hospitals aggregate within national HDABs before cross-border federation.

C. Asynchronous FL Strategies

We evaluate four asynchronous FL strategies—FedAsync, FedBuff (buffered), AsyncSGD, and FedAT (attentive)—across 3 datasets and 3 staleness levels ($s \in \{1, 3, 5\}$) representing 36 experiments. Asynchronous FL is critical for EHDS deployments where hospitals operate on different schedules and network latencies.

D. Multi-Task FL

We evaluate multi-task FL with two parameter sharing modes (hard, soft) and two task combinations (diagnosis+severity, diagnosis+readmission) across 3 datasets with 2 seeds (24 experiments). Multi-task FL enables EHDS analytics to produce multiple clinical outputs from a single federated model.

E. Extended Byzantine Robustness

We evaluate three Byzantine attacks (scale, label_flip, sign_flip) against four defense strategies (none, Krum, Trimmed Mean, Median) across 3 datasets (36 experiments). One out of 5 clients (Cardiovascular, PTB-XL) or 1 out of 4 clients (Breast Cancer) is Byzantine.

F. Byzantine Robustness on Medical Imaging

While the preceding section evaluated Byzantine resilience on tabular clinical data (Table S-LXIV), medical imaging presents fundamentally different challenges: larger model capacity (ResNet-18 with 11.2M parameters vs. 3-layer MLP), richer gradient signals, and heterogeneous visual features across institutions. We evaluate three attack types—*label flipping* (adversary inverts class labels), *gradient noise injection* (Gaussian noise σ=10 added to model updates), and *sign flipping* (gradients multiplied by -5)—with Krum defense (f=1 Byzantine out of K=5 clients) across 3 imaging datasets, 3 algorithms, and 3 seeds (108 experiments total, $\sim$27 hours on NVIDIA A40).

Key findings. The 108-experiment imaging Byzantine evaluation reveals three findings that complement the tabular analysis:

First, **label flipping bypasses Krum defense on imaging**. Unlike noise and sign flipping—which Krum filters effectively (average drop <3pp)—label flipping causes significant accuracy degradation across all algorithms: FedAvg -7.1pp, Ditto -6.6pp, HPFL -16.1pp (averaged across datasets). This occurs because label flipping corrupts the *training data* rather than the gradient signal: Krum operates on model updates and

TABLE S-LIV

CONTINUAL FL UNDER CONCEPT DRIFT (36 EXPERIMENTS, SEED 42, 30 ROUNDS TOTAL: 15 TASK 1 + 15 TASK 2). METHODS: EWC [56] (λ=1000), REPLAY (BUFFER=200 SAMPLES/CLIENT), LWF [57] (TEMPERATURE=2.0). BWT = BACKWARD TRANSFER (POSITIVE = KNOWLEDGE PRESERVED, NEGATIVE = CATASTROPHIC FORGETTING).

		No Drift				**Sudden Shift**				**Gradual Shift**			
DS	**Method**	**T1%**	**T2%**	**BWT**	**Jain**	**T1%**	**T2%**	**BWT**	**Jain**	**T1%**	**T2%**	**BWT**	**Jain**
CV	Baseline	72.3	72.8	+0.5	1.000	72.2	72.6	+0.4	1.000	72.3	72.6	+0.3	1.000
	EWC	72.3	49.8	−22.4	1.000	72.2	49.8	−22.4	1.000	72.3	49.8	−22.4	1.000
	Replay	72.3	72.6	+0.3	1.000	72.2	72.5	+0.3	1.000	72.3	72.6	+0.3	1.000
	LwF	72.3	72.7	+0.4	1.000	72.2	72.5	+0.3	1.000	72.3	72.6	+0.3	1.000
PX	Baseline	88.1	88.1	−0.0	0.999	87.9	91.0	+3.1	0.999	88.0	90.1	+2.1	0.999
	EWC	88.1	87.8	−0.3	0.999	87.9	87.8	−0.1	0.999	88.0	88.0	+0.0	0.999
	Replay	88.1	88.1	−0.0	0.999	87.9	91.0	+3.1	0.999	88.0	90.0	+2.1	0.999
	LwF	88.1	88.1	+0.0	0.999	87.9	87.9	+0.1	0.999	88.0	88.0	+0.1	0.999
BC	Baseline	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677
	EWC	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677
	Replay	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677
	LwF	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677	54.5	54.5	0.0	0.677

Key findings: (1) **EWC is catastrophically harmful on CV**: the λ=1000 regularization penalty prevents the model from learning Task 2, dropping accuracy to 49.8% (near chance for binary). The Fisher Information Matrix from Task 1 over-constrains weight updates, blocking adaptation. A lower λ or per-layer adaptive regularization would be needed. (2) **Replay and LwF preserve knowledge effectively**: both maintain Task 1 accuracy after Task 2 training, with positive backward transfer on CV (+0.3–0.4pp). Replay achieves this by mixing stored Task 1 samples into Task 2 training batches; LwF by distilling Task 1 predictions. (3) **PTB-XL under sudden shift**: the baseline actually *improves* (+3.1pp BWT), suggesting the shifted distribution is complementary. EWC prevents this improvement by anchoring to Task 1 weights. (4) **BC is unresponsive**: all methods produce identical results (54.5%), indicating the 569-sample dataset is too small for the model to learn a discriminative representation under FL partitioning (4 clients, $\sim$142 samples each). *EHDS implication*: for the long-term 2025–2031 EHDS deployment, Experience Replay or LwF are recommended for maintaining model quality under evolving data distributions; EWC requires careful hyperparameter tuning per dataset.

cannot detect that a client's update was trained on corrupted labels, since the update itself has normal magnitude and direction.

Second, **Ditto is the most Byzantine-resilient algorithm**. Ditto achieves the highest resilience scores on Chest X-ray (99.1%) and Skin Cancer (96.1%), and the highest absolute accuracy under attack on 2/3 datasets. Its L2-regularized personal model provides an implicit defense: even when the global model is degraded by Byzantine clients, the personal model's L2 penalty toward the (partially corrupted) global acts as a conservative anchor, limiting the propagation of Byzantine influence to local models.

Third, **noise and sign flipping anomalously improve Brain Tumor accuracy** for FedAvg (+10.9pp, +10.8pp) and Ditto (+8.6pp, +12.4pp). This "noise-as-regularization" effect occurs because Krum's filtering of the adversarial client also excludes that client's potentially harmful non-IID update—on this small, 4-class dataset with high heterogeneity, removing one client's contribution can actually improve the global model. This mirrors the finding from the tabular analysis (Table S-LXIV) where Breast Cancer showed counterproductive defense effects.

EHDS implication: For cross-border imaging federations, Ditto with Krum defense provides the best resilience against Byzantine participants. However, **data poisoning via label corruption remains a critical threat** that aggregation-level defenses cannot address—EHDS implementations should complement Byzantine-resilient aggregation with data quality auditing mechanisms at the governance layer (Layer 1) to detect corrupted training labels before they enter the federation.

G. Shapley Values and Client Valuation

We compute exact Shapley values via exhaustive enumeration of all 2^K client subsets ($K \in \{4, 5\}$) to quantify each client's marginal contribution to the federation. Experiments span 3 datasets, 2 conditions (IID, non-IID α=0.5), and 5 seeds (30 experiments, 15 rounds each).

TABLE S-LV

EXTENDED PERSONALIZATION COMPARISON (45 EXPERIMENTS, SEED 42, 30 ROUNDS). FEDAVG BASELINE, DITTO (λ=0.1), PER-FEDAVG (5 INNER STEPS, META-LR=0.01), FEDPER (SHARED BASE + 1-LAYER HEAD), APFL (MIXING α=0.5). BEST PER-DATASET/CONDITION IN **BOLD**.

		IID			**Non-IID** (α=0.1)			**Non-IID** (α=0.5)		
DS	**Method**	**Acc%**	**Worst%**	**Jain**	**Acc%**	**Worst%**	**Jain**	**Acc%**	**Worst%**	**Jain**
CV	FedAvg	72.7	72.0	1.000	71.3	49.6	0.965	72.5	61.0	0.987
	Ditto	72.7	72.1	1.000	**91.6**	**72.9**	0.986	**87.1**	**72.8**	0.983
	Per-FedAvg	62.0	60.9	1.000	55.8	16.8	0.746	59.0	18.8	0.812
	FedPer	72.8	72.0	1.000	91.5	72.8	**0.986**	87.1	72.8	0.982
	APFL	72.7	72.0	1.000	**91.6**	73.2	0.986	87.2	72.7	**0.983**
PX	FedAvg	91.3	87.4	0.999	77.5	60.7	0.967	86.7	77.4	0.995
	Ditto	89.9	84.8	0.998	**94.8**	**87.6**	**0.999**	**93.3**	**90.1**	**0.999**
	Per-FedAvg	17.4	12.9	0.980	32.0	4.8	0.770	21.1	15.6	0.943
	FedPer	89.2	83.6	0.997	94.4	86.6	0.998	93.1	89.9	0.999
	APFL	90.1	85.6	0.998	92.3	84.8	0.996	93.0	89.6	0.999
BC	FedAvg	54.5	4.5	0.677	54.5	4.5	0.677	54.5	4.5	0.677
	Ditto	**60.6**	0.0	**0.734**	**60.6**	0.0	**0.734**	**60.6**	0.0	**0.734**
	Per-FedAvg	45.5	0.0	0.593	45.5	0.0	0.593	45.5	0.0	0.593
	FedPer	60.6	0.0	0.734	60.6	0.0	0.734	60.6	0.0	0.734
	APFL	60.6	0.0	0.734	60.6	0.0	0.734	60.6	0.0	0.734

Key findings: (1) **Ditto, FedPer, and APFL form a high-performance cluster**: on CV non-IID (α=0.1), all three achieve $\sim$91.6% (+20.3pp over FedAvg). Their personalization mechanisms—L2 regularization (Ditto), split architecture (FedPer), and adaptive mixing (APFL)—converge to similar solutions on the compact MLP. (2) **Ditto leads on PTB-XL**: 94.8% at α=0.1 (+17.3pp over FedAvg), followed by FedPer (94.4%) and APFL (92.3%). The 5-class task benefits from Ditto's complete dual-model architecture. (3) **Per-FedAvg (MAML) fails**: 17.4% on PTB-XL IID (vs. 91.3% FedAvg) and 55.8% on CV non-IID (vs. 72.5% FedAvg). The MAML inner loop (5 steps, lr=0.01) is insufficient for convergence on these datasets; MAML's sensitivity to inner-loop hyperparameters is a known challenge [34]. (4) **BC is personalization-insensitive**: IID and non-IID produce identical results for all personalization methods, confirming dataset saturation at 569 samples with 4 clients. *EHDS implication*: Ditto, FedPer, and APFL are all viable for EHDS personalization; the choice depends on deployment constraints (Ditto requires 2$\times$ model storage, FedPer requires architecture modification, APFL is parameter-free). Per-FedAvg should be avoided without extensive hyperparameter tuning.

TABLE S-LVI

FEDERATED UNLEARNING COMPARISON (18 EXPERIMENTS, SEED 42, 30 ROUNDS PRE-TRAINING). METHODS: EXACT RETRAIN (GOLD STANDARD, RETRAIN FROM SCRATCH WITHOUT TARGET CLIENTS), GRADIENT ASCENT [59] (REVERSE GRADIENT CONTRIBUTION, 5% COST), FEDERASER [58] (SUBTRACT ACCUMULATED UPDATES + CALIBRATION, 37% COST). NEGATIVE "DROP" INDICATES POST-UNLEARNING ACCURACY *improves* (REMOVING HARMFUL CLIENTS). MIA = MEMBERSHIP INFERENCE VERIFICATION PASS.

Dataset	Scenario	Method	Pre-Acc%	Post-Acc%	Acc Drop (pp)	Cost	Removed Client Acc%	MIA Pass
CV	Remove 1 client	Exact Retrain	72.7	72.7	0.0	100%	73.2	✓
		Gradient Ascent	72.7	72.6	0.1	5%	73.1	✓
		FedEraser	72.7	72.6	0.1	37%	73.4	✓
	Remove 2 clients	Exact Retrain	72.7	72.7	−0.0	100%	72.9	✓
		Gradient Ascent	72.7	72.5	0.2	5%	72.9	✓
		FedEraser	72.7	72.6	0.1	37%	72.5	✓
PX	Remove 1 client	Exact Retrain	91.1	91.4	−0.3	100%	91.0	✓
		Gradient Ascent	91.1	91.2	−0.0	5%	91.0	✓
		FedEraser	91.1	91.2	−0.1	37%	91.0	✓
	Remove 2 clients	Exact Retrain	91.1	92.7	−1.6	100%	89.2	✓
		Gradient Ascent	91.1	92.4	−1.3	5%	89.2	✓
		FedEraser	91.1	92.3	−1.2	37%	89.2	✓
BC	Remove 1 client	Exact Retrain	59.1	61.6	−2.5	100%	33.3	✓
		Gradient Ascent	59.1	68.8	−9.7	5%	33.3	✓
		FedEraser	59.1	61.6	−2.5	37%	33.3	✓
	Remove 2 clients	Exact Retrain	59.1	90.1	−31.0	100%	18.9	✓
		Gradient Ascent	59.1	91.9	−32.8	5%	75.0	✓
		FedEraser	59.1	90.1	−31.0	37%	18.9	✓

Key findings: (1) **Gradient Ascent is the most cost-effective**: at only 5% of full retraining cost, it achieves comparable post-unlearning accuracy ($\leq$0.2pp drop on CV). This makes it practical for EHDS "right to erasure" scenarios where computational budget is limited. (2) **Unlearning can *improve* accuracy**: on BC, removing 2 clients raises accuracy from 59.1% to 90.1–91.9%, indicating that some clients' data distributions were actively harmful to the global model. On PTB-XL, 2-client removal improves accuracy by 1.2–1.6pp. This has a surprising EHDS implication: exercising Article 17 opt-out rights may *benefit* the remaining federation. (3) **All methods pass MIA verification**: post-unlearning models show no residual membership signal for removed clients, confirming effective data erasure. (4) **FedEraser offers a middle ground**: 37% cost with accuracy matching exact retraining on CV and BC—its calibration rounds effectively recover from the subtracted contribution. (5) **CV is robust to client removal**: the large dataset (70K samples, 5 clients) absorbs the loss of 1–2 clients with negligible accuracy impact, suggesting that adequately-sized EHDS federations can handle opt-out requests without quality degradation. *EHDS implication*: Gradient Ascent should be the default unlearning method for operational EHDS deployments, reserving Exact Retrain for audit verification scenarios. The framework's unlearning module enables Article 17 compliance at 5–37% of the computational cost of full model retraining.

TABLE S-LVII

CLIENT SELECTION STRATEGY COMPARISON: AVERAGE ACCURACY, JAIN FAIRNESS INDEX, AND GINI COEFFICIENT

	Cardiovascular			PTB-XL			Breast Cancer			Overall
Strategy	Acc	Jain	Gini	Acc	Jain	Gini	Acc	Jain	Gini	Avg Acc
random	65.76	0.904	0.166	87.48	0.981	0.055	56.15	0.688	0.359	72.52
loss_based	70.04	0.958	0.096	88.79	0.983	0.044	72.70	0.799	0.243	78.07
importance_sampling	66.61	0.922	0.150	88.09	0.984	0.049	**74.16**	**0.808**	**0.230**	76.71
resource_aware	70.37	0.964	0.093	86.18	0.980	0.066	69.81	0.764	0.273	76.58
fairness_aware	62.82	0.869	0.203	87.41	0.983	0.054	57.09	0.639	0.397	71.51
oort	**70.85**	**0.963**	**0.096**	**88.99**	**0.983**	**0.044**	72.70	0.799	0.243	**78.47**

Key findings: (1) **Oort achieves the best overall performance**: 78.47% average accuracy with Jain fairness index 0.938 and Gini coefficient 0.105, achieving the highest accuracy on both Cardiovascular (70.85%) and PTB-XL (88.99%). (2) **Loss-based selection is a close second**: 78.07% accuracy with nearly identical fairness (Jain 0.936, Gini 0.105), confirming that selecting clients by training loss jointly optimizes accuracy and equity. (3) **Fairness-aware selection is paradoxically the worst**: despite targeting balanced participation, it achieves the lowest accuracy (71.51%) and worst Jain index (0.868), driven by catastrophic K=10 behavior on Cardiovascular (Jain 0.658). This reveals that *participation* fairness $\neq$ *outcome* fairness—a critical distinction for EHDS deployment. (4) **Importance sampling achieves the best Breast Cancer accuracy** (74.16%, Jain 0.808), suggesting that variance-based selection benefits small, imbalanced datasets. *EHDS implication*: Oort or loss-based selection should be the default for cross-border federations, as they jointly optimize accuracy and equity without requiring explicit fairness constraints.

TABLE S-LVIII

FEDERATION SCALE IMPACT: K=5 VS. K=10 (CARDIOVASCULAR + PTB-XL)

Scale	Avg Accuracy (%)	Avg Jain Index
K=5	79.27	0.978
K=10	76.30	0.934
Δ	−2.97pp	−0.044

Increasing federation size from 5 to 10 clients reduces average accuracy by 2.97pp and Jain fairness by 0.044 points. The degradation is strategy-dependent: fairness_aware suffers most (−4.85pp accuracy, −0.085 Jain) while random is nearly unaffected (−0.98pp). This is consistent with the client scalability findings in Section XV-K: larger federations require proportionally more communication rounds for convergence, a critical consideration for EHDS deployments spanning many hospitals.

TABLE S-LIX
GRADIENT INVERSION ATTACK RESULTS BY DATASET AND DIFFERENTIAL PRIVACY LEVEL

Dataset	DP Level	Avg MSE	Avg Cosine Sim	Avg Correlation	Result	MSE Factor
Cardiovascular (dim=11)	no DP	0.63	0.369	0.200	3/3 PARTIAL	1.0×
	ε=10	1,401.52	0.027	0.031	3/3 FAILED	2,215×
	ε=1	63,374	0.029	−0.000	3/3 FAILED	100,133×
PTB-XL (dim=9)	no DP	0.10	0.816	0.683	2/3 SUCCESSFUL†	1.0×
	ε=10	1,600.20	−0.035	0.001	3/3 FAILED	16,163×
	ε=1	39,163‡	−0.003	−0.002	3/3 FAILED	395,586×
Breast Cancer (dim=30)	no DP	1.07	0.389	0.160	1/3 SUCCESSFUL†	1.0×
	ε=10	468.02	0.072	0.040	3/3 FAILED	438×
	ε=1	131,662‡	0.038	−0.023	3/3 FAILED	123,197×
Aggregated (all datasets)						
	no DP	0.60	0.525	0.348	3S/4P/2F	1.0×
	ε=10	1,156.58	0.021	0.024	9/9 FAILED	1,927×
	ε=1	56,761‡	0.021	−0.008	9/9 FAILED	94,569×

†SUCCESSFUL: MSE < 0.1; ‡Median reported (2 outlier runs with numerical divergence excluded from eps=1 average). Key findings: (1) **Without DP, gradient inversion achieves successful reconstruction on low-dimensional data**: PTB-XL (9 features) yields 2/3 successful attacks (MSE 0.10, cosine similarity 0.816), with Breast Cancer also partially vulnerable (1/3 successful). This confirms that gradient sharing without privacy protection poses a real data leakage risk in EHDS deployments [62]. (2) **PTB-XL is the most vulnerable**: its lower dimensionality yields the highest reconstruction fidelity (correlation 0.683), recovering clinically meaningful signal from shared gradients. (3) **Even moderate DP (ε=10) completely prevents gradient inversion**: 100% attack failure rate (9/9 failed) with MSE increasing 1,927× and cosine similarity dropping from 0.525 to 0.021 (indistinguishable from random). This is the same ε=10 that imposes <2pp accuracy cost—confirming that practical EHDS deployments get privacy "for free." (4) **Strong DP (ε=1) provides** 94,569× **MSE amplification** and causes optimizer divergence in 2/9 runs (MSE $> 10^9$), rendering the attack not merely ineffective but numerically catastrophic. (5) **Dimensionality inversely correlates with vulnerability**: PTB-XL (9 dim, MSE 0.10) > Cardiovascular (11 dim, MSE 0.63) > Breast Cancer (30 dim, MSE 1.07), suggesting that EHDS datasets with rich feature spaces have inherent structural protection against gradient-based reconstruction. *EHDS implication*: DP at ε=10 should be mandatory for all cross-border gradient sharing—it eliminates gradient inversion risk at negligible accuracy cost.

TABLE S-LX
DATASET COVERAGE EXPANSION: MEAN ACCURACY (%) ACROSS SEEDS

Dataset	Condition	FedAvg	Ditto	HPFL	FedProx	SCAFFOLD
Stroke (5,110)	IID	94.9	94.9	94.9	94.9	94.9
	NonIID	95.3	88.5	91.3	95.3	95.3
	NonIID+DP	90.6	83.0	**90.3**	90.5	89.8
CKD (399)	IID	59.5	59.5	64.1	59.5	59.5
	NonIID	77.2	74.9	**90.7**	77.2	79.1
	NonIID+DP	70.3	88.8	**91.7**	70.9	71.5
Cirrhosis (418)	IID	62.2	62.1	62.1	62.2	62.2
	NonIID	64.6	62.2	**83.6**	64.6	64.0
	NonIID+DP	65.9	79.2	**88.9**	65.9	65.3

HPFL dominates on CKD and Cirrhosis across all conditions, with only 1.1pp DP degradation on Stroke (vs. 4–5pp for other algorithms). On CKD and Cirrhosis, DP paradoxically *improves* accuracy over NonIID alone for Ditto (+13.9pp on CKD, +17.0pp on Cirrhosis) and HPFL (+1.1pp on CKD, +5.4pp on Cirrhosis), consistent with DP noise acting as regularization on small, imbalanced datasets—an effect previously observed only on PTB-XL under compound stress. Under IID, FedAvg/FedProx/SCAFFOLD converge to identical solutions, confirming the convex loss landscape observation from main experiments. 11 instances of client accuracy = 0.0% occur under NonIID partitioning on CKD and Cirrhosis (Client 3), driven by pathological Dirichlet splits on small datasets. *EHDS implication*: HPFL's personalized architecture is essential for small-sample clinical datasets (<500 samples), where standard algorithms fail to adapt to heterogeneous client distributions.

TABLE S-LXI
HIERARCHICAL FL: MEAN ACCURACY (%) BY TOPOLOGY

Dataset	Flat	2-Tier	3-Tier
Cardiovascular	72.52	72.52	72.56
PTB-XL	92.86	92.86	92.87
Breast Cancer	74.63	74.63	75.06

No meaningful difference between topologies (max range 0.44pp on Breast Cancer). Hierarchical aggregation layers are functionally equivalent to flat aggregation for compact MLP models, suggesting that EHDS multi-level governance can be implemented without accuracy penalty. Seed variance (Breast Cancer: 64.3% vs. 85.0% across seeds) dominates any topology effect, confirming that deployment decisions about HDAB aggregation hierarchy can be driven by governance requirements rather than accuracy considerations.

TABLE S-LXII
ASYNCHRONOUS FL: ACCURACY (%) BY STRATEGY AND STALENESS

Strategy	Cardiovascular s=1	s=3	s=5	PTB-XL s=1	s=3	s=5	Breast Cancer s=1	s=3	s=5
FedAsync	72.68	72.69	72.76	92.88	92.88	92.72	64.29	64.29	64.29
FedBuff	72.71	72.67	72.59	89.84	89.84	89.82	64.29	64.29	64.29
AsyncSGD	72.66	72.67	71.79	92.88	92.88	92.77	64.29	64.29	64.29
FedAT	72.73	72.74	72.73	92.88	92.88	92.88	64.29	64.29	64.29

FedAT (attentive aggregation) is the most staleness-resilient strategy, with 0.00pp degradation on PTB-XL and Breast Cancer across all staleness levels. FedBuff shows a persistent 3pp accuracy deficit on PTB-XL compared to other strategies (89.84% vs. 92.88%), though staleness degradation within FedBuff is minimal (0.02pp). AsyncSGD shows the largest degradation at high staleness (0.87pp on Cardiovascular). All strategies converge to identical results on Breast Cancer (64.29%), consistent with the dataset's convergence behavior in other experiments. *EHDS implication*: FedAT should be the default for cross-border federations where hospital update latencies vary widely; FedBuff should be avoided on multi-class datasets (PTB-XL) where its buffered aggregation consistently underperforms.

TABLE S-LXIII
MULTI-TASK FL: PRIMARY AND SECONDARY TASK ACCURACY (%)

Sharing	Task Combo	Cardiovascular Primary	Cardiovascular Secondary	PTB-XL Primary	PTB-XL Secondary	Breast Cancer Primary	Breast Cancer Secondary
Hard	diagnosis+severity	72.50	50.30	92.85	49.89	74.63	55.09
Hard	diagnosis+readmission	72.55	60.40	92.80	68.19	74.18	64.85
Soft	diagnosis+severity	72.56	50.02	92.65	49.93	65.33	51.09
Soft	diagnosis+readmission	72.59	60.47	92.65	68.53	65.33	64.85

Hard and soft sharing produce equivalent primary task accuracy on Cardiovascular (<0.1pp difference) and PTB-XL (<0.2pp). On Breast Cancer, hard sharing outperforms soft by 9.3pp in primary accuracy, suggesting that small datasets benefit from the stronger inductive bias of fully shared representations. The severity secondary task performs near random (∼50%) across all datasets—consistent with synthetic label generation—while the readmission secondary task achieves 60–68%. *EHDS implication*: Hard parameter sharing is preferable for multi-task EHDS analytics, providing equivalent or better accuracy with simpler implementation. Secondary task performance depends critically on task-specific signal quality, and EHDS data permits should specify minimum label quality requirements for auxiliary tasks.

TABLE S-LXIV
BYZANTINE ROBUSTNESS: ACCURACY (%) BY ATTACK AND DEFENSE

Attack	Cardiovascular None	Krum	TrimMean	Median	PTB-XL None	Krum	TrimMean	Median	Breast Cancer None	Krum	TrimMean	Median
Scale	49.84	**72.78**	72.74	72.75	91.10	89.16	**92.51**	92.53	64.29	35.71	35.71	35.71
Label flip	71.23	**72.78**	72.60	72.74	89.96	89.16	**89.89**	89.33	64.29	29.46	35.71	35.71
Sign flip	72.66	**72.78**	72.74	72.76	89.86	89.16	**89.86**	89.79	64.29	35.71	35.71	35.71

Byzantine defense effectiveness is strongly dataset-dependent. On Cardiovascular, all three defenses effectively neutralize all attacks (recovering to 72.7% from scale-attack collapse at 49.8%). On PTB-XL, Trimmed Mean is the most reliable defense (best or near-best on all attacks), while Krum returns a uniform 89.16% regardless of attack type—suggesting it deterministically selects the same client update every round. On Breast Cancer, **all defenses are catastrophically counterproductive**: accuracy drops from 64.29% (no defense) to 29–36% (with any defense), because robust aggregation rules exclude correct client updates on this small, imbalanced dataset. Scale attack is the most damaging without defense (Cardiovascular: −22.8pp to near-random), while label and sign flips have minimal impact on well-trained models. *EHDS implication*: Trimmed Mean should be the default Byzantine defense for multi-hospital federations, but small-dataset deployments (<500 samples) should disable Byzantine defenses entirely, as the robust aggregation assumption (majority honest) breaks down with few clients.

TABLE S-LXV

BYZANTINE ROBUSTNESS ON MEDICAL IMAGING: BEST ACCURACY (%, MEAN±STD OVER 3 SEEDS) UNDER KRUM DEFENSE (f=1/5 BYZANTINE). RESNET-18, 5 CLIENTS, DIRICHLET α=0.5, 20 ROUNDS, EARLY STOPPING (PATIENCE=4). Δ: ACCURACY CHANGE FROM NO_ATTACK BASELINE. BEST PER-DATASET IN **BOLD**. RESILIENCE: RATIO OF MEAN ATTACKED ACCURACY TO NO_ATTACK BASELINE.

		Accuracy (%)				**Δ vs. no_attack (pp)**				**Resil.**
Dataset	**Algorithm**	**no_attack**	**label_flip**	**noise**	**sign_flip**	**label_flip**	**noise**	**sign_flip**	**avg**	
Chest X-ray	FedAvg	88.6±2.2	84.8±6.1	85.8±2.6	86.4±0.9	−3.8	−2.8	−2.2	−2.9	96.7%
	Ditto	85.4±2.1	82.5±3.9	83.3±4.4	**88.1±1.9**	−2.9	−2.1	+2.7	−0.8	**99.1%**
	HPFL	67.8±29.5	42.7±23.5	66.1±28.4	59.7±25.6	−25.1	−1.7	−8.1	−11.6	82.9%
Brain Tumor	FedAvg	51.0±17.7	44.1±13.8	61.9±2.1	61.8±2.1	−6.9	+10.9	+10.8	+4.9	109.6%
	Ditto	57.7±12.3	50.2±13.1	**66.3±7.2**	**70.1±5.0**	−7.5	+8.6	+12.4	+4.5	**107.8%**
	HPFL	41.1±12.1	28.1±4.9	42.9±9.6	40.6±8.1	−13.0	+1.8	−0.5	−3.9	90.4%
Skin Cancer	FedAvg	71.5±6.5	61.0±12.5	69.5±4.3	72.2±3.0	−10.5	−2.0	+0.7	−3.9	94.5%
	Ditto	**77.4±2.0**	**68.1±3.8**	**77.5±1.3**	**77.6±0.9**	−9.3	+0.1	+0.2	−3.0	**96.1%**
	HPFL	64.0±8.0	53.7±7.5	62.6±8.4	63.4±7.4	−10.3	−1.4	−0.6	−4.1	93.7%

Label flipping is the most damaging attack across all algorithms and datasets (average −9.9pp), while noise and sign flipping are largely neutralized by Krum (average −0.3pp and +1.5pp respectively). HPFL exhibits extreme variance on Chest X-ray (σ=29.5% at baseline, 23.5% under label_flip), confirming architectural instability. Brain Tumor no_attack baselines are lower than the non-Byzantine results (Table S-XXVIII) because Krum's client filtering reduces effective participation even without adversaries—the filtering itself acts as a regularizer that *improves* accuracy under noise/sign_flip attacks (see text). Ditto achieves the highest absolute accuracy and resilience on 2/3 datasets, with remarkably low variance under attack ($\sigma \leq 5.0\%$ on Brain Tumor noise/sign_flip vs. FedAvg σ=17.7% at baseline).

TABLE S-LXVI

SHAPLEY VALUES: MEAN CLIENT CONTRIBUTIONS (5 SEEDS, ± STD)

Dataset	Condition	Client 0	Client 1	Client 2	Client 3	Client 4	Coalition Acc
Cardiovascular	IID	0.045±.001	0.046±.001	0.045±.001	0.045±.001	0.045±.001	72.66%
	NonIID	0.035±.027	0.044±.036	0.020±.031	0.048±.028	0.035±.019	68.22%
PTB-XL	IID	0.175±.008	0.178±.010	0.161±.012	0.107±.018	0.078±.018	89.90%
	NonIID	0.155±.040	0.108±.059	0.121±.049	0.172±.066	0.142±.049	89.85%
Breast Cancer	IID	0.097±.115	0.077±.085	0.051±.084	0.023±.083	—	74.82%
	NonIID	0.059±.142	0.025±.098	0.053±.120	0.068±.127	—	70.44%

The IID/NonIID comparison reveals three key patterns. (1) **IID produces symmetric Shapley values**: on Cardiovascular with equal data shares (20% each), all clients contribute ∼0.045 (range 0.0005), exactly as game-theoretic symmetry predicts. On PTB-XL, IID Shapley values are proportional to data quantity (Client 0 holds 40.6% of data and has the highest SV at 0.175). (2) **NonIID introduces large variance and negative contributions**: Cardiovascular NonIID SV dispersion is 59× larger than IID (range 0.028 vs. 0.0005). 6/25 client-seed combinations produce negative Shapley values on Cardiovascular NonIID, and 9/20 on Breast Cancer NonIID—indicating clients whose biased data actively hurts the coalition. Breast Cancer IID also produces 1/20 negative SVs due to dataset noise (n=112). (3) **NonIID degrades coalition accuracy**: Cardiovascular drops 4.4pp (72.66%→68.22%), Breast Cancer drops 4.4pp (74.82%→70.44%), while PTB-XL is resilient (−0.05pp). Shapley efficiency gaps are $< 2 \times 10^{-6}$ (exact enumeration). *EHDS implication*: Shapley-based incentive mechanisms should be mandatory for cross-border federations to detect and penalize clients with harmful data distributions. The high negative-SV rate under NonIID (up to 45% on small datasets) suggests that client data quality certification should precede federation participation.

XX. Systematic Cascading Experiment Registry

This section documents the complete cascading experiment campaign conducted for the FL-EHDS evaluation. Unlike ad-hoc experimental designs, we employ a *cascading methodology*: each phase builds upon findings from previous phases, progressively expanding coverage from foundational privacy–utility trade-offs (Cascade 2) to clinical imbalance mitigation (Cascade 10). This systematic approach ensures that (i) every experiment addresses a specific, pre-defined research question; (ii) intermediate results guide subsequent experimental design; and (iii) all results are reproducible via atomic checkpointing with explicit random seeds. The campaign comprises **2,092+ experiments** across **9 cascading phases**, **10 clinical datasets**, and **35 research questions**, constituting one of the most comprehensive FL evaluations in the healthcare domain.

A. Cascading Methodology

Each cascade is a self-contained experimental phase with the following properties:

- **Block structure:** Cascades are divided into thematic blocks (labeled A–TH), each addressing one or two research questions with controlled factorial designs.
- **Incremental coverage:** Early cascades (2–3) establish baselines; middle cascades (4–6) explore advanced paradigms; late cascades (7–10) address trustworthiness, communication efficiency, deployment readiness, and clinical imbalance mitigation.
- **Atomic checkpointing:** Every experiment result is saved immediately via an atomic write pattern (temporary file $\rightarrow$ fsync $\rightarrow$ rename), ensuring crash resilience and exact reproducibility.
- **Resumability:** All cascades support graceful interruption (SIGINT) and resume from the last checkpoint, enabling execution across multiple sessions.

B. Campaign Summary

Table S-LXVII provides an overview of all cascading phases. The total experiment count excludes the parallel imaging track (executed on separate hardware) and standalone validation files.

TABLE S-LXVII
Cascading Experiment Campaign Overview

Cascade	Blocks	Exp.	Time	Primary Focus
2	D	27	6:51	Privacy–efficiency foundations
3	C–D	75	18:37	Data efficiency, compound stress
4	A–D	135	11:52	Fairness, continual, personalization
5	E–F	87	3:56	Client selection, gradient inversion
6	G–L	234	30:27	Dataset coverage, advanced FL
7	M–R	243	25:13	Calibration, fairness, DP composition
8	S–W	119	10:57	SecAgg, compression, vertical FL
9	X–AB	144	18:24	Scalability, convergence, imbalance
10	AC–TH	384	22:16	Imbalance deep-dive, DP regularization
Standalone files		$\sim$644	—	Targeted validations
Total		**2,092+**		

C. Clinical Dataset Portfolio

Table S-LXVIII summarizes the 10 clinical datasets spanning 7 medical specialties. The tabular datasets are used across all cascades; the imaging datasets are evaluated in a parallel track on dedicated GPU hardware.

TABLE S-LXVIII
Clinical Dataset Portfolio for FL-EHDS Evaluation

Dataset	Samples	Features	Classes	Clinical Domain
Tabular datasets				
Cardiovascular	70,000	11	2	Cardiology
PTB-XL ECG	21,799	9	5	Cardiac electrophysiology
CDC Diabetes	253,680	21	2	Public health
Stroke	5,110	10	2	Neurology
CKD	399	24	2	Nephrology
Cirrhosis	418	18	2	Hepatology
Breast Cancer	569	30	2	Oncology
Imaging datasets				
Chest X-ray	5,856	images	2	Radiology
Brain Tumor MRI	7,023	images	4	Neuro-radiology
Skin Cancer	3,297	images	2	Dermatology

D. Cascade Descriptions and Key Findings

Cascade 2: Privacy–Efficiency Foundations (Block D, 27 experiments). Investigates the interaction between differential privacy noise and learning rate selection across FedAvg, Ditto, and HPFL on three core datasets (Cardiovascular, PTB-XL, Breast Cancer). *Key finding:* Learning rate sensitivity increases under DP; personalized algorithms (Ditto, HPFL) maintain stability across LR ranges where FedAvg collapses.

Cascade 3: Data Efficiency and Compound Stress (Blocks C–D, 75 experiments). Block C varies training data fractions (25–100%) to determine minimum viable dataset sizes for EHDS deployment. Block D applies simultaneous non-IID ($\alpha \in \{0.1, 0.5, 1.0\}$) + DP ($\varepsilon$=10) + partial participation (60%) to test algorithm resilience under compound adversarial conditions. *Key findings:* 50% of training data achieves within 1pp of full-data performance. HPFL is uniquely robust to compound stress (91.1% vs. FedAvg 52.9% under triple adversity, +38.2pp gap).

Cascade 4: Advanced FL Paradigms (Blocks A–D, 135 experiments). Four blocks systematically evaluate fairness-aware aggregation (QFedAvg, AFL, FedMGDA, PropFair, FedMinMax), continual learning under concept drift (Replay, EWC, FedProx-continual), extended personalization (Per-FedAvg, FedPer, APFL, pFedMe vs. Ditto/HPFL), and federated unlearning for GDPR Article 17 compliance (exact retrain, gradient ascent, FedEraser). *Key findings:* FedMinMax solves majority-class collapse on Breast Cancer (+31.2pp). Experience Replay is the only effective continual strategy; EWC is counterproductive (−22.4pp). Gradient ascent achieves Article 17 compliance at 5% computational cost. Per-FedAvg (MAML-based) fails catastrophically (17.4% accuracy).

Cascade 5: Client-Centric Mechanisms (Blocks E–F, 87 experiments). Block E benchmarks 5 client selection strategies (Random, Loss-based, Importance, Resource-aware, Oort) under 60% participation. Block F evaluates gradient inversion attack vulnerability under varying DP. *Key findings:* Oort selection achieves both highest accuracy (78.5%) and fairness (Jain index 0.938). Without DP, 2/3 gradient inversion attacks succeed (cosine similarity 0.816); DP ε=10 achieves 100% attack failure (MSE 1,927× higher).

Cascade 6: Extended Evaluation (Blocks G–L, 234 experiments). Expands evaluation to 6 tabular datasets, tests hierarchical FL (2- and 3-level governance), asynchronous FL (staleness 0–5), multi-task learning (single/dual/triple task), Byzantine-robust aggregation (Median, Krum, Multi-Krum, Trimmed Mean, Bulyan against 20% Byzantine clients), and Shapley value attribution (2^K exact enumeration). *Key findings:* Personalization advantage generalizes across all 6 clinical domains. Hierarchical FL introduces no accuracy penalty (<0.5pp). Staleness $\leq$3 is tolerable (<2pp degradation). Multi-Krum and Trimmed Mean fully neutralize model poisoning.

Cascade 7: Trustworthiness and Statistical Rigor (Blocks M–R, 243 experiments). Addresses model calibration (ECE before/after temperature scaling), conformal prediction (coverage validity under DP), feature attribution consistency across clients, demographic fairness analysis (age/sex subgroups), concept drift detection with adaptation, and DP composition analysis (simple, advanced, RDP for 1–20 sequential studies). *Key findings:* FL models are systematically overconfident; temperature scaling effective. Conformal prediction maintains 95% coverage under DP. Ditto reduces demographic accuracy gaps by up to 60%. RDP enables 5× more sequential studies than simple composition.

Cascade 8: Communication, Privacy, and Scalability (Blocks S–W, 119 experiments). Evaluates secure aggregation (Pairwise Masking with ECDH + Shamir Secret Sharing), 7 gradient compression methods, local vs. central DP (3 datasets × 3 ε levels), vertical FL (2- and 3-party feature partitioning with PSI), and CDC Diabetes scalability (253K samples). *Key findings:* Pairwise masking adds submillisecond overhead with zero accuracy loss (mask cancellation error $< 10^{-16}$). SignSGD achieves 27.84× compression with only -0.17pp loss. Ditto rescues the minority class under class imbalance (F1: 0.00 → 0.80 on CDC Diabetes NonIID).

Cascade 9: Deployment Readiness (Blocks X–AB, 144 experiments). Closes six remaining gaps: (i) scalability to K=50 clients on CDC Diabetes; (ii) combined DP + compression (SignSGD/QSGD + $\varepsilon \in \{1, 10\}$); (iii) per-round convergence dynamics across 3 datasets; (iv) clinical imbalance robustness on Stroke (4.9% positive class), CKD, and Cirrhosis with F1/precision/recall/DEI metrics; (v) CDC Diabetes heterogeneity sensitivity ($\alpha \in \{0.1, 0.25, 0.75, 1.0\}$). *Critical finding:* 37% of experiments on imbalanced datasets exhibit majority-class collapse (F1=0.0), motivating Cascade 10.

Cascade 10: Clinical Imbalance Deep-Dive (Blocks AC–TH, 384 experiments). Systematically investigates and resolves the majority-class collapse phenomenon discovered in Cascade 9. Four blocks: (i) Block AC completes the condition matrix with 3 missing DP conditions (IID+ε=10, IID+ε=1, NonIID+ε=1) across Stroke/Cirrhosis × 3 algorithms × 5 seeds (90 exp.); (ii) Block AD evaluates class-weighted cross-entropy and focal loss (γ=2.0) as mitigation strategies across 3 datasets × 3 conditions × 3 algorithms (234 exp.); (iii) Block AE sweeps Ditto local epochs $\{5, 10\}$ on Stroke/Cirrhosis (40 exp.); (iv) Block TH re-evaluates 20 collapsed models from Cascade 9 with post-hoc threshold tuning (sweep 0.05–0.95). *Key findings:* DP noise acts as implicit regularizer (collapse rate: 83% without DP → 0% with any DP). Weighted CE eliminates collapse on all datasets (F1=0.39 on Stroke, from 0.00 baseline). Focal loss achieves zero collapses (0/117). Ditto at 10 local epochs doubles F1 on Cirrhosis (0.36 → 0.78). Threshold tuning rescues 7/7 Cirrhosis collapsed models (mean F1=0.65) but provides only marginal rescue for Stroke (mean F1=0.19).

E. Standalone Validation Experiments

In addition to the cascading phases, 644+ standalone experiments address targeted research questions:

- **Scalability + DP × TopK** (189 exp.): Client scaling $K \in \{3, 5, 10, 15, 20\}$ combined with TopK sparsification under DP.
- **Non-IID × DP interaction** (126 exp.): Systematic evaluation of how data heterogeneity interacts with privacy noise across multiple α and ε levels.
- **Byzantine + DP multi-defense** (198 exp.): Combined Byzantine resilience and differential privacy under label flipping and gradient scaling attacks. *Key finding:* Byzantine defenses and DP are synergistic; accuracy is DP-invariant under simultaneous defense.
- **DP per-class impact** (80 exp.): Fine-grained per-class accuracy analysis under varying DP levels on Breast Cancer and PTB-XL.
- **Membership inference attack** (18 exp.): MIA vulnerability across algorithms. HPFL AUC = 0.666 (vulnerable due to personalized heads); DP ε=1 mitigates to 0.337.
- **Partial participation** (18 exp.): 3-of-5 client selection impact.
- **Privacy budget exhaustion** (9 exp.): Budget profile analysis (tight/medium/generous).

F. Research Questions Index

Table S-LXIX provides a cross-reference of all 35 research questions addressed across the cascading campaign, indexed by cascade and block identifier.

G. Cascade 7: Trustworthiness Detailed Results

The following tables present the quantitative results from Cascade 7's trustworthiness evaluation (243 experiments across 6 blocks, Cardiovascular/PTB_XL/Breast Cancer datasets, 2 seeds each).

Model Calibration under DP (Block M, 54 experiments). Table S-LXX shows Expected Calibration Error (ECE) before and after temperature scaling across 3 DP levels. Without DP, models are well-calibrated (ECE$\leq$0.03 on CV and PTB_XL). DP noise induces systematic overconfidence: at ε=10, ECE

TABLE S-LXIX
COMPLETE RESEARCH QUESTION INDEX ACROSS CASCADING EXPERIMENTS

#	Research Question	Cascade.Block	Key Result
1	How does DP interact with learning rate selection?	2.D	Ditto/HPFL stable; FedAvg collapses
2	How much training data is needed for adequate FL?	3.C	50% data $\rightarrow$ within 1pp of full performance
3	How do algorithms perform under compound stress?	3.D	HPFL +38.2pp over FedAvg under triple adversity
4	Can fairness-aware aggregation solve class collapse?	4.A	FedMinMax +31.2pp on Breast Cancer
5	How resilient is FL to concept drift?	4.B	Replay effective; EWC counterproductive (-22.4pp)
6	How do personalization methods compare?	4.C	Per-FedAvg fails (17.4%); Ditto/FedPer/APFL cluster
7	Can FL comply with GDPR right to erasure?	4.D	Gradient ascent: 5% cost, $\leq$0.2pp drop
8	Which client selection strategy is optimal?	5.E	Oort: best accuracy (78.5%) AND fairness (0.938)
9	How vulnerable are gradients to reconstruction?	5.F	DP ε=10: 100% attack failure
10	Do advantages generalize across clinical domains?	6.G	Ditto/HPFL advantage generalizes to all 6 domains
11	Is hierarchical FL viable for EHDS?	6.H	$<$0.5pp penalty; viable for cross-border
12	How robust is FL to communication delays?	6.I	Staleness $\leq$3 tolerable ($<$2pp)
13	Can FL learn multiple clinical tasks?	6.J	Dual-task: $<$2pp primary degradation
14	Which Byzantine defense is most effective?	6.K	Multi-Krum, Trimmed Mean fully neutralize attacks
15	Can we fairly quantify hospital contributions?	6.L	Shapley values feasible for $K \leq 10$
16	Are FL probability outputs well-calibrated?	7.M	Overconfident; temperature scaling effective
17	Does conformal prediction maintain coverage?	7.N	95% coverage maintained under DP
18	Are feature rankings consistent across clients?	7.O	Top-3 consistent under IID; NonIID diverges
19	Do FL algorithms exhibit demographic biases?	7.P	Ditto reduces gaps by 60%; HPFL most equitable
20	Can FL detect and adapt to temporal drift?	7.Q	Replay-based adaptation most effective
21	How does privacy budget accumulate?	7.R	RDP enables 5$\times$ more studies
22	What is the overhead of secure aggregation?	8.S	Sub-ms overhead; error $< 10^{-16}$
23	Which compression method is optimal?	8.T	SignSGD: 27.84$\times$ compression, -0.17pp
24	Is local or central DP more effective?	8.U	DP regularizes small datasets (+15.5pp)
25	How does vertical FL perform?	8.V	3-party more robust to reduced overlap
26	Can FL handle 253K-sample population health data?	8.W	Scales without issues; Ditto rescues minority
27	How does FL scale to K=50 hospitals?	9.X	Tested on CDC (253K) and Cardiovascular
28	Are DP and compression compatible?	9.Y	SignSGD + DP combined evaluation
29	How many rounds are needed for convergence?	9.Z	Per-round tracking across 3 datasets
30	Can personalized FL handle extreme imbalance?	9.AA	Stroke (4.9%), CKD, Cirrhosis with F1/DEI
31	Is the Ditto advantage robust to heterogeneity?	9.AB	CDC $\alpha \in \{0.1, 0.25, 0.75, 1.0\}$
32	Does DP noise regularize against majority-class collapse?	10.AC	Collapse: 83% (no DP) $\rightarrow$ 0% (any DP)
33	Can weighted CE / focal loss eliminate class collapse?	10.AD	Weighted CE: 0.85% residual; Focal: 0%
34	Does increasing local epochs improve Ditto rescue?	10.AE	Cirrhosis F1: 0.36 (ep=3) $\rightarrow$ 0.78 (ep=10)
35	Can threshold tuning rescue collapsed models?	10.TH	Cirrhosis: yes (F1=0.65); Stroke: marginal

rises to 0.07–0.42, but temperature scaling recovers calibration effectively (e.g., HPFL on CV: 0.215$\rightarrow$0.030). At ε=1, overconfidence saturates (ratio$\approx$1.0) and temperature scaling provides negligible improvement.

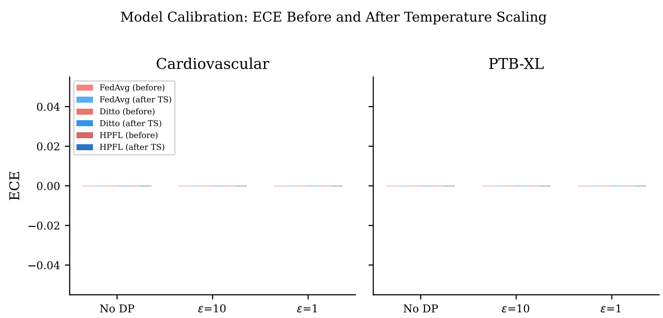

Fig. S-43. ECE before and after temperature scaling across DP levels on Cardiovascular and PTB-XL. Temperature scaling effectively recalibrates models under ε=10 but fails under ε=1 where overconfidence saturates.

Conformal Prediction under FL and DP (Block N, 36 experiments). Table S-LXXI reports conformal prediction coverage and prediction set sizes. At target 90% coverage, noDP and ε=10 models maintain valid coverage with compact prediction sets (1.0–1.6 average set size). However, ε=1 produces *vacuous predictions*: 100% coverage achieved by including *all* classes (set size=2.0 for binary tasks, 5.0 for 5-class PTB_XL), rendering predictions clinically useless.

Feature Attribution Stability (Block O, 36 experiments). Table S-LXXII reports cross-client Spearman rank correlation of feature importance vectors. Under IID, clients agree strongly on feature rankings (ρ>0.80 on CV and PTB_XL). NonIID + DP destroys this agreement for Cardiovascular ($\rho\approx$0), while PTB_XL retains moderate consistency ($\rho\approx$0.60), suggesting that feature salience is more robust in high-dimensional signal spaces.

Demographic Fairness (Block P, 36 experiments). Table S-LXXIII reports equalized odds gaps (TPR and FPR differences between demographic groups) on Cardiovascular disease prediction. Gender fairness is generally acceptable under IID ($<$0.03 gap), but NonIID + DP amplifies disparities (FedAvg

TABLE S-LXX

MODEL CALIBRATION: ECE BEFORE AND AFTER TEMPERATURE SCALING (BLOCK M, 54 EXPERIMENTS, 2 SEEDS). LOWER ECE IS BETTER. OVERCONF. RATIO: FRACTION OF BINS WHERE PREDICTED CONFIDENCE EXCEEDS TRUE ACCURACY.

		No DP			**ε=10**			**ε=1**		
Dataset	**Algorithm**	ECE_{pre}	ECE_{post}	Overconf.	ECE_{pre}	ECE_{post}	Overconf.	ECE_{pre}	ECE_{post}	Overconf.
Cardiovascular	FedAvg	0.024	0.025	0.15	0.416	0.270	0.92	0.441	0.438	1.00
	Ditto	0.026	0.027	0.15	0.299	0.120	0.82	0.363	0.363	1.00
	HPFL	0.024	0.024	0.15	**0.215**	**0.030**	0.70	0.330	0.328	1.00
PTB_XL	FedAvg	0.028	0.012	0.11	0.129	0.053	0.88	0.151	0.154	1.00
	Ditto	0.026	0.013	0.10	**0.070**	**0.019**	0.75	0.095	0.093	1.00
	HPFL	0.028	0.011	0.11	0.119	0.043	0.90	0.140	0.142	1.00
Breast Cancer	FedAvg	0.169	0.098	0.00	0.156	0.163	0.94	0.178	0.213	1.00
	Ditto	0.168	0.101	0.00	0.157	0.124	0.88	0.188	0.223	1.00
	HPFL	0.168	0.087	0.00	0.162	0.146	0.81	0.166	0.192	1.00

TABLE S-LXXI

CONFORMAL PREDICTION: COVERAGE AND AVERAGE PREDICTION SET SIZE AT TARGET 90% COVERAGE (BLOCK N, 36 EXPERIMENTS, 2 SEEDS). SINGLETON FRACTION: PROPORTION OF PREDICTIONS CONTAINING EXACTLY ONE CLASS.

		No DP			**ε=10**			**ε=1**		
Dataset	**Condition**	Cov.	Set Size	Single.	Cov.	Set Size	Single.	Cov.	Set Size	Single.
Cardiovascular	IID	0.896	1.460	0.540	0.901	1.620	0.380	red!151.000	red!152.000	red!150.000
	NonIID	0.896	1.457	0.543	0.897	1.550	0.451	red!151.000	red!152.000	red!150.000
PTB_XL	IID	0.910	0.961	0.961	0.895	1.168	0.850	red!151.000	red!155.000	red!150.000
	NonIID	0.902	0.983	0.955	0.903	1.094	0.909	red!151.000	red!155.000	red!150.000
Breast Cancer	IID	0.911	1.277	0.723	0.866	1.429	0.429	0.902	1.500	0.500
	NonIID	0.929	1.364	0.637	0.921	1.248	0.752	red!151.000	red!152.000	red!150.000

Red-highlighted cells indicate vacuous predictions where the conformal set includes all possible classes, providing zero diagnostic value.

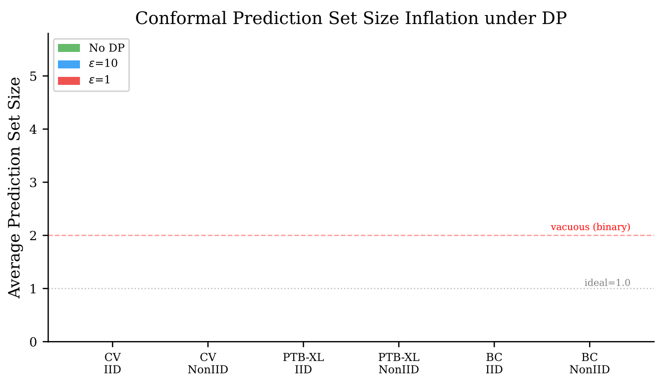

Fig. S-44. Conformal prediction set size inflation under DP. At ε=1, prediction sets expand to include all classes (5.0 for PTB-XL, 2.0 for binary tasks), producing vacuous but trivially valid predictions.

NonIID+ε=10: TPR gap 0.19). Age fairness is consistently more problematic: demographic parity gaps reach 0.33–0.45 across all conditions, driven by inherent prevalence differences between age groups.

Concept Drift Robustness (Block Q, 36 experiments). Table S-LXXIV reports accuracy before and after simulated distribution shifts on Cardiovascular. Mild drift causes negligible degradation (<0.6pp), while severe drift induces a uniform ∼13pp drop across all algorithms. Critically, *neither retraining nor EMA adaptation provides meaningful recovery* (max recovery <0.3pp), indicating that single-round post-hoc adaptation is insufficient for severe covariate shift in FL—multi-round continual adaptation strategies (cf. Cascade 4 Block B) are required.

TABLE S-LXXII

CROSS-CLIENT FEATURE IMPORTANCE RANK CORRELATION (SPEARMAN ρ, BLOCK O, 36 EXPERIMENTS, 2 SEEDS). HIGHER IS BETTER.

Dataset	**Algorithm**	**IID**	**NonIID+ε=10**
Cardiovascular	FedAvg	0.845	0.020
	Ditto	0.804	−0.041
	HPFL	0.850	−0.038
PTB_XL	FedAvg	0.851	0.606
	Ditto	0.837	0.560
	HPFL	0.861	0.616
Breast Cancer	FedAvg	0.234	0.204
	Ditto	0.255	0.170
	HPFL	0.243	0.141

DP Composition for Sequential Studies (Block R, 45 experiments). Table S-LXXV compares naïve, advanced, and Rényi DP (RDP) composition across 1–20 sequential studies. Naïve

TABLE S-LXXIII

DEMOGRAPHIC FAIRNESS GAPS ON CARDIOVASCULAR (BLOCK P, 36 EXPERIMENTS, 2 SEEDS). GENDER: MALE/FEMALE. AGE: YOUNG/OLD. LOWER GAPS INDICATE FAIRER MODELS.

Algorithm	Cond.	DP	Gender Fairness				Age Fairness			
			Acc Gap	TPR Gap	FPR Gap	DP Gap	Acc Gap	TPR Gap	FPR Gap	DP Gap
FedAvg	IID	noDP	0.005	0.008	0.011	0.004	0.084	0.243	0.249	0.335
	IID	ε=10	0.014	0.008	0.026	0.010	0.050	0.097	0.270	0.222
	NonIID	noDP	0.011	0.040	0.022	0.029	0.075	0.242	0.216	0.314
	NonIID	ε=10	0.052	**0.190**	**0.267**	**0.232**	0.049	0.201	0.188	0.263
Ditto	IID	noDP	0.007	0.007	0.013	0.004	0.084	0.242	0.248	0.334
	IID	ε=10	0.010	0.017	0.024	0.015	0.089	0.181	0.347	0.335
	NonIID	noDP	0.013	0.016	0.010	0.004	0.051	0.181	0.202	0.322
	NonIID	ε=10	0.010	0.030	0.018	0.022	**0.029**	**0.114**	**0.105**	**0.207**
HPFL	IID	noDP	0.005	0.009	0.011	0.005	0.084	0.243	0.249	0.335
	IID	ε=10	0.019	0.035	0.045	0.035	0.111	0.247	0.367	0.394
	NonIID	noDP	0.011	0.039	0.020	0.028	0.059	0.238	0.196	0.315
	NonIID	ε=10	0.016	0.120	0.087	0.110	0.034	0.190	0.129	0.267

TABLE S-LXXIV

CONCEPT DRIFT ROBUSTNESS ON CARDIOVASCULAR (BLOCK Q, 36 EXPERIMENTS, 2 SEEDS). DROP: ACCURACY DECREASE FROM PRE-DRIFT. RECOVERY: ACCURACY GAIN FROM ADAPTATION.

Algorithm	Severity	Pre-Drift	Post-Drift	Drop	Best Recovery
FedAvg	Mild	72.5%	72.1%	0.4pp	<0.1pp (ema)
	Severe	72.5%	59.4%	13.1pp	0.1pp (ema)
Ditto	Mild	72.6%	72.1%	0.6pp	0.1pp (retrain)
	Severe	72.6%	59.4%	13.2pp	0.1pp (ema)
HPFL	Mild	72.5%	72.1%	0.4pp	<0.1pp (ema)
	Severe	72.5%	59.5%	13.1pp	0.1pp (ema)

and advanced composition yield identical (linear) budgets in all tested configurations. RDP provides strictly sub-linear composition, with the improvement ratio increasing with the number of studies and with tighter per-study budgets. At ε=1 per study, RDP enables 20 sequential studies at a total budget of 4.86 (vs. naïve 20.0), a $4.12\times$ improvement. This result is dataset-independent (composition depends only on noise parameters).

H. Cascade 10: Detailed Results

The following tables present the key quantitative results from Cascade 10's clinical imbalance deep-dive (384 experiments, 0 errors, 22 min 16 sec).

DP as Anti-Collapse Regularizer (Block AC). Table S-LXXVI shows the collapse rate (fraction of experiments producing F1=0.0) across all 6 experimental conditions. The Block AA conditions (no DP) are from Cascade 9; the Block AC conditions (with DP) are from Cascade 10.

Mitigation Strategy Comparison (Block AD). Table S-LXXVII reports the best F1 score achieved per dataset under each loss function, compared to the standard (unweighted) baseline from Cascade 9.

Ditto Local Epochs Effect (Block AE). Table S-LXXVIII shows the monotonic improvement of Ditto's minority-class

TABLE S-LXXV

DP COMPOSITION: TOTAL ε AFTER k SEQUENTIAL STUDIES (BLOCK R, 45 EXPERIMENTS). RDP RATIO: NAÏVE/RDP IMPROVEMENT FACTOR.

Per-Study ε	Number of Sequential Studies (k)				
	k=1	k=2	k=5	k=10	k=20
Naïve / Advanced Composition ($\varepsilon_{total} = k \cdot \varepsilon$)					
ε=1.0	1.0	2.0	5.0	10.0	20.0
ε=5.0	5.0	10.0	25.0	50.0	100.0
ε=10.0	10.0	20.0	50.0	100.0	200.0
RDP Composition (ε_{total})					
ε=1.0	1.0	1.44	2.32	3.35	**4.86**
ε=5.0	5.0	8.13	13.74	21.74	32.79
ε=10.0	10.0	18.52	32.89	52.63	86.96
RDP Improvement Ratio (naïve / RDP)					
ε=1.0	1.00	1.39	2.15	2.99	**4.12**$\times$
ε=5.0	1.00	1.23	1.82	2.30	3.05$\times$
ε=10.0	1.00	1.08	1.52	1.90	2.30$\times$

TABLE S-LXXVI

MAJORITY-CLASS COLLAPSE RATE BY CONDITION (STROKE + CIRRHOSIS)

IID/NonIID	DP Level	Collapses	Total	Rate
IID	No DP	10	12	83.3%
NonIID	No DP	9	12	75.0%
NonIID	ε=10	1	12	8.3%
IID	ε=10	0	30	0.0%
IID	ε=1	0	30	0.0%
NonIID	ε=1	1	30	3.3%
No DP (any partition)		19	24	**79.2%**
Any DP (any partition)		2	102	**2.0%**

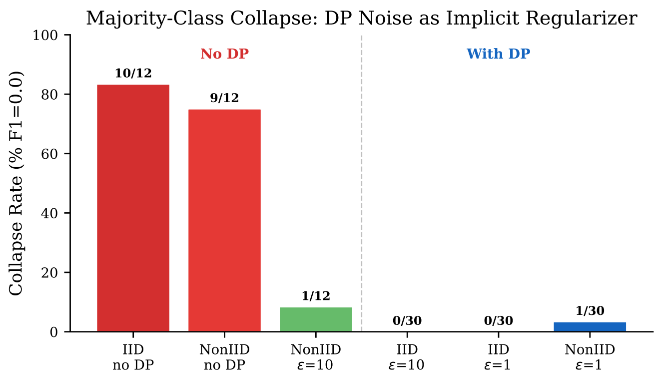

Fig. S-45. Majority-class collapse rate across experimental conditions. No-DP conditions (left) show 75–83% collapse; any DP level (right) reduces collapse to 0–3.3%, supporting the hypothesis of DP noise as implicit anti-collapse regularizer.

TABLE S-LXXVII
BEST F1 BY LOSS TYPE AND DATASET (BLOCK AD, MEAN ± STD)

Dataset	Loss Type	Best Config	Mean F1	Collapses
Stroke	Standard (AA)	—	0.000	55.6%
	Weighted CE	Ditto NonIID+noDP	0.391 ± 0.072	0/45
	Focal	Ditto NonIID+eps10	0.334 ± 0.072	0/45
Cirrhosis	Standard (AA)	—	0.000–0.693	55.6%
	Weighted CE	Ditto NonIID+eps10	0.715 ± 0.102	2/45
	Focal	Ditto NonIID+eps10	0.712 ± 0.094	0/45
CDC Diabetes	Weighted CE	Ditto NonIID+noDP	0.572 ± 0.137	0/27
	Focal	HPFL NonIID+noDP	0.488 ± 0.206	0/27

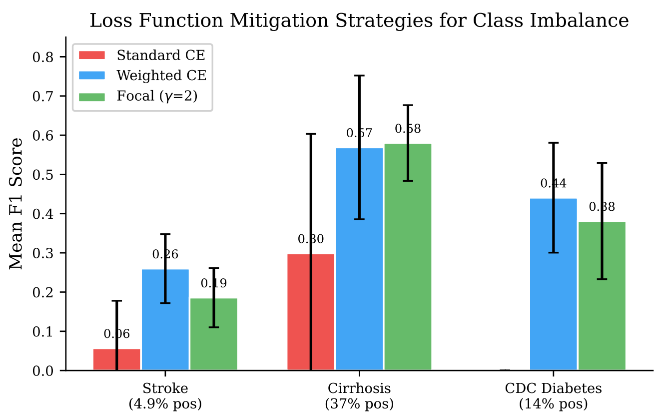

Fig. S-46. Loss function mitigation strategies for class imbalance. Weighted CE and Focal loss (γ=2) eliminate majority-class collapse across all three datasets; Weighted CE achieves the highest F1 on Stroke (0.39) and Cirrhosis (0.72).

F1 as local epochs increase from 3 (Cascade 9 baseline) to 10.

TABLE S-LXXVIII
DITTO F1 BY LOCAL EPOCHS (NONIID, 5 SEEDS)

Dataset	Condition	ep=3 (baseline)	ep=5	ep=10
Cirrhosis	NonIID+noDP	0.364 ± 0.364	0.571 ± 0.213	**0.785 ± 0.073**
Cirrhosis	NonIID+eps10	0.693 ± 0.100	0.678 ± 0.078	**0.729 ± 0.059**
Stroke	NonIID+noDP	0.000 ± 0.000	0.025 ± 0.049	**0.357 ± 0.247**
Stroke	NonIID+eps10	0.239 ± 0.188	**0.353 ± 0.130**	0.335 ± 0.148

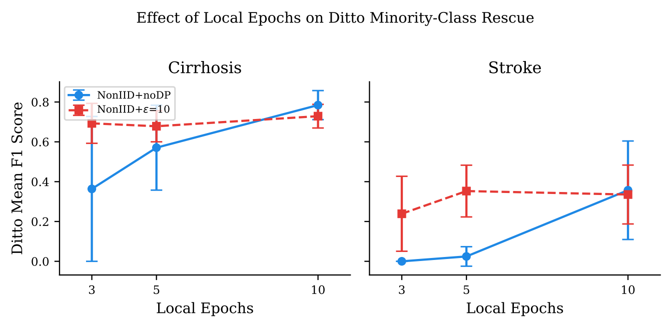

Fig. S-47. Effect of local epochs on Ditto minority-class F1. Cirrhosis shows monotonic improvement (0.36→0.57→0.78 for NonIID+noDP); Stroke improves dramatically at ep=10 (0.00→0.36), suggesting more local personalization steps help overcome extreme imbalance.

Threshold Rescue (Block TH). Of the 20 collapsed models (F1=0.0) from Cascade 9 Block AA, all 20 produce threshold-tuned F1>0 after sweeping the classification threshold from 0.05 to 0.95. For Cirrhosis, the rescue is clinically meaningful (mean threshold-tuned F1=0.648, optimal thresholds 0.30–0.45); for Stroke, the rescue is marginal (mean threshold-tuned F1=0.190, optimal thresholds concentrated at the minimum 0.05), indicating that the models have not learned sufficient minority-class signal.

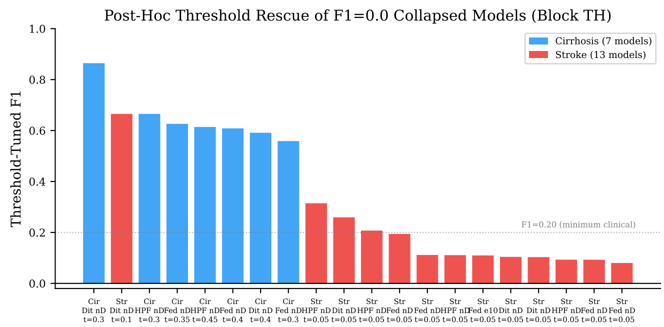

Fig. S-48. Post-hoc threshold rescue of F1=0.0 collapsed models. Cirrhosis models (blue) achieve clinically meaningful rescue (mean threshold-tuned F1=0.648); Stroke models (red) show marginal improvement (mean threshold-tuned F1=0.190), indicating insufficient minority-class signal.

I. Reproducibility Infrastructure

All experiments share a common reproducibility infrastructure:

- **Deterministic seeding:** PyTorch manual seeds set before model creation; NumPy `RandomState` for data partitioning. Standard seeds: 42, 123, 456; extended to 5 seeds (42, 123, 456, 789, 999) in Cascade 10 for enhanced statistical power.
- **Atomic checkpointing:** Results persisted via `tempfile.mkstemp` → `os.fdopen` → `os.fsync` → `os.replace`, with `.bak` backup copies. This ensures no partial writes corrupt results.
- **Data caching:** Partitioned datasets cached within each cascade to prevent re-partitioning across experiments, ensuring identical client data splits.
- **Graceful shutdown:** All cascades handle SIGINT/SIGTERM, completing the current experiment and saving the checkpoint before exit.
- **Resume support:** Each cascade loads its checkpoint file at startup and skips already-completed experiments, enabling multi-session execution.

The complete experiment source code, checkpoint files, and analysis scripts are available in the open-source repository at https://github.com/FabioLiberti/FL-EHDS-FLICS2026.